

TEL AVIV UNIVERSITY

The Iby and Aladar Fleischman Faculty of Engineering

Department of Electrical Engineering

**Deep Learning Architectures for
Two-Hop Relay Communication:
A Comparative Study of Classical and
Neural Network Relay Strategies**

A thesis submitted toward the degree of
Master of Science in Electrical and Electronic Engineering
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This research work was carried out at Tel-Aviv University
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under the supervision of Dr. Anatoly Khina and Prof. Raja Giryes

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Abstract

This thesis compares classical and learned relay strategies for two-hop communication. One configuration is fixed throughout the core study and never departed from: a single-relay SISO link, independent and identically distributed Rayleigh fast fading applied identically on both hops, complex baseband, Gray-coded QPSK, and uncoded transmission, with the relay function as the only variable. Nine relay strategies are implemented, spanning amplify-and-forward (AF), symbol-wise decode-and-forward (DF), a multilayer perceptron (MLP), a hybrid switch, generative models (a variational autoencoder and a conditional generative adversarial network), and sequence models (Transformer, Mamba-S6 and Mamba-2 SSD); eight are evaluated in the final comparison, the conditional generative adversarial network being excluded on cost grounds. All results are Monte Carlo estimates reported with 95% confidence intervals, and the simulator is first validated against closed-form BPSK error probabilities on an additive white Gaussian noise channel, which serves only as that calibration reference; symbol-wise DF on the canonical Rayleigh channel is independently checked the same way, against the closed-form two-hop composition, and matches to within one percent at every measured SNR.

Under the canonical benchmark, the best learned relays outperform AF at low signal-to-noise ratio and attain BER comparable to symbol-wise DF at medium and high SNR, while DF achieves the lowest BER of the relays evaluated at no parameter cost. This should not be read as a general optimality claim for DF. It is an empirical statement about uncoded, per-symbol relaying over the SNR range measured here; the information-theoretic results for block-based DF concern coded transmission and asymptotic blocklengths, and neither they nor the reported measurements establish that DF cannot be beaten by a coded or soft-information relay; a coded, hard-decision relay is measured directly below, though a soft-information one remains open. A complexity study indicates that performance is governed by capacity rather than architecture at high SNR, where an MLP of 169 trainable parameters attains the BER of models two orders of magnitude larger; at low SNR the architectures separate, the feedforward relays tracking DF while the sequence models trail.

A further study is reported separately because it requires a configuration the canonical one cannot absorb. It measures the coding caveat left open above directly: an outer rate-1/2

convolutional code with a genuine block-DF relay beats the uncoded DF baseline above roughly 6–8 dB but is worse below that threshold, which sharpens rather than closes for stronger codes (constraint length swept up to 7). Against that coded classical baseline the learned relays lose clearly at low-to-mid SNR and win at high SNR, and instrumenting the relay output identifies why, which is not that a network out-decodes Viterbi: Viterbi places roughly half as many symbol errors on the air, but block-DF re-encodes after decoding, so its residual errors arrive as valid-but-wrong codewords the destination cannot repair, while a relay that only denoises leaves the code’s redundancy intact and has 57–74% of its errors repaired downstream. Forwarding soft rather than hard decisions improves the learned relay at every SNR on identical weights, but does *not* rescue block-DF, which localizes the liability precisely: it is consuming the code’s redundancy at the relay, not quantizing there. Held to equal spectral efficiency rather than equal E_s/N_0 , moreover, the rate-1/2 code does not pay for itself below 20 dB at all, and where it does the gain is $1.35\times$ while costing $20\times$ the buffering latency and $50\times$ the arithmetic of the learned relay. Letting the code rate itself adapt to the channel, over a grid spanning both constellations and rates 1/2 through 3/4 plus uncoded transmission, bounds the relay-decoding question further: maximizing delivered goodput shows rate selection matters an order of magnitude more than which relay strategy decodes the code, and the optimal choice at high SNR is to switch coding off entirely, delivering under a third of the channel’s unconstrained Shannon capacity even there, a gap set by the finite constellation and residual frame-error rate rather than by relay strategy. That bound is itself conditional on the link being free to choose any blocklength: because a relay that decodes and re-encodes must buffer a full frame twice against a denoising relay’s once, imposing a round-trip latency budget locks block-DF out of coding sooner and reopens the relay-strategy gap, reversing it by up to $1.5\times$ at SNRs where the unconstrained comparison had favored the classical relay.

The principal contribution is organised as a ladder of four layers, each removing exactly one assumption from the layer above while the classical–learned comparison is repeated unchanged. At layer 1, the canonical setup, classical processing wins: symbol-wise DF is not improved upon, which is a property of the problem rather than of the training, since on a memoryless matched channel hard slicing is the optimal per-symbol relay function. Layer 2 adds channel memory in the form of an unknown three-tap intersymbol-interference filter and holds channel knowledge fixed. Here the memoryless classical relays fail outright, their error rate saturating and then rising with transmit power as hard decisions propagate the interference, while a windowed learned relay of 169 parameters restores the link; given the same filter, a Viterbi maximum-likelihood sequence detector remains 1 to 1.5 dB ahead of it, so what this layer separates is the function class, and what the learned relay buys is constant cost per symbol against a trellis cost exponential in channel memory. Layer 3 degrades the channel estimate to a finite pilot budget, and the classical receiver’s advantage is found to be real but purchased: it holds down to roughly

twenty pilots and is lost by ten. Layer 4 removes the channel family altogether, redrawing the channel every block, and the learned relay leads the best blind classical equaliser while avoiding the instability of decision-directed sequence estimation.

The results delineate operating regimes rather than declaring a winner. Where the assumed channel model matches the true channel, classical processing is not improved upon and should be used. Where the channel is unknown, or its estimate is degraded below roughly ten pilot symbols, or no per-block estimate is available at all, the learned relay is the option that continues to work, at fixed cost per symbol and with no identification stage – generalizing across realizations of the trained-on channel family rather than to channel families never presented during training.

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Chapter 1

Introduction

Relay communication is a key technique in contemporary wireless systems that uses intermediate nodes between a source and a destination to extend coverage, enhance reliability, and increase throughput. Classical relay strategies such as amplify-and-forward (AF) and decode-and-forward (DF) have been extensively analyzed and are widely used as baseline protocols; in several canonical relay-channel models, block-based DF is capacity-achieving or capacity-approaching under asymptotically long codewords [1, 2, 3, 4, 5]. As clarified in Remark 4.2, these results concern coded block operation and do not directly characterize the uncoded, symbol-wise setting studied in this thesis.

Recent advances in machine learning have motivated the application of neural networks to physical-layer communication problems, including detection, decoding, and end-to-end system optimization. Neural models can approximate nonlinear mappings that are analytically intractable and have been reported to offer advantages in low-SNR or non-ideal regimes.

This motivates the central question of this thesis: *under what conditions can neural network-based relays improve upon or complement classical relay strategies in two-hop relay communication?*

This thesis answers that question within a unified simulation framework in which the relay function is the only variable. Nine relay methods are evaluated: two classical relay strategies (AF and DF) and seven neural relays spanning supervised learning (MLP, SNR-adaptive Hybrid relay), generative modeling (variational autoencoder VAE and conditional GAN, cGAN, with WGAN-GP), and sequence-based architectures (Transformer, Mamba-S6, and Mamba-2 SSD with structured state-space duality).

The objective is not to propose a new relay design but to answer the question above with a boundary rather than a verdict: to say where learning helps, where it does not, and why. Existing studies typically vary several things at once, a set of architectures and a set of

channel models together, which makes it difficult to attribute a difference in error rate to model structure, to parameter count, or to the channel.

Method: one assumption at a time. This thesis answers the question by holding the comparison fixed and relaxing the receiver’s assumptions in a monotone sequence. The same relays, the same channel, the same metric and the same simulation budget are used at every step; the only thing that changes from one step to the next is a single assumption that is withdrawn. Four such layers are examined, and reading their outcomes in order *is* the answer to the central question.

At **layer 1** the channel is memoryless and perfectly known, which is the canonical setup of Chapter 5. Classical processing is not improved upon here, and the reason is structural rather than a matter of training effort: on a memoryless matched channel the optimal per-symbol relay function is hard slicing, which is precisely what decode-and-forward already does at no parameter cost. **Layer 2** (Chapter 6) adds channel memory and holds channel knowledge fixed. The memoryless classical relays now fail outright rather than degrade, and a windowed learned relay restores the link, but a sequence detector given the true channel is still ahead of it; what this layer isolates is therefore the *function class*, together with the cost of realizing it. **Layer 3** degrades the channel estimate to a finite pilot budget, and the classical receiver’s advantage turns out to be real but purchased: it survives while enough pilots remain and is lost when they do not. **Layer 4** removes knowledge of the channel family altogether, redrawing the channel for every block, and here the learned relay leads.

The shape of that sequence is the contribution. What degrades from layer to layer is not the classical algorithms, which remain optimal for whatever channel they are handed, but their *preconditions*. The learned relay never becomes a better detector; it becomes the only one whose requirements are still satisfied. Hypotheses H1–H5 are settled at layer 1 and its modulation branch, and H6 at layers 2–4.

Scope of this thesis. This thesis studies one communication problem, held unchanged throughout, the classical source–relay–destination (two-hop) link, under *one canonical setup*: SISO on both hops, i.i.d. Rayleigh fading, complex baseband, QPSK, uncoded BER. The only axis varied is the *relay function*. BPSK appears solely on the AWGN channel used to calibrate the simulator against closed-form error probabilities. A dedicated unknown-channel contribution (Chapter 6) is included; all other departures from the canonical setup are identified as future work in Chapter 7:

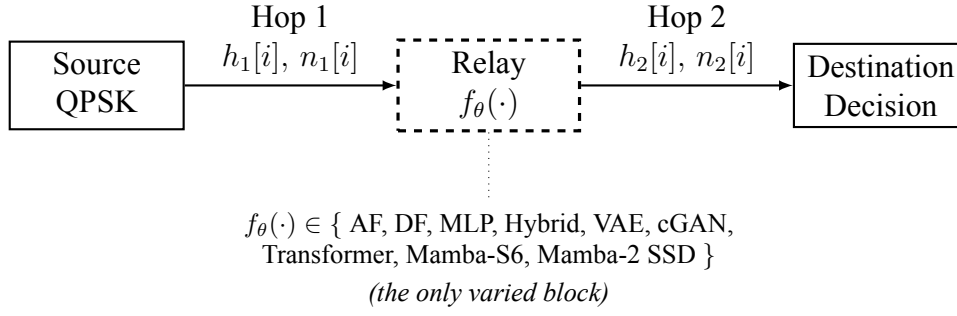


Figure 1.1: Canonical two-hop relay benchmark used throughout the thesis. The source, the channel model on both hops, and the destination are the same in every experiment; only the relay processing function $f_{\theta}(\cdot)$ (dashed) is varied among the nine candidate strategies. The fading itself is not static: $h_1[i]$ and $h_2[i]$ are redrawn every symbol. Observed BER differences are therefore attributable to the relay strategy alone.

Table 1.1: Scope of the thesis: the canonical setup (QPSK on Rayleigh), the 16-QAM extension, and the unknown-channel contribution. No other topology, relay protocol, or performance metric is evaluated in the main line of this work; all such variations are deferred to future work.

Axis	Canonical value (future work in parentheses)
Topology	SISO on both hops (<i>future work: MIMO second hop</i>)
Channel	i.i.d. Rayleigh fast fading (canonical), applied identically on both hops; AWGN solely as the noise term of the canonical model, the simulator being calibrated instead against the closed-form two-hop DF composition on the canonical channel itself (Table 5.2); unknown, mismatched, and time-varying channels are studied in the dedicated contribution (Chapter 6) (<i>future work: Rician and other channel families</i>)
Relay Function (<i>the studied axis</i>)	<i>Classical:</i> AF, DF. <i>Learned:</i> MLP, Hybrid, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD
Modulation	QPSK is paired with the canonical Rayleigh channel throughout (Chapter 5) and is additionally re-checked under the unknown-ISI channel of Chapter 6 (<i>future work: 16-PSK, 64-QAM and beyond</i>)
Performance Metric	Uncoded BER vs. SNR, Monte-Carlo averaged

61 Only the two classical relay strategies evaluated in this thesis, amplify-and-forward and
 62 symbol-wise decode-and-forward, are treated as part of the studied system. Material in the
 63 literature review that lies outside the configuration above, such as the general relay channel
 64 with a direct link and its capacity bounds, is background context and is not evaluated here.

1.1 System Model

System model studied in this thesis. Throughout this work, a single canonical setup underlies the main line of argument: a half-duplex two-hop link with a single relay and *no* direct source-to-destination path, in which the source S transmits during a listening phase to the relay R , and the relay then transmits during a transmission phase to the destination D . Both hops are single-input single-output (SISO) complex-baseband channels with i.i.d. Rayleigh *fast* fading, a fresh realization $h \sim \mathcal{CN}(0, 1)$ per channel use, known perfectly at the corresponding receiver. The modulation is QPSK, and the performance metric is the *uncoded* bit-error rate (BER) as a function of SNR, averaged by Monte Carlo over the fading ensemble. Because the channel is complex, so is the constellation: QPSK places one bit on each axis, and each receiver applies coherent compensation ($y \cdot h^* / |h|$) per axis, so both dimensions the channel acts in are used and nothing is discarded. The learned component of the system is confined to the *relay processing function* $f_\theta(\cdot)$, the block for which AF and DF are the classical alternatives; the source, the channel model, and the destination demodulator are the same in every experiment, so observed BER differences are attributable to the relay strategy alone. The simulator is calibrated on the canonical channel itself: symbol-wise DF’s measured BER matches the closed-form two-hop composition to within one percent at every tabulated SNR (Table 5.2), once the QPSK constellation’s E_s/N_0 -to- E_b/N_0 conversion is applied. Beyond the canonical core, the thesis makes two further contributions of unequal weight. A smaller, supplementary study (Sections 5.4–5.4.5) revisits the uncoded/symbol-wise scope of the canonical setup itself: it adds a genuine information-theoretic block-DF baseline, coded with a rate-1/2 convolutional code and swept over constraint length on both QPSK and 16-QAM, together with soft-decision counterparts of both the classical and the learned relay, to measure – rather than only assert, as Remark 4.2 does – where coding does and does not change the standing of symbol-wise DF against the learned relays. Its conclusion is architectural rather than algorithmic: a relay that decodes the code spends the code’s redundancy before it transmits, so its residual errors cannot be repaired at the destination, which is why a relay that merely denoises can end up ahead of one built on an optimal decoder. That advantage is then bounded, not erased, once the code rate is itself allowed to adapt to the channel: maximizing goodput over a modulation-and-coding grid shows the choice of rate dominates the choice of relay strategy by an order of magnitude or more, so the relay-decoding question this study opened matters far less once the link is doing the more basic thing correctly – and the envelope this adaptation reaches stays well short of the channel’s own ergodic Shannon capacity throughout, under a third of it even at the top of the swept range, confirming the achieved goodput is limited by the finite constellation and coding grid rather than by anything left on the table for relay strategy to claim. Imposing a hard latency deadline on top of that adaptive-rate picture reopens the question the rate adaptation had

closed: because a relay that decodes and re-encodes must buffer a full frame twice (once to decode, once again at the destination) against a denoising relay’s once, a tight enough deadline locks the block-DF relay out of coding altogether while the denoising relay still has headroom, reversing the relay-strategy ordering by up to $1.5\times$ at operating points where the unconstrained comparison had favored the classical relay. The principal experimental contribution (Chapter 6) is a systematic study of learned relaying under *unknown and mismatched* channels that deliberately violate the assumptions of the canonical memoryless model, mapping, along every axis relevant to practical deployment, precisely when a minimal learned relay adds value over the strongest classical baselines and when it does not (H6). That chapter also revisits, under an unknown-ISI channel, whether the QPSK behaviour established on the canonical benchmark still holds: the memoryless-relay failure mode does, but the ordering between the learned relay and genie-CSI Viterbi MLSE does not (Section 6.1.2). All other departures (other channel families, a MIMO second hop, learned constellations) are identified as future work in Chapter 7. Capacity-theoretic results and the cooperative-diversity order are quoted in Chapter 2 as classical *background only*; they are not the operating point evaluated in the experiments.

Figure 1.1 summarizes this canonical benchmark: the source, the channel model, and the destination are identical in every experiment, while only the relay function $f_\theta(\cdot)$ is varied. ”Identical” here refers to the model, not to any particular channel realization: the fading is *fast*, so the coefficients $h_1[i]$ and $h_2[i]$ are redrawn independently for every symbol. Nothing in this thesis is a static or quasi-static link.

This chapter states the canonical setup only at the level needed to follow the argument; the complete system model – per-hop channel equations in both their raw complex and coherently-compensated forms, the SNR convention, and two terminology remarks the rest of this thesis relies on throughout (on the causality of the relay’s observation window, and on the symbol-wise versus information-theoretic sense of “decode-and-forward”) – is given once, precisely, in Section 4.1 rather than repeated here.

Chapter 2

Background and Related Work

Scope relative to the canonical model. Unless stated otherwise, the relay-processing methods reviewed in this chapter are discussed in the context of the canonical memoryless channel model adopted throughout Chapter 5. Channel-memory effects, sequence detection, and channel models with intersymbol interference (ISI) are intentionally deferred to the unknown-channel study of Chapter 6.

2.1 Classical Relay Strategies

The fundamental question that motivates this thesis is: can a learned relay function $f_\theta(\cdot)$ outperform the classical relay functions (AF scaling, DF regeneration) by exploiting patterns in the received signal that analytical methods do not capture? This question is particularly relevant at low SNR, where both AF (noise amplification) and DF (decoding errors) have well-known limitations, and where the non-linear decision boundary learned by a neural network may provide a potential advantage.

2.1.1 Amplify-and-Forward (AF)

The AF relay amplifies the received signal by a gain factor G that normalizes the output power: (Equation 2.1)

$$G = \sqrt{\frac{P_{\text{target}}}{\mathbb{E}[|y_R|^2]}} \quad (2.1)$$

The end-to-end SNR for AF relay is the standard harmonic-mean expression [3, 6]: (Equation 2.2)

$$\text{SNR}_{\text{eff}}^{\text{AF}} = \frac{\text{SNR}_1 \cdot \text{SNR}_2}{\text{SNR}_1 + \text{SNR}_2 + 1} \quad (2.2)$$

This form assumes *CSI-assisted variable-gain* AF, in which the relay knows its instantaneous first-hop gain and normalizes its output power per symbol (Equation 2.1); this is the variant implemented throughout this thesis. The additive +1 in the denominator originates from the relay’s own noise being re-amplified along with the signal, and it is the term that distinguishes this convention from the fixed-gain (blind, long-term-power-normalized) variant, whose end-to-end SNR takes a slightly different form.

This expression reveals a fundamental limitation: even when one hop has high SNR, the effective SNR is bottlenecked by the weaker hop. Moreover, the relay amplifies both signal and noise from the first hop, leading to noise accumulation at the destination.

2.1.2 Decode-and-Forward (DF)

The DF relay demodulates the received signal, recovers the transmitted bits, and re-modulates clean symbols:

$$\hat{b}_R = \text{demod}(y_R), \quad x_R = \text{mod}(\hat{b}_R) \quad (2.3)$$

The end-to-end BER for DF is: (Equation 2.4)

$$P_e^{\text{DF}} = P_{e,1}(1 - P_{e,2}) + (1 - P_{e,1}) \cdot P_{e,2} \quad (2.4)$$

where $P_{e,1}$ and $P_{e,2}$ are the BER of the first and second hops, respectively: an end-to-end bit error occurs exactly when *one* of the two hops flips the bit (two flips cancel), so this expands to $P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2}$, the same expression used in the AWGN analysis (Equation 2.27). For BPSK modulation over AWGN (real-valued noise with variance $1/\text{SNR}$), each hop’s BER is $P_e = Q(\sqrt{\text{SNR}})$. DF provides clean regeneration at high SNR but suffers from error propagation when the first-hop BER is non-negligible. As discussed in Remark 4.2, the “DF” baseline used here is the uncoded symbol-wise slicer, not information-theoretic block-DF; its optimality claims are therefore confined to the uncoded setting.

2.1.3 Other Classical Relay Techniques

AF and DF represent two canonical extremes of the relay processing spectrum: AF performs minimal processing (linear scaling) while DF performs maximal classical processing (full regeneration). Additional techniques (e.g. CF, EF, CoF) occupy intermediate

points along this spectrum. In this thesis, the AI-based relay strategies are designed to *learn* the relay processing function from data, potentially discovering strategies that subsume or outperform these fixed classical schemes. The focus on AF and DF as classical baselines is therefore deliberate: on the canonical matched-channel benchmark they bracket the classical design space and serve as representative reference baselines against which the learned strategies are evaluated; they should not be read as strict lower and upper performance bounds, since under channel mismatch a learned relay can outperform DF (Chapter 6). These additional relay strategies are reviewed for completeness; they are not simulated in this thesis because the studied system assumes a two-hop link without a direct source-destination path.

2.2 Machine Learning in Wireless Communication

The application of machine learning to physical-layer wireless communication has gained significant momentum in recent years, driven by the ability of neural networks to learn complex non-linear mappings directly from data. Deep learning has been applied to channel estimation [7], signal detection [8], autoencoder-based end-to-end communication [9], and resource allocation [10]. These approaches learn complex mappings from data, potentially outperforming hand-crafted algorithms in scenarios where analytical solutions are intractable or suboptimal.

2.2.1 Theoretical Basis: Universal Approximation and Denoising

The theoretical justification for applying neural networks to relay signal processing rests on two pillars. First, the **universal approximation theorem** [11] guarantees that a feedforward network with a single hidden layer and a non-linear activation function can approximate any continuous function on a compact domain to arbitrary accuracy, given sufficient width. For relay denoising, the target function maps noisy observations to clean transmitted symbols:

$$f^* : \mathbb{R}^{2w+1} \rightarrow [-1, 1], \quad f^*(y_{i-w}, \dots, y_{i+w}) = \mathbb{E}[x_i \mid y_{i-w}, \dots, y_{i+w}] \quad (2.5)$$

This conditional expectation f^* is the Bayes-optimal denoiser: it minimizes the mean squared error (MSE) over all possible estimators. For BPSK symbols corrupted by AWGN, f^* reduces to the posterior mean:

$$f^*(y) = \tanh\left(\frac{y}{\sigma^2}\right) \quad (2.6)$$

78

79 which is a smooth sigmoid-like function that approaches the hard-decision signum func-
 80 tion as $\sigma^2 \rightarrow 0$ (high SNR). A neural network with a single hidden layer and tanh output
 81 can represent this posterior *exactly for a fixed* σ^2 , a single tanh unit with the appropriate
 82 input scaling suffices. The deployed relay, however, is trained at $\text{SNR} \in \{5, 10, 15\}$ dB
 83 and evaluated over 0–20 dB, so one set of weights must approximate an entire *family* of
 84 posteriors indexed by σ^2 (and, on the fading channel, a mixture over random effective
 85 SNRs). That a 169-parameter network approximates this family closely enough to track
 86 the classical baselines is an empirical finding of this thesis, not a consequence of the exact
 87 single- σ^2 representation.

88 Second, the **bias-variance decomposition** provides a framework for understanding model
 89 complexity:

$$\mathbb{E}[(\hat{x} - x)^2] = \text{Bias}^2(\hat{x}) + \text{Var}(\hat{x}) + \sigma_{\text{irreducible}}^2 \quad (2.7)$$

90

91 The irreducible noise $\sigma_{\text{irreducible}}^2$ represents the minimum achievable MSE, determined by
 92 the channel noise. Increasing model complexity (more parameters) reduces bias but in-
 93 creases variance. For the relay denoising task, the target function f^* is simple (essentially
 94 a soft threshold), so the bias term is already small for modest networks. Adding param-
 95 eters beyond this point primarily increases variance (overfitting), explaining the U-shaped
 96 relationship between model size and BER observed in this thesis.

97 2.2.2 Prior Work in Deep Learning for Physical-Layer Processing

98 Ye et al. [7] demonstrated that deep neural networks can jointly perform channel estimation
 99 and signal detection in OFDM systems, achieving near-optimal performance with lower
 100 complexity than separate estimation and detection stages. Their key insight was that the
 101 DNN implicitly learns the channel’s statistical structure, eliminating the need for explicit
 102 pilot-based estimation.

103 Samuel et al. [8] proposed DetNet, an unfolded projected gradient descent network for
 104 MIMO detection. By unrolling iterative optimization into a fixed number of neural net-
 105 work layers, DetNet achieves near-maximum-likelihood detection performance with $O(N_t^2)$
 106 complexity instead of the exponential complexity of exhaustive ML search. This demon-
 107 strates the power of incorporating domain knowledge (the iterative structure of the detec-
 108 tion problem) into the network architecture.

109 Dorner et al. [9] proposed treating the entire communication system: modulator, channel,
 110 and demodulator: as an autoencoder, trained end-to-end to minimize bit error rate. This

approach jointly optimizes all components, potentially discovering novel modulation and coding schemes that outperform separately designed modules. The autoencoder paradigm is conceptually related to the relay processing task: both involve learning an encoder-decoder pair that maps through a noisy channel.

Sun et al. [10] applied deep learning to the NP-hard problem of resource allocation in interference networks, demonstrating that a DNN can learn near-optimal power control policies orders of magnitude faster than conventional iterative algorithms.

In the context of relay-assisted communication systems, most existing works employ neural networks primarily for *relay-selection* and *resource-allocation*, particularly in decode-and-forward (DF) relaying scenarios, where learning-based classifiers or predictors are used to identify the optimal relay based on channel state information, signal-to-noise ratios, or energy constraints, while the underlying relay operation remains conventional [12].

A broader perspective on the role of machine learning in wireless communications is provided in [13], which highlights the potential of learning-based methods across multiple protocol layers, including cooperative communications.

Another related line of work exploits the structural similarities between amplify-and-forward (AF) relay networks and neural networks, treating relay nodes as neurons with nonlinear transfer functions and using deep-learning optimization tools to tune relay gains and exploit hardware-induced nonlinearities [14, 15]. While these approaches demonstrate significant gains over classical linear AF schemes, they focus on analog gain optimization rather than symbol-level detection.

Classical cooperative communication frameworks based on AF and DF relaying remain well understood and widely adopted [5], but they rely on fixed linear forwarding or hard-decision rules. Consequently, despite substantial progress, relatively little attention has been devoted to learning the relay's symbol-level input-output mapping itself, motivating neural network-based detection at the relay as a generalized, data-driven alternative to conventional AF and DF processing.

2.2.3 Neural Network Relay Processing

For relay processing specifically, a supervised learning approach trains a neural network f_θ to minimize the empirical mean-squared error: (Equation 2.8)

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i)^2, \quad \hat{x}_i = f_\theta(y_{i-w:i+w}) \quad (2.8)$$

where $\hat{x}_i = f_\theta(y_{i-w:i+w})$ is the network output based on a sliding window of $2w + 1$

received symbols, and x_i is the clean transmitted symbol. This window-based approach provides temporal context that enables the network to exploit statistical dependencies in the noise-corrupted signal.

A well-known example of $f_\theta(\cdot)$ is the *multilayer perceptron (MLP)*, a feedforward artificial neural network composed of an input layer, one or more fully connected hidden layers with nonlinear activation functions, and an output layer, trained using backpropagation to model nonlinear input–output relationships. The MLP equation is shown in equation 2.9:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1 \mathbf{w} + \mathbf{b}_1), \quad \hat{x} = \tanh(\mathbf{W}_2 \mathbf{h} + \mathbf{b}_2) \quad (2.9)$$

The sliding window formulation ($i \in [i - w : i + w]$) is motivated by the observation that adjacent received symbols share common noise statistics and channel conditions. Under the canonical model (i.i.d. fast fading, white noise), adjacent symbols are statistically independent, so a single-symbol estimator ($w = 0$) is already a sufficient statistic and a window confers no theoretical benefit. The window is therefore not motivated by temporal correlation in the canonical setting; it is retained because it is required by the unknown-channel study of Chapter 6, where a 3-tap ISI channel couples adjacent symbols and the relay must span that memory. Keeping a fixed windowed architecture across all experiments isolates the effect of the channel assumptions from the network structure.

Multi-SNR training is a key design choice: by training on data generated at multiple SNR levels (5, 10, 15 dB in this thesis), the network learns a denoising function that generalizes across operating conditions rather than specializing to a single noise level. This is analogous to training a robust estimator that operates well across a range of noise variances, a concept related to minimax estimation in statistical decision theory.

A critical question in applying neural networks to relay processing is the relationship between model complexity and performance. Prior work has generally assumed that larger models yield better performance, but this assumption has not been rigorously tested in the relay communication context. The bias–variance framework predicts that for a low-complexity target function like relay denoising, performance should plateau or degrade beyond a modest model size: a prediction that this thesis confirms empirically. Understanding this relationship is essential for practical deployment, where relay nodes may have limited computational resources.

2.3 Generative Models for Signal Processing

Generative models offer an alternative paradigm for relay signal processing. Rather than directly learning a denoising function (discriminative approach), generative models learn

the distribution of clean signals $p(\mathbf{x})$ and use this knowledge to reconstruct the transmitted signal from noisy observations via Bayes' rule:

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \quad (2.10)$$

The generative approach is theoretically appealing because it separates the modeling of the signal prior from the noise model, potentially enabling better generalization to unseen noise conditions.

2.3.1 Variational Autoencoders

Variational Autoencoders (VAEs) [16] learn a latent representation by maximizing the evidence lower bound (ELBO). The generative model posits that data $\mathbf{x}$ is generated from a latent variable $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a decoder $p_\theta(\mathbf{x}|\mathbf{z})$. Since the true posterior $p(\mathbf{z}|\mathbf{x})$ is intractable, an encoder network $q_\phi(\mathbf{z}|\mathbf{x})$ approximates it. The ELBO objective is derived from the log-evidence decomposition: (Equation 2.11)

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi} \left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right]}_{\text{ELBO}(\theta, \phi; \mathbf{x})} + \underbrace{D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z}|\mathbf{x}))}_{\geq 0} \quad (2.11)$$

Since the KL divergence is non-negative, the ELBO is a lower bound on the log-evidence. Maximizing the ELBO simultaneously trains the encoder to approximate the true posterior and the decoder to reconstruct the data. The ELBO decomposes into a reconstruction term and a regularization term:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction quality}} - \underbrace{\beta \cdot D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{Latent space regularity}} \quad (2.12)$$

The reconstruction term encourages the decoder to accurately reproduce the input from the latent code, while the KL term regularizes the latent space to be close to the standard Gaussian prior $p(\mathbf{z})$. The β parameter (β -VAE) [17] in Equation 2.12 controls the trade-off between reconstruction quality and latent space smoothness: $\beta < 1$ prioritizes reconstruction (beneficial for the relay task where accurate signal recovery is paramount), while $\beta > 1$ encourages disentangled latent representations.

The *reparameterization-trick* enables gradient-based optimization through the stochastic sampling layer:

$$\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \quad (2.13)$$

201 Instead of sampling from equation 2.13, the sample is expressed as:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon} \quad (2.14)$$

202

203 where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This moves the stochasticity outside the computational graph, allow-
204 ing backpropagation through the encoder.

205 For relay signal processing, the VAE learns a compressed latent representation of the clean
206 signal manifold. During inference, the noisy received signal is encoded into the latent
207 space, and the decoder maps it back to a denoised estimate. The regularized latent space
208 provides implicit denoising: noisy inputs that map to regions far from the learned signal
209 manifold are pulled back toward high-probability regions during decoding. However, the
210 stochastic sampling introduces additional variance into the estimate, which can be harmful
211 for the deterministic BPSK denoising task: a trade-off that manifests as the consistent VAE
212 under-performance observed in this thesis.

213 2.3.2 Conditional Generative Adversarial Networks

214 **Conditional GANs (cGANs)** [18] learn a denoising map by adversarial training: a gener-
215 ator conditioned on the noisy observation produces a clean estimate, and a discriminator
216 is trained to separate generated from true clean symbols, the two optimizing a minimax
217 objective. The original formulation is unstable to train, because the divergence the discrim-
218 inator implicitly minimizes saturates when the two distributions do not overlap and stops
219 supplying useful gradients. The Wasserstein GAN with gradient penalty (WGAN-GP)
220 [19] replaces that objective with the Wasserstein distance, estimated through a critic con-
221 strained to be approximately 1-Lipschitz by penalizing the norm of its gradient at points
222 interpolated between real and generated samples. This is the variant implemented in this
223 thesis (Section 4.2).

224 The cGAN is described here and in Chapter 4 for completeness, but it is *excluded from ev-*
225 *ery reported comparison* on cost grounds: training it took roughly two GPU-hours against
226 the minimal MLP's few seconds on a CPU, which is not a defensible expenditure for a
227 relay-denoising target this simple. No claim in this thesis rests on it, and the theory above
228 is therefore kept deliberately brief.

229 2.3.3 Generative vs. Discriminative Paradigms for Relay Processing

230 The application of generative models to relay processing has, to the best of our knowledge,
231 not been extensively studied. The key question is whether the generative inductive bias
232 (learning the data distribution rather than just the input-output mapping) provides any ad-

vantage for the relay denoising task. For BPSK, the clean signal distribution is trivially simple (a discrete distribution on $\{-1, +1\}$), suggesting that the generative overhead may not be justified. This thesis compares VAE-based relay processing against classical and supervised learning methods on a common benchmark, and the results show that the generative paradigm provides no significant advantage for this particular task. A conditional GAN relay is implemented and described in Chapter 4 but is excluded from the reported comparison on cost grounds, so no claim is made about it here. On the canonical setup the VAE tracks the supervised MLP and symbol-wise DF rather than trailing them, reaching 0.00972 at 20 dB against DF's 0.00972 and the MLP's 0.00992 (Table 5.2), and it stays inside the feedforward group at every SNR in the equal-parameter comparison as well (Table 5.3). The generative bias is therefore neither an advantage nor a handicap here: it costs parameters and training time to arrive where a 169-parameter regressor already is. One qualification remains, unchanged by this: the relay samples $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$ stochastically at inference, and the deterministic variant that substitutes the posterior mean $\boldsymbol{\mu}$ at test time was not evaluated, so nothing here bears on whether that variant would do better.

2.4 Sequence Models: Transformers, State Space Models

Recent advances in sequence modeling have produced two competing paradigms with distinct computational properties and inductive biases. Both have achieved state-of-the-art results in natural language processing, but their relative merits for physical-layer signal processing (where sequences are real-valued temporal signals rather than discrete tokens) have, to the best of our knowledge, not been previously investigated.

2.4.1 Transformers and the Attention Mechanism

Transformers [20] use multi-head self-attention to capture global dependencies in sequences. The core attention mechanism computes a weighted combination of value vectors, where the weights are derived from the compatibility of query and key vectors:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (2.15)$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d_k}$ and $\mathbf{V} \in \mathbb{R}^{n \times d_v}$ are obtained from the input via learned linear projections W^Q, W^K, W^V . The scaling factor $1/\sqrt{d_k}$ prevents the dot products from growing too large in magnitude, which would push the softmax into saturation regions with vanishing gradients.

Multi-head attention extends this by running h parallel attention heads, each with inde-

pendent projections, and concatenating their outputs: (Equation 2.16)

$$\text{MultiHead}(\mathbf{X}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O, \text{head}_i = \text{Attention}(\mathbf{X}W_i^Q, \mathbf{X}W_i^K, \mathbf{X}W_i^V) \quad (2.16)$$

265

Each head can attend to different aspects of the input: for signal processing, one head might focus on the immediate neighbors (local denoising) while another captures longer-range patterns. The total parameter count for multi-head attention is:

$$3d_{\text{model}}^2 + d_{\text{model}}^2 = 4d_{\text{model}}^2 \quad (2.17)$$

269

for Q, K, V projections and the output projection. The attention matrix:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{n \times n} \quad (2.18)$$

271

computes pairwise interactions between all n positions, yielding a time and memory complexity of: $O(n^2)$. For the 11-symbol relay window used in this thesis, this is negligible ($11^2 = 121$ entries). However, the quadratic cost becomes prohibitive for longer sequences (e.g., $n = 1000$ symbols), motivating linear-time alternatives.

Positional encoding is necessary because the attention mechanism is permutation-equivariant (it treats input positions symmetrically). This thesis uses sinusoidal positional encoding: (Equation 2.19)

$$PE_{(pos, 2k)} = \sin\left(\frac{pos}{10000^{2k/d}}\right), \quad PE_{(pos, 2k+1)} = \cos\left(\frac{pos}{10000^{2k/d}}\right) \quad (2.19)$$

279

which injects position information into the embeddings, enabling the model to distinguish between symbols at different positions in the window.

2.4.2 Structured State Space Models

Structured state space models (SSMs) [21] treat a sequence model as a discretized linear dynamical system. A continuous system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$, $y(t) = \mathbf{C}\mathbf{x}(t) + Du(t)$ is discretized with a step size Δ (zero-order hold, $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$) to give the recurrence

$$\mathbf{x}_k = \bar{\mathbf{A}}\mathbf{x}_{k-1} + \bar{\mathbf{B}}u_k, \quad y_k = \mathbf{C}\mathbf{x}_k + Du_k \quad (2.20)$$

which is a linear recursive filter with a learned state transition. Two ingredients make this competitive with attention. The HiPPO framework supplies a principled initialization for $\mathbf{A}$ under which the state maintains a compressed polynomial approximation of the input history, and S4 [21] shows that a structured (diagonal or diagonal-plus-low-rank) $\mathbf{A}$ makes the recurrence cheap to evaluate. The resulting model is linear-time in sequence length, in contrast to attention’s quadratic cost.

2.4.3 Mamba and Mamba-2

Mamba [22] makes the SSM parameters *input-dependent*, turning the time-invariant system into a time-varying one:

$$\Delta_k = \text{softplus}(W_\Delta u_k + b_\Delta), \quad \mathbf{B}_k = W_B u_k, \quad \mathbf{C}_k = W_C u_k \quad (2.21)$$

With $\mathbf{A}$ diagonal and negative, Δ_k acts as a learned gate: a large Δ_k drives $\bar{\mathbf{A}}_k = \exp(\Delta_k \mathbf{A})$ toward zero, resetting the state so the new input dominates, while a small Δ_k leaves $\bar{\mathbf{A}}_k \approx \mathbf{I}$, preserving the state and ignoring the input. This selectivity is the property of interest for relay processing: in principle the model can retain signal structure while suppressing noise, deciding per sample which of the two an observation carries. The selective layer is wrapped in a gated block with an expand-factor-2 projection and a short causal convolution.

Remark 2.1. These adaptive-filtering connections become directly relevant only in the unknown-channel study of Chapter 6; on the canonical memoryless channel the relay reduces to a per-symbol function and there is no temporal structure for a state to track.

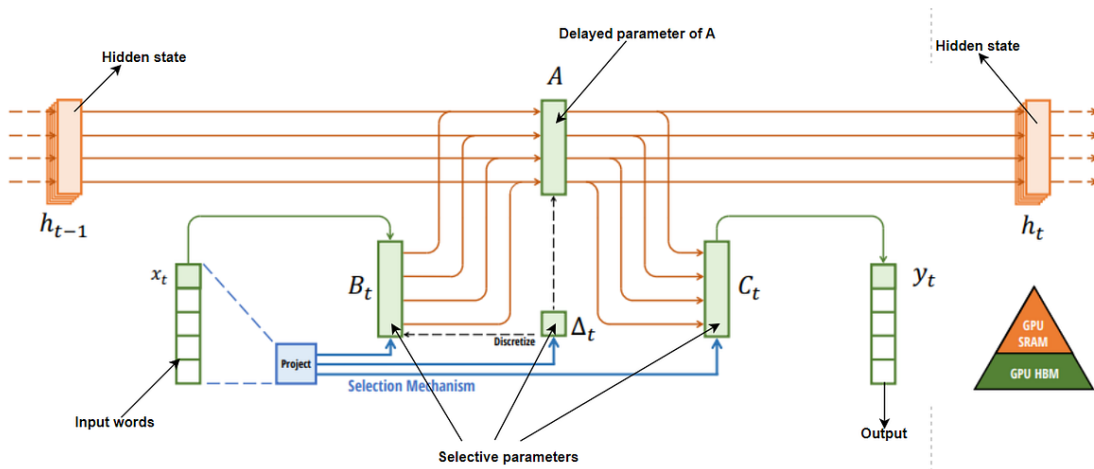


Figure 2.1: Structured SSMs map each channel independently through a higher-dimensional latent state; the selective variant makes the transition input-dependent so the state is reset or held per sample.

Mamba-2 (SSD) [23] reformulates the same computation through an algebraic duality between the linear recurrence and a structured matrix multiplication: the output can be written $\mathbf{y} = M \cdot (\mathbf{B} \odot \mathbf{u})$ for a lower-triangular semi-separable matrix M . The practical consequence is that the sequence can be processed in chunks, with a batched matrix multiply inside each chunk and a single state hand-off between chunks, so the number of sequential steps falls from one per sample to one per chunk. Mamba-2 also forms the state-space parameters at the start of the block rather than sequentially along it. Both variants are evaluated here; at the short relay windows this thesis uses, the distinction is architectural rather than a source of measured advantage.

2.4.4 State Space Models for Signal Processing

The application of state space models to physical-layer signal processing is, to the best of our knowledge, largely unexplored. Classical signal processing relies extensively on LTI state space models (Kalman filters, Wiener filters), and the SSM framework provides a natural neural extension of these classical tools. The connection is direct: a trained SSM with fixed (input-independent) parameters implements a learnable linear filter, while the selective Mamba variant implements a learnable *adaptive* filter. For the relay denoising task, this means Mamba can potentially learn the optimal filter structure from data, without requiring manual specification of filter order, cutoff frequencies, or adaptation algorithms. This thesis presents, to the best of our knowledge, the first comparison of Mamba-S6, Mamba-2 SSD, and Transformers for relay communication, demonstrating that state space models are well suited to this domain and that the SSD formulation of Mamba-2 yields significant speed advantages at longer context lengths.

2.5 Theoretical Foundations: Channel Models

This section presents the theoretical BER analysis for each channel model used in this study, derives the closed-form expressions, and validates them against Monte Carlo simulation. The theoretical-simulative comparison serves two purposes: (i) it verifies the correctness of the simulation framework, and (ii) it establishes the baseline performance that AI relays must beat.

2.5.1 AWGN Channel: Theoretical Analysis

The AWGN channel adds zero-mean Gaussian noise to the transmitted signal: (Equation 2.22)

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{P_s}{\text{SNR}_{\text{linear}}} \quad (2.22)$$

where $P_s = \mathbb{E}[|x|^2] = 1$ for unit-power BPSK symbols. The noise variance is inversely proportional to the linear SNR.

2.5.1.1 Single-hop BER

For BPSK ± 1 symbols and real-valued AWGN with variance $\sigma^2 = 1/\text{SNR}$, an error occurs when the noise pushes the received sample past the decision boundary at the origin. The theoretical BER is: (Equation 2.23)

$$P_e^{\text{AWGN}} = Q\left(\frac{1}{\sigma}\right) = Q\left(\sqrt{\text{SNR}}\right) = \frac{1}{2} \text{erfc}\left(\sqrt{\frac{\text{SNR}}{2}}\right) \quad (2.23)$$

where $Q(x) = \frac{1}{2} \text{erfc}(x/\sqrt{2})$ is the Gaussian Q-function and $\text{SNR} = P_s/\sigma^2$ is the ratio of signal power to noise variance.

Remark 2.2 (Noise convention). The standard textbook result $P_e = Q(\sqrt{2E_b/N_0})$ assumes that the one-sided noise PSD is N_0 , giving a baseband noise variance of $N_0/2$ per real dimension. In our AWGN implementation the noise is purely real with variance $\sigma^2 = P_s/\text{SNR}$, so $N_0 = 2\sigma^2 = 2P_s/\text{SNR}$ and $E_b/N_0 = \text{SNR}/2$. Substituting yields $Q(\sqrt{2 \cdot \text{SNR}/2}) = Q(\sqrt{\text{SNR}})$, which is the expression used throughout this work.

By contrast, the fading channels add **complex** Gaussian noise with total power $1/\text{SNR}$ (i.e. $\sigma_I^2 = \sigma_Q^2 = 1/(2 \text{SNR})$) and extract the real part after ZF equalization, halving the effective noise variance per decision dimension. This naturally introduces a factor of 2 inside the Q-function argument for all fading-channel BER expressions (the fading-channel analysis above), making them consistent with the standard textbook forms.

2.5.1.2 Two-hop AF BER

For amplify-and-forward with equal-SNR hops, the effective end-to-end SNR is:

$$\text{SNR}_{\text{eff}}^{\text{AF}} = \frac{\text{SNR}_1 \cdot \text{SNR}_2}{\text{SNR}_1 + \text{SNR}_2 + 1} = \frac{\gamma^2}{2\gamma + 1} \quad (2.24)$$

where $\gamma = \text{SNR}$ is the per-hop SNR. The resulting BER is

$$P_e^{\text{AF}} = Q\left(\sqrt{\text{SNR}_{\text{eff}}}\right) \quad (2.25)$$

under the same real-noise convention as the single-hop case (Remark 2.2). At high SNR,

$$\text{SNR}_{\text{eff}} \approx \frac{\gamma}{2} \quad (2.26)$$

364

365 confirming the well-known 3 dB penalty of AF relaying.

366 2.5.1.3 Two-hop DF BER

367 For a decode-and-forward relay with equal-SNR hops, an error occurs at the destination
368 if exactly one hop introduces an error:

$$P_e^{\text{DF}} = P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2} \quad (2.27)$$

369

370 With equal hops (i.e. $P_{e,1} = P_{e,2} = P_e^{\text{AWGN}}$), this becomes

$$P_e^{\text{DF}} = 2P_e(1 - P_e) \quad (2.28)$$

371

372 At high SNR, $P_e \ll 1$ and

$$P_e^{\text{DF}} \approx 2P_e \quad (2.29)$$

373 i.e., the two-hop penalty is approximately a factor of 2 in BER.

374 2.5.2 Rayleigh Fading Channel: Theoretical Analysis

375 The Rayleigh fading channel models non-line-of-sight (NLOS) propagation where the sig-
376 nal undergoes multiplicative fading:

$$y = hx + n, \quad h \sim \mathcal{CN}(0, 1), \quad n \sim \mathcal{CN}(0, \sigma^2) \quad (2.30)$$

377

378 The fading coefficient h is a circularly-symmetric complex Gaussian random variable, so
379 its magnitude $|h|$ follows a Rayleigh distribution:

$$f_{|h|}(r) = 2r \cdot e^{-r^2}, \quad r \geq 0 \quad (2.31)$$

380

381 with $\mathbb{E}[|h|^2] = 1$ (unit average power). The instantaneous SNR after equalization ($\hat{x} =$
382 y/h) becomes $\gamma_h = |h|^2 \cdot \text{SNR}$, which is exponentially distributed.

383 2.5.2.1 Single-hop BER

384 Averaging the conditional BER $P_e(\gamma_h) = Q(\sqrt{2\gamma_h})$ over the exponential distribution of
 385 γ_h yields the closed-form [24, Eq. 14-4-15] (Equation 2.32):

$$P_e^{\text{Rayleigh}} = \frac{1}{2} \left(1 - \sqrt{\frac{\bar{\gamma}}{1 + \bar{\gamma}}} \right) \quad (2.32)$$

386

387 where $\bar{\gamma} = \text{SNR}$ is the average SNR. At high SNR ($\bar{\gamma} \gg 1$):

$$P_e^{\text{Rayleigh}} \approx \frac{1}{4\bar{\gamma}} \quad (2.33)$$

388

389 This $1/\text{SNR}$ decay is fundamentally slower than the exponential decay of AWGN ($Q(\sqrt{\gamma}) \sim$
 390 $e^{-\gamma/2}$), explaining why Rayleigh fading is significantly more challenging. The channel is
 391 **diversity-limited**: deep fades (where $|h| \approx 0$) cause errors regardless of the average SNR.
 392 Combating this diversity limitation is the primary motivation for the relay architectures
 393 studied in this thesis.

394 2.5.2.2 Two-hop DF BER

395 The two-hop DF relay over Rayleigh fading follows the same composition rule as AWGN:

$$P_e^{\text{DF, Rayleigh}} = 2P_e^{\text{Rayleigh}}(1 - P_e^{\text{Rayleigh}}) \quad (2.34)$$

396

Chapter 3

Research Objectives

3.1 Main Objective

To systematically evaluate and compare classical and AI-based relay strategies for two-hop communication on a single canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER) and to determine the most effective relay strategy as a function of SNR regime and computational constraints.

3.2 Research Gap and Motivation

Despite growing interest in AI for wireless communication, a systematic comparison of relay processing paradigms (spanning classical signal processing, supervised learning, generative modeling, and modern sequence architectures) is, to the best of our knowledge, absent from the literature. The following specific gaps motivate this thesis:

- Gap 1: No cross-paradigm relay comparison.** Existing work on AI-based relay processing focuses on individual architectures in isolation. Ye et al. [7] demonstrated DNNs for OFDM detection, Dorner et al. [9] explored autoencoders for end-to-end communication, and Samuel et al. [8] applied unfolded networks to MIMO detection. However, no study systematically compares supervised feedforward networks, variational autoencoders, adversarial networks, attention-based Transformers, and state space models (Mamba) for the same relay denoising task under controlled conditions. Without such a comparison, practitioners cannot make informed architecture selections.
- Gap 2: Model complexity-performance relationship is unknown.** It is commonly assumed that larger neural networks yield better performance. However, the relay denoising task (recovering a BPSK symbol from a noisy observation) is a low-dimensional mapping problem. The theoretical minimum description length for this

mapping is small (a soft threshold function), suggesting that large models may overfit. No prior work has rigorously characterized the complexity-performance trade-off for relay processing, nor has a normalized (equal-parameter) comparison been conducted to isolate architectural effects from capacity effects.

3. **Gap 3: State space models for physical-layer signal processing.** The Mamba architecture [22] and its Mamba-2 SSD variant [23] have demonstrated strong performance in NLP and genomics, but their application to relay-level physical-layer signal processing remains, to the best of our knowledge, unexplored. The theoretical connection between SSMs and classical adaptive filters makes this a natural research direction, but empirical validation is needed.

This thesis addresses all three gaps through a unified framework that implements and compares nine relay strategies on the canonical setup, with both original and normalized parameter counts, full statistical analysis, and a dedicated complexity study. The following table summarizes the positioning of this work relative to the literature:

Table 3.1: Research positioning: comparison of this thesis against prior work in deep-learning-based relay communication.

Aspect	Prior Work	This Thesis
Relay strategies compared	1–2 (typically AF vs. DF, or one AI method)	9 (2 classical + 7 AI)
AI paradigms	Single (e.g., supervised only)	4 (supervised, generative, adversarial, sequential)
Evaluation discipline	Single ad-hoc configuration	One canonical setup (SISO, Rayleigh, QPSK), chosen in advance and never varied, with AWGN analytical calibration
Complexity analysis	None	Normalized ~3K-parameter comparison + complexity study (U-shaped BER vs. model size)
State space models	None for relay/physical-layer	Mamba-S6, Mamba-2 SSD, with crossover benchmark
Statistical rigor	Often missing	Wilcoxon test, 95% CI, 10 trials per point

Relationship between the canonical study and the extension study. Hypotheses H1–H5 concern the canonical relay scenario introduced in Chapter 1: a memoryless two-hop channel with white noise and uncoded transmission. Hypothesis H6 studies a deliberately different regime in which the canonical model no longer represents the true channel. It therefore serves as a robustness extension rather than a direct continuation of the canonical experiments.

3.3 Research Hypotheses

Based on the theoretical background of Chapter 2, this thesis tests the following hypotheses. H1–H4 are posed and answered on the canonical setup. H6 is the central hypothesis of the thesis’s principal contribution (Chapter 6), posed on channels *outside* the canonical family:

1. **H1 (AI advantage over AF at low SNR).** On the canonical setup, some AI-based relay strategies achieve lower BER than AF at low SNR (0–4 dB), where the non-linear denoising learned by the neural network provides an advantage over AF’s noise amplification.

Rationale: At low SNR, the MMSE-optimal (posterior-mean) denoiser of a BPSK symbol from a single AWGN observation, $f^*(y) = \mathbb{E}[x \mid y] = \tanh(y/\sigma^2)$, is a smooth non-linear mapping that differs substantially from AF’s linear scaling; a neural network can approximate f^* closely, whereas AF amplifies noise together with the signal. Note that f^* is optimal in the mean-square sense for the relay’s local estimate; it is not claimed to be the end-to-end BER-optimal relay function, which is why H1 is posed as an empirical hypothesis rather than a theorem.

2. **H2 (DF dominance at high SNR).** The symbol-wise DF relay (Remark 4.2) achieves the lowest BER at medium-to-high SNR (≥ 6 dB), outperforming all AI methods.

Rationale: At high SNR, $f^*(y) \approx \text{sign}(y)$, which is exactly the symbol-wise DF operation. AI relays introduce a small but non-zero approximation error, so DF should be optimal among the per-symbol relay functions considered in this uncoded setting. This dominance is asserted only within the uncoded, symbol-wise family evaluated here and is not a claim of information-theoretic optimality (cf. Remark 4.2).

3. **H3 (U-shaped complexity curve).** There exists an optimal model size for relay denoising beyond which performance degrades due to overfitting; equivalently, BER is U-shaped in model size (U-shaped in accuracy). Specifically, models with ~ 100 – 200 parameters achieve performance comparable to models with 10 – $100\times$ more parameters. Only the second, weaker part of this hypothesis is testable with the protocol adopted here: establishing *overfitting* as the mechanism behind any upturn

would require seed replication and train/test-gap measurement, which Section 7.4 records as not performed.

Rationale: The bias-variance analysis (Section 2.2.1) predicts that the low-complexity target function f^* requires minimal model capacity. Excess capacity increases variance without reducing bias.

4. **H4 (Architecture convergence at an equal parameter budget).** When all AI models are normalized to the same parameter count ($\sim 3,000$), the performance differences between architectures narrow significantly, indicating that parameter count is a more important factor than architectural choice. This is an equal-*budget* comparison rather than an isolation of architecture from capacity: as Table 4.2 records, meeting the budget required adjusting width, depth, state dimension and input window size together, so those degrees of freedom co-vary with architecture and are not separately controlled.

Rationale: All architectures are universal approximators and the target function is simple. At equal capacity, architectural inductive biases provide diminishing returns.

Unlike H1–H4, which assume the canonical memoryless channel throughout, H6 deliberately introduces impairments not present in the system model of Chapter 1, including channel memory, ISI, and nonlinear distortion.

1. **H6 (Principal contribution, unknown-channel mitigation).** On channels with structure unknown to the receiver chain, memory (intersymbol interference) or asymmetry (biased nonlinearity), the *memoryless* classical relays (AF, symbol-wise DF) fail to relay reliably, exhibiting error floors or severe degradation, whereas the minimal 169-parameter MLP relay restores reliable relaying by learning the impairment from data, approaching the performance of the optimal model-aware detector without requiring knowledge of the channel family.

Rationale: AF forwards the channel impairment untouched, and symbol-wise DF applies a fixed memoryless zero-threshold decision; neither can compensate structure they do not model. Classical theory does provide a remedy *once the channel family is known*, estimate-then-MLSE (pilot-based channel estimation followed by Viterbi equalization) is optimal for a linear FIR channel and is included as a benchmark. The hypothesis is therefore not that learning beats classical theory, but that a single fixed learned architecture can approach model-aware performance with no channel-family knowledge, no explicit identification stage, and complexity independent of channel memory, and that it remains applicable on impairments (e.g., nonlinear bias) for which linear MLSE is mis-specified. How far this holds, and on which impairment structures it breaks down, is determined empirically in Chap-

ter 6. Conversely, on channels that satisfy the classical assumptions, H2 predicts no learned relay can beat DF, so H6 also delineates the precise boundary of the learned relay's value.

3.4 Specific Objectives

1. **Implement and compare nine relay strategies** spanning four learning paradigms: no learning (AF, DF), supervised learning (MLP, Hybrid), generative modeling (VAE, cGAN), and sequence modeling (Transformer, Mamba-S6, Mamba-2 SSD).
2. **Evaluate on the canonical setup:** SISO Rayleigh fast fading with QPSK, validated against closed-form theory (with AWGN/BPSK as the analytical calibration limit), and probe modulation generalizability in a scoped 16-QAM extension.
3. **Investigate the complexity–performance trade-off** by testing model architectures ranging from 0 parameters (classical) to 26,179 parameters (Mamba-2 SSD), and by conducting a normalized comparison at approximately 3,000 parameters.
4. **Determine whether state space models outperform attention mechanisms** for relay signal processing by comparing Mamba-S6 and Transformer architectures at both original and normalized parameter counts, and by benchmarking at extended context lengths.
5. **Identify practical deployment recommendations** for selecting the appropriate relay strategy given specific operational constraints (SNR range, computational budget, channel environment).

3.5 Scope and limitations

The following limitations define the scope of this study, which is confined entirely to the canonical setup (SISO, Rayleigh fast fading, complex baseband, QPSK, uncoded):

- **Modulation:** The experiments use QPSK on the canonical channel throughout. Higher-order constellations (16-QAM, 16-PSK, 64-QAM and denser), together with the joint 2D N -class relay formulation they require, are deferred to future work.
- **Channel knowledge:** *In the canonical core* (Chapter 5), perfect CSI is assumed at the receiver and channel estimation errors are not modeled. This limitation is deliberately lifted in the principal contribution: Chapter 6 models finite-pilot least-squares channel estimation (a 200-pilot Viterbi baseline and a swept pilot budget) and, in the blind regime, a fully zero-CSI setting with no channel prior at all.
- **Relay topology:** Single relay, two-hop, half-duplex. Multi-relay and full-duplex configurations are excluded.

- 145 • **Topology:** SISO on both hops. Multi-antenna (MIMO) configurations are deferred
146 to future work.
- 147 • **Training regime:** Offline training on synthetic data. Online adaptation is not im-
148 plemented.

149 These delimitations are chosen to enable a clean comparison of relay processing functions
150 in isolation, without confounding factors from protocol-level or system-level complexity.

Chapter 4

Methods

4.1 System Model

The system under study is a two-hop relay network with a single relay node. All methods in this chapter are described for the canonical setup of Chapter 1: SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER. QPSK on the canonical Rayleigh channel is the only pairing used; it is set out in Table 4.1.

Scope of the canonical system model. The system model defined in this chapter serves as the canonical benchmark used throughout the core experiments of the thesis (Chapter 5). It assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. Accordingly, all relay comparisons in the canonical study concern per-symbol processing and do not require sequence detection: the current observation $y_R[i]$ is already a sufficient statistic for detecting $x[i]$, and the classical AF and DF baselines operate per symbol. Some neural architectures nevertheless receive a finite window of neighboring observations (Section 4.2); this is an architectural input convention for the evaluated feedforward and sequence models, and under the canonical i.i.d. assumptions it neither introduces channel memory nor supplies additional information about $x[i]$. Channel memory, intersymbol interference (ISI), pilot-based channel estimation, blind equalization, and sequence-detection techniques such as Viterbi MLSE are introduced only later in Chapter 6 as deliberate departures from the canonical assumptions in order to study robustness under model mismatch.

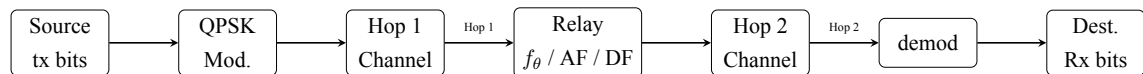


Figure 4.1: Two-hop relay system model: source transmits through two channels with a neural network relay in between. Each hop applies the channel operator followed by coherent compensation (no equalizer is used in the canonical setup)

Source bits are QPSK-modulated on the canonical setup, passed through Hop 1 (SISO Rayleigh), processed by the relay (classical or neural), transmitted over Hop 2 (SISO Rayleigh), and demodulated at the destination; no equalization is involved. The same chain carries BPSK on the AWGN calibration channel and 16-QAM in the extension, with only the modulator and demodulator changing. Both hop channels are known perfectly at their respective receivers, per the conventions of Chapter 1.

$$\text{Source} \xrightarrow{\text{Hop 1}} \text{Relay} \xrightarrow{\text{Hop 2}} \text{Destination} \quad (4.1)$$

4.1.1 Modulation

Three constellations appear in this thesis. The base case is BPSK, which maps a single bit to a real-valued symbol,

$$x = 1 - 2b, \quad b \in \{0, 1\} \implies x \in \{-1, +1\} \quad (4.2)$$

with hard-decision demodulation $\hat{b} = \mathbb{K}[\hat{x} < 0]$ at the destination. QPSK is defined in Section 4.5. Both are normalized to unit average symbol energy, $E_s = 1$, so they differ in bits per symbol rather than in transmit power: $k = 1$ and 2 respectively, giving $E_b = E_s/k$. Higher-order constellations are future work.

4.1.2 Hop Model

In the half-duplex two-hop model studied in this thesis, the relay cannot transmit and receive simultaneously, and there is no direct source–destination link. The communication proceeds in two time slots. On the canonical Rayleigh fast-fading channel each hop applies an i.i.d. complex fading coefficient followed by complex AWGN:

Slot 1 (Source $\rightarrow$ Relay), listening phase:

$$y_R[i] = h_1[i] x[i] + n_1[i], \quad h_1[i] \sim \mathcal{CN}(0, 1), \quad n_1[i] \sim \mathcal{CN}(0, \sigma^2) \quad (4.3)$$

Slot 2 (Relay $\rightarrow$ Destination), transmission phase:

$$y_D[i] = h_2[i] x_R[i] + n_2[i], \quad h_2[i] \sim \mathcal{CN}(0, 1), \quad n_2[i] \sim \mathcal{CN}(0, \sigma^2) \quad (4.4)$$

where $x[i]$ is the complex symbol transmitted by the source at time i (QPSK on this canonical channel; the real BPSK alphabet $\{-1, +1\}$ applies only to the AWGN calibration special case below), $y_R[i]$ is the signal received at the relay, $x_R[i]$ is the signal transmitted by the relay, and $y_D[i]$ is the signal received at the destination. The relay output is written generally as

$$x_R[i] = f_\theta(\mathcal{Y}_{R,i}),$$

where $\mathcal{Y}_{R,i}$ denotes the observation supplied to the relay-processing function; for AF and symbol-wise DF, $\mathcal{Y}_{R,i} = y_R[i]$.

Each hop applies a channel function followed by optional equalization (Equation 4.5):

$$y = \mathcal{H}(x, \text{SNR}) + n \quad (4.5)$$

where $\mathcal{H}(\cdot)$ denotes the canonical Rayleigh fast-fading channel operator (the AWGN special case is retained for analytical calibration of the simulator). The calligraphic symbol distinguishes the channel *operator* from the scalar fading *coefficient* h used throughout the rest of this thesis.

Both hops use the same channel. $\mathcal{H}(\cdot)$ is applied identically on Hop 1 and Hop 2 at the same SNR: a configuration never mixes channel families across hops. A configuration is therefore specified completely by the pair (constellation, channel), and it is that pair which names each experiment below.

Coherent compensation. Because the fading coefficient is known at each receiver, coherent compensation ($y \cdot h^*/|h|$ followed by $\Re\{\cdot\}$) reduces each hop to an equivalent real channel with a random per-symbol effective gain $|h[i]|$,

$$\tilde{y}_R[i] = |h_1[i]| x[i] + \tilde{n}_1[i], \quad \tilde{y}_D[i] = |h_2[i]| x_R[i] + \tilde{n}_2[i], \quad (4.6)$$

with $\tilde{n}[i] \sim \mathcal{N}(0, \sigma^2)$ real per axis. Because the channel is complex, the canonical constellation is two-dimensional: QPSK places one bit on each axis, so both dimensions the channel acts in are used and the relay processes the pair; the real reduction above then applies per axis rather than discarding one. BPSK, being one-dimensional, is never transmitted over the complex fading channel, which would require discarding the quadrature component; it appears only in the closed-form expressions of Chapter 2. The *AWGN limit* referenced in those closed forms is the special case $|h[i]| \equiv 1$, i.e. $y_R[i] = x[i] + n_1[i]$ and $y_D[i] = x_R[i] + n_2[i]$.

The noise terms are independent and identically distributed with variance

$$\sigma^2 = \frac{P_s}{\gamma} \quad (4.7)$$

76

77 where $P_s = \mathbb{E}[|x|^2]$ is the transmit symbol power (unit for every constellation studied, per
 78 the normalization of Section 4.1.1) and γ denotes the per-hop linear signal-to-noise ratio
 79 (SNR), equal on both hops.

80 4.1.3 Evaluated Configurations

81 This thesis evaluates one (constellation, channel) configuration, listed in Table 4.1. The
 82 governing principle is that **a real channel is paired with a real constellation and a com-**
 83 **plex channel with a complex constellation**: the signal space the constellation lives in
 84 and the signal space the channel acts on are the same space. Rayleigh with QPSK is the
 85 complex case, both living in the plane, where the fading coefficient $h \sim \mathcal{CN}(0, 1)$ rotates
 86 as well as scales and the constellation has a phase for it to act on.

87 The pairings excluded by this principle are precisely the ones that would force the simula-
 88 tion to patch over a mismatch. A real constellation on the complex Rayleigh channel must
 89 discard the imaginary part of the equalized sample after dividing by h , throwing away
 90 half the plane the channel acted in; a complex constellation on AWGN is not wrong, but
 91 it exercises none of what makes the complex channel interesting, since there is no phase
 92 rotation to undo. Neither is reported here.

Table 4.1: The evaluated configurations. The channel shown applies to both hops.

Constellation	Channel	Role	Why this pairing
QPSK (complex)	Rayleigh (complex)	Canonical	The complex case, and the operating point this thesis draws its conclusions on. QPSK places one bit on each axis, so both dimensions the fading coefficient acts in carry information and none is discarded.
BPSK (real)	Unknown 3-tap ISI	Extension	Used only in Chapter 6, where the impairment under study is channel <i>memory</i> rather than fading, and the one-dimensional alphabet keeps the trellis small enough for an exact Viterbi benchmark.

AWGN appears in this thesis only as the noise term of the channel model and in the closed-form expressions quoted as analytical background (Chapter 2); no relay comparison is run on it, so no conclusion is drawn from a channel the thesis does not study.

A consequence worth stating plainly. The SNR argument sets the noise power from the average *symbol* energy, so it denotes E_s/N_0 in every configuration. Since $E_b = E_s/k$, E_b/N_0 is 3.01 dB below the stated value for QPSK. Where a BER figure is read against a BPSK closed form, this conversion is stated explicitly at the point of use.

Power Normalization. All relay strategies normalize their output power to ensure fair comparison:

$$x_R \leftarrow x_R \cdot \sqrt{\frac{P_{\text{target}}}{P_{\text{current}}}} \quad (4.8)$$

4.1.4 Relay Processing

On Hop 1 the relay’s single receive antenna observes, as shown in Equation 4.9:

$$\tilde{y}_R[i] = |h_1[i]| x[i] + \tilde{n}_1[i], \quad \tilde{n}_1[i] \sim \mathcal{N}(0, \sigma^2) \quad (4.9)$$

where the effective observation follows coherent compensation of the Rayleigh fading coefficient (Section 4.1). Note that the per-symbol effective SNR is $|h_1[i]|^2 \gamma$ (a random, exponentially distributed quantity) not a constant; the AWGN calibration limit is recovered by setting $|h_1[i]| \equiv 1$. The relay applies its processing function to a sliding window of received samples and transmits a cleaner estimate $\hat{x}_R = f_\theta(\tilde{y}_{R,i-w:i+w})$; classical AF/DF use $w = 0$. This is purely a noise-removal task: in the canonical SISO setup there is no inter-stream interference at any stage, and the destination demodulates the Hop-2 output directly with no equalizer.

Remark 4.1 (Realizability of the symmetric window). The canonical *channel* of this chapter is memoryless: $y_R[i]$ is already a sufficient statistic for $x[i]$, so $w = 0$ loses nothing, and the classical AF and symbol-wise DF baselines do use $w = 0$ and are strictly causal. The learned relays nevertheless take a symmetric window $y_R[i - w : i + w]$ with $w > 0$ everywhere they appear, the canonical experiments included: the MLP uses $w = 2$ (a 5-sample window, Section 4.2) and the sequence models $w = 5$. This is an *architectural input convention*, not a claim about the channel. Under the canonical i.i.d. assumptions the extra samples are redundant — they carry no additional information about $x[i]$ — and the convention is adopted so that the identical architecture carries over unchanged to the three-tap intersymbol-interference channel of Chapter 6, where the window is what spans

the memory and is no longer redundant. Holding the input format fixed across both settings is what allows the channel assumptions to be varied without also varying the network structure.

Two consequences follow, and both apply to the canonical experiments rather than only to Chapter 6. First, a symmetric window is not causal: it reads w samples ahead. It is admissible in the half-duplex protocol only because the relay buffers the whole listening-phase block before transmitting, and it costs a fixed w -symbol processing latency. Second, that non-causality is a limitation of the learned relays, recorded as such rather than justified: a strictly causal past-only window $y_R[i - 2w : i]$ is what a streaming deployment would require, and it is not evaluated in this thesis. Because the window is redundant on the canonical channel, none of the canonical results depend on the look-ahead; the cost is real only where the window is doing work, which is Chapter 6.

4.2 Relay Strategies

Nine relay strategies were implemented, spanning four learning paradigms. The selection of these nine strategies is designed to systematically explore the relay design space along three axes:

1. the degree of processing (from no learning to deep generative models).
2. the type of inductive bias (feedforward, recurrent, attention, generative).
3. the model capacity (from 0 to 26K parameters).

This section describes each strategy’s architecture, training procedure, and the design rationale motivating each choice.

4.2.1 Selection and Classification of Relay Strategies

The selected nine relay strategies are drawn from four distinct architectural concepts:

(i) classical methods (AF and DF), (ii) supervised-learning models (MLP and hybrid MLP–DF), (iii) generative models (VAE and cGAN), and (iv) sequence models (Transformer, Mamba-S6, and Mamba-2 SSD).

1. **Classical AF:** Gain amplification as shown in Equation 2.1. AF serves as the minimal-processing classical baseline. It performs no learned processing and simply rescales the received signal, preserving both signal and noise. Its end-to-end effective SNR is given in Equation 2.2.
2. **Classical DF:** Demodulates, recovers bits, and re-modulates clean symbols (QPSK on the canonical channel, applied per axis). DF serves as the maximal-processing

classical baseline. It performs full signal regeneration through symbol-wise demodulation and re-modulation. On the canonical matched channel and in the uncoded setting studied here, DF achieves the lowest BER among the classical baselines at medium-to-high SNR (see Remark 4.2). Its theoretical BER follows the error-composition formula of Equation 2.4.

3. **Supervised MLP (Minimal):** A two-layer feedforward neural network (multilayer perceptron, MLP) with 169 parameters following the relay architecture defined in Equation 2.9,

where $\mathbf{w} \in \mathbb{R}^5$ is a sliding window of received symbols ($w = 2$ neighbors on each side), and the hidden layer has 24 neurons. Parameters: $(5 \times 24 + 24) + (24 \times 1 + 1) = 169$.

Architecture note: The MLP relay is a standard discriminative two-layer perceptron trained with supervised learning (MSE loss on input–output pairs). In contrast, the generative models in this study are the VAE and cGAN (Section 4.2), which learn to *sample from* or *approximate* the data distribution. The MLP relay simply learns a deterministic mapping $f : \mathbb{R}^5 \rightarrow [-1, +1]$ from a noisy observation window to a denoised symbol estimate: a classical regression task.

Design rationale: The tanh output activation naturally constrains the output to $[-1, +1]$, matching the BPSK symbol range. The ReLU hidden layer provides the non-linearity needed to approximate the Bayes-optimal soft threshold $\tanh(y/\sigma^2)$. With 24 hidden neurons and a 5-dimensional input, the network has approximately 34 parameters per input dimension: sufficient for the low-complexity denoising task while avoiding overfitting. He initialization is used for ReLU layers to maintain proper gradient flow.

Training uses MSE loss with multi-SNR training data (i.e., 5, 10, and 15 dB), 25,000 samples, and 100 epochs with a learning rate of 0.01.

4. **Hybrid:** SNR-adaptive relay that switches between MLP (low SNR) and DF (high SNR) based on a learned threshold. Combines the AI advantage at low SNR with DF’s low-BER, zero-parameter operation at high SNR. Same 169 parameters as MLP.

Design rationale: The Hybrid relay is motivated by the observation (confirmed empirically) that AI relays outperform DF only at low SNR, while DF achieves the lowest BER among the evaluated symbol-wise relay strategies at high SNR. By introducing a switching threshold, the Hybrid relay automatically selects the better strategy for each operating condition. The threshold is determined empirically from the training data by finding the SNR at which MLP and DF BER curves cross. This

approach requires no additional parameters beyond those of the MLP sub-network.

Training points at the relay node: SNR = **5, 10, 15 dB** (same trained MLP sub-network as the Minimal MLP relay).

5. **Generative VAE:** Probabilistic relay with encoder $q_\phi(\mathbf{z}|\mathbf{x})$ mapping to a latent space and decoder $p_\theta(\mathbf{x}|\mathbf{z})$ reconstructing the signal. Architecture: encoder ($7 \rightarrow 32 \rightarrow 16 \rightarrow \mu, \sigma^2(8)$), decoder ($8 \rightarrow 16 \rightarrow 32 \rightarrow 1$). Total: 1,777 parameters. Trained with β -VAE loss ($\beta = 0.1$) for 100 epochs.

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The low $\beta = 0.1$ prioritizes reconstruction quality over latent space regularization, appropriate for a task where accurate signal recovery is paramount. The 8-dimensional latent space provides sufficient representational capacity for the BPSK signal manifold while maintaining a tractable KL divergence. The encoder window of 7 symbols provides local context.

6. **Generative cGAN (WGAN-GP):** Adversarial relay with a generator conditioned on the noisy signal and a critic providing the training signal. Generator: ($7 + 8 \rightarrow 32 \rightarrow 32 \rightarrow 16 \rightarrow 1$), Critic: ($1 + 7 \rightarrow 32 \rightarrow 16 \rightarrow 1$). Total: 2,946 parameters. Trained with Wasserstein loss, gradient penalty ($\lambda = 10$), and L1 reconstruction loss ($\lambda_{L1} = 100$) for 200 epochs.

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The high $\lambda_{L1} = 100$ ensures that the generator is primarily supervised by the reconstruction loss, with the adversarial term acting as a regularizer that encourages outputs to lie on the clean signal manifold. The noise vector $\mathbf{z} \in \mathbb{R}^8$ provides the stochastic input needed by the GAN framework. The 5:1 critic-to-generator update ratio follows the WGAN-GP recommendation for stable critic training. The doubled epoch count (200 vs. 100) compensates for the slower per-step convergence of adversarial training.

7. **Sequential models: Transformer:** Multi-head self-attention over a window of 11 symbols. Architecture: $d_{\text{model}} = 32$, 4 attention heads, 2 encoder layers, feedforward dimension 128. Total: 17,697 parameters. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The Transformer is included as the dominant sequence architecture in modern deep learning. The 11-symbol window provides the same temporal context as the Mamba-S6/SSD models. With $d_{\text{model}} = 32$ and 4 heads, each head operates in an 8-dimensional subspace, which is sufficient for capturing the local noise structure. The feedforward dimension of 128 ($4 \times d_{\text{model}}$) follows the standard

Transformer expansion ratio.

8. **Sequential models: Mamba-S6** Selective state space model with input-dependent state transitions. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 Mamba blocks with residual connections. Total: 24,001 parameters. Each block applies: LayerNorm $\rightarrow$ expand ($32 \rightarrow 64$) $\rightarrow$ split to Conv1D/SiLU/S6 main branch and SiLU gate $\rightarrow$ contract ($64 \rightarrow 32$) $\rightarrow$ residual. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The expand factor of 2 ($32 \rightarrow 64$) doubles the internal dimension during the S6 scan, providing richer state dynamics. The state dimension $N = 16$ means each S6 layer implements a 16th-order adaptive filter: substantially more expressive than classical Wiener or matched filters. LayerNorm and residual connections prevent gradient degradation across layers. The SiLU gating provides multiplicative interactions that help the model learn sharp decision boundaries.

9. **Sequential models: Mamba-2 SSD** Structured State Space Duality model that replaces the sequential S6 recurrence with a chunk-parallel structured matrix multiply. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, chunk size 8, 2 Mamba-2 blocks with SiLU gating and residual connections. Total: 26,179 parameters. Each block applies: LayerNorm $\rightarrow$ parallel gate/SSD branches $\rightarrow$ SiLU gate $\rightarrow$ contract ($64 \rightarrow 32$) $\rightarrow$ residual. The SSD layer builds a lower-triangular causal kernel M per chunk and applies it via batched matmul, with inter-chunk state passing for continuity. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$) and gradient clipping ($\|\nabla\| \leq 1$).

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The chunk size of 8 is chosen to balance parallelism (larger chunks = fewer sequential inter-chunk passes) against memory cost (the $L \times L$ SSM matrix scales quadratically with chunk size). At the 11-symbol window, this yields 2 chunks ($8 + 3$), with a single inter-chunk state pass. Gradient clipping at norm 1.0 prevents the exponential blowup of gradients through the cumulative matrix products in the SSD computation. The S4D-style initialization $A_{\log} = \log(1, 2, \dots, N)$ provides geometrically spaced decay rates, enabling the model to simultaneously capture short-range and medium-range dependencies.

Remark 4.2 (Terminology: symbol-wise DF vs. information-theoretic DF). The term “decode-and-forward” is used in two distinct senses in the literature, and the distinction matters for interpreting the results of this thesis. In its *information-theoretic* sense [1, 2], DF is a *block* strategy: the source protects each message block with a strong (ideally capacity-achieving) channel code, the relay decodes the entire block, and only then re-encodes and forwards the recovered message. This block-DF is inherently *not* memoryless

and, over a single link, can operate reliably at any rate below the first-hop capacity.

The system studied in this thesis is *uncoded*: one QPSK symbol carries two information bits, one per axis, and no outer error-correction code is applied. Consequently, the “DF” baseline throughout this work refers to *symbol-wise slicing*: per-symbol demodulation followed by re-modulation, i.e., $f_{\text{DF}}(y_R[i]) = \text{sign}(y_R[i])$ with $w = 0$. This is the appropriate classical counterpart to the learned relays under the uncoded BER metric used here, but it is strictly weaker than block-DF with coding; the reported DF results should not be read as bounds on coded block-DF performance. This caveat is measured directly, not left as a standing assumption: Section 5.4 adds a real information-theoretic block-DF baseline (a rate-1/2 convolutional code with soft-decision Viterbi decoding, swept over constraint length and both canonical constellations) and reports where it does and does not beat the symbol-wise DF used elsewhere in this thesis.

4.3 Simulation Framework

4.3.1 Monte Carlo BER Estimation

BER is estimated through Monte Carlo simulation, which provides an unbiased estimate of the true BER at each SNR point. The simulation parameters are:

- **Bits per trial:** 10,000
- **Trials per SNR:** 10 (independent random seeds)
- **Total bits per SNR point:** 100,000
- **SNR range:** 0 to 20 dB (step: 2 dB), yielding 11 evaluation points
- **Random seed control:** Each trial uses a unique seed for bit generation and noise realization

BER Computation:

$$\hat{P}_e = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(b_i \neq \hat{b}_i) \quad (4.10)$$

where $\mathbb{I}(\cdot)$ is the indicator function and $N = 10,000$ is the number of bits per trial. By the central limit theorem, for large N and moderate BER (say $P_e \geq 10^{-3}$), the per-trial BER estimate is approximately normally distributed with variance $P_e(1 - P_e)/N$.

Confidence Intervals. The 95% confidence interval is computed from the $M = 10$ independent trial estimates as:

$$\text{CI}_{95\%} = \bar{P}_e \pm t_{0.025, M-1} \cdot \frac{s}{\sqrt{M}} \quad (4.11)$$

295

296 where $\bar{P}_e$ is the sample mean BER across trials, s is the sample standard deviation, and
 297 $t_{0.025,9} = 2.262$ is the critical value of the Student's t -distribution with 9 degrees of free-
 298 dom. At low BER ($\hat{P}_e \lesssim 10^{-4}$), the normal approximation may become unreliable due to
 299 the small number of observed errors; however, in this regime the relay methods converge
 300 and the relative ranking is stable.

301 **BER resolution.** With $N \cdot M = 100,000$ total bits per SNR point, the minimum detectable
 302 BER is approximately $1/N = 10^{-4}$ per trial, or 10^{-5} for the aggregate. BER values below
 303 this threshold are reported as 0.

304 **Per-experiment simulation budgets.** The budget above ($M = 10$ trials $\times N = 10,000$
 305 bits = 100,000 bits per SNR point) applies to the canonical relay comparison and the
 306 parameter-normalization/complexity study (E2–E3). Other experiments use different bud-
 307 gets, stated in their respective sections: the main unknown-ISI study, the flat-channel
 308 control, and the QPSK generalization check (Chapter 6, Sections 6.1, 6.1.1, 6.1.2) use
 309 $10 \times 100,000 = 1,000,000$ bits per SNR point, matching the project-wide $M = 10$ stan-
 310 dard; the composite cascade (Section 6.1.3) likewise uses $10 \times 100,000$; and the blind and
 311 partial-posterior sub-studies (Sections 6.1.5–6.1.4) use $50 \times 20,000 = 1,000,000$ bits per
 312 SNR or operating point. The larger trial count in those last two is deliberate rather than
 313 incidental: their hop-1 channel redraws its ISI, amplifier and phase realization for every
 314 block, so a trial is a draw from the impairment family and the ensemble average is limited
 315 by the trial count, not by bits per trial. Increasing bits per trial sharpens the estimate for
 316 one particular channel; only increasing trials averages over the family the study is about.
 317 Every experiment therefore uses $M \geq 10$ trials, so the Wilcoxon signed-rank test is at-
 318 tainable throughout (at $M = 5$ its smallest two-sided p is $2^{-4} = 0.0625$, which could not
 319 reach $p < 0.05$).

320 4.3.2 Statistical Significance Testing

321 Differences between relay methods are assessed using the **Wilcoxon signed-rank test**, a
 322 non-parametric paired test. For each pair of relay strategies (A, B) at each SNR point, the
 323 test compares the $M = 10$ paired BER observations (Equation 4.12):

$$H_0 : \text{median}(P_{e,A} - P_{e,B}) = 0 \quad \text{vs.} \quad H_1 : \text{median}(P_{e,A} - P_{e,B}) \neq 0 \quad (4.12)$$

324

325 The Wilcoxon test is preferred over the parametric paired t -test for two reasons: (i) BER
 326 distributions are bounded ($[0, 0.5]$) and potentially skewed, violating the normality assump-
 327 tion, and (ii) the test is robust to outlier trials. The significance level is set at $\alpha = 0.05$

328 *per comparison*; no multiple-comparison correction (e.g. Holm or Bonferroni, or false-
 329 discovery-rate control) is applied across the many relay-pair and SNR-point comparisons
 330 made throughout this thesis, so the reported significance claims should be read at the per-
 331 comparison level rather than as holding jointly across the full set of tests performed.

332 4.3.3 Training Protocol

333 All AI relays are trained **once** at the beginning of each experiment using: - **Multi-SNR**
 334 **training data:** Signals generated at SNR = 5, 10, 15 dB in equal proportions - **Training**
 335 **samples:** 25,000 (supervised), 20,000 (generative), 10,000 (sequence models) - **Single**
 336 **training per channel:** The same trained model is evaluated across all SNR points

337 The multi-SNR training protocol is critical: training at a single SNR would produce a
 338 model that performs well at that SNR but poorly at others (overfitting to a specific noise
 339 level). By training at three representative SNR values spanning the low-to-moderate range,
 340 the network learns a robust denoising function that generalizes across operating conditions.
 341 The SNR values 5, 10, 15 dB were chosen to cover the regime where AI relays provide
 342 the most benefit (low-to-medium SNR).

343 **Impact of sparse SNR training grid.** Because models are trained at only three SNR
 344 points ($\{5, 10, 15\}$ dB) but evaluated over eleven points (0 to 20 dB, step 2), performance
 345 reflects two regimes: (i) **interpolation** within the trained span (approximately 4–16 dB),
 346 where BER is generally stable, and (ii) **extrapolation** at the edges (0–2 dB and 18–20 dB),
 347 where mild degradation may occur due to noise-statistics mismatch. In the present results,
 348 this sparse-grid effect is secondary: the relay ranking remains consistent across the full
 349 evaluated SNR range.

350 **Training method (what is trained, where it is trained).** Training is performed **offline**
 351 before BER sweeps. Only the seven AI relays are optimized (MLP, Hybrid’s MLP sub-
 352 network, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD); AF and DF are analyti-
 353 cal baselines and have no trainable parameters. Training data are synthetically generated
 354 source/channel pairs under the multi-SNR protocol, with model-specific losses (MSE for
 355 supervised relays, ELBO for VAE, WGAN-GP + L1 for cGAN, Adam-based supervised
 356 loss for sequence models). Compute placement follows implementation: NumPy models
 357 (MLP/Hybrid) train on CPU, while PyTorch models train on CPU or CUDA depending
 358 on device availability/configuration. After training, weights are checkpointed and reused
 359 across all SNR evaluations; BER curves are then produced in **inference-only** mode with-
 360 out further parameter updates.

361 **Weight initialization.** He initialization [25] is used for all layers with ReLU activation
 362 ($W \sim \mathcal{N}(0, 2/n_{\text{in}})$), while Xavier initialization is used for layers with tanh activation
 363 ($W \sim \mathcal{N}(0, 1/n_{\text{in}})$). These initialization schemes maintain proper gradient magnitude

through the network layers, preventing both vanishing and exploding gradients during training.

Optimizer settings. All PyTorch-based models use the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ (defaults). The MLP and Hybrid relays use a simple SGD implementation in NumPy with constant learning rate $\eta = 0.01$.

4.3.4 Reproducibility and Weight Management

All experiments use controlled random seeds at both the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results. Specifically, for seed s and trial t , the random state is initialized as $\text{seed}(s \cdot 1000 + t)$, ensuring that each trial is independent while the overall experiment is deterministic.

A **weight checkpoint system** saves trained model parameters to disk after training completes, enabling experiment resumption without retraining. This is particularly important for the cGAN (2-hour training time) and sequence models (24–37 minutes). The checkpoint system supports: - Automatic save after training with architecture metadata - Resume from saved weights with architecture validation - Seed-specific weight directories (e.g., `trained_weights/seed_42/`)

The framework includes 159 automated tests (pytest) covering all modules: channels, modulation, coding, relay strategies, simulation, statistics, and weight management, with 100% pass rate.

Code and reproducibility archive. The complete simulation framework (relaynet), every experiment script, and the committed result files each figure and table is generated from are maintained under version control (git) in the Gilzuk/relaynet2 repository; each numerical claim in this thesis traces to a specific committed script and result file rather than to an unarchived intermediate run, and `verify_thesis_tables.py` in that repository re-derives every numerical table from its source file and flags any mismatch. The exact commit this thesis was typeset from is recorded in the repository’s history rather than reproduced here, since stating a fixed hash in the text it describes is self-referential; the repository’s tags and commit log identify the revision corresponding to any given PDF build.

4.4 Normalized Parameter Comparison

A fundamental confound in comparing AI architectures is that different models have vastly different parameter counts: from 169 (MLP) to 26,179 (Mamba-2 SSD): a ratio of 155:1. Performance differences between these models could reflect either (a) the superiority of one architecture’s inductive bias, or (b) the simple advantage of having more learnable

parameters. To reduce this confound, all seven AI models were scaled to approximately 3,000 parameters, giving a comparison at an equal parameter budget. The control is over capacity alone: as Table 4.2 shows, meeting a common budget required changing width, depth, state dimension and, for the feedforward models, the input window size, so those factors co-vary with architecture and the comparison does not isolate architectural inductive bias from them. The MLP-3K in particular sees an 11-sample window against its canonical 5, so it is not fed the same context as the models whose windows were left unchanged.

Choice of target parameter count. The target of $\sim 3,000$ parameters was chosen as a compromise: it is large enough that all architectures can express a reasonable denoising function (avoiding underfitting), yet small enough that the normalization represents a meaningful reduction for the larger models (Transformer: $5.9\times$ reduction, Mamba-S6: $7.9\times$, Mamba-2 SSD: $8.7\times$). The MLP model is scaled **up** from 169 to 3,004 parameters, testing whether additional capacity benefits the simple feedforward architecture.

Scaling methodology. Each architecture’s hyperparameters were adjusted to reach the target while preserving its essential structure:

Table 4.2: Normalized 3K-parameter model configurations: parameter count and scaling method for each architecture.

Model	Parameters	Scaling Method
MLP-3K	3,004	Increased window (5→11) and hidden (24→231)
Hybrid-3K	3,004	Same as MLP-3K + DF switching
VAE-3K	3,037	Increased window (7→11), adjusted latent/hidden
cGAN-3K	3,004	Increased window (7→11), adjusted hidden layers
Transformer-3K	3,007	Reduced d_{model} (32→18), heads (4→2), layers (2→1)
Mamba-S6-3K	3,027	Reduced d_{model} (32→16), d_{state} (16→6), layers (2→1)
Mamba-2-SSD-3K	3,004	Reduced d_{model} (32→15), d_{state} (16→6), layers (2→1)

Parameter counting convention. All learnable parameters are counted, including biases, normalization parameters (LayerNorm gain/bias), and projection matrices. Non-learnable buffers (e.g., positional encoding tables, fixed $\mathbf{A}$ matrices) are excluded. The counts are verified programmatically via `sum(p.numel() for p in model.parameters() if p.requires_grad)`.

Unified input window. All 3K models use a window size of 11 (vs. 5 for original MLP/Hybrid, and 7 for original VAE/cGAN), providing a common input context. This ensures that differences in performance reflect architectural inductive biases rather than differences in the amount of input information available.

This normalization removes parameter count as an explanation for any residual gap, which is what makes the comparison informative: if performance converges at equal parameter budgets, the conclusion is that capacity (not architecture) was the dominant factor in the unnormalized comparison. If significant gaps persist, the conclusion is weaker than “architectural inductive bias provides an advantage”, because the scaling changed more than the architecture label: it is that something other than raw parameter count accounts for the gap, with architecture, depth, width, state dimension and window size still confounded among themselves.

4.5 Modulation: QPSK

The experiments use QPSK, the canonical constellation. This section defines it and the I/Q-splitting technique that lets real-valued relays process a complex constellation.

4.5.1 QPSK: Gray-Coded Quadrature Phase-Shift Keying

Quadrature Phase-Shift Keying maps pairs of bits (b_0, b_1) to complex symbols [24]:

$$x = \frac{(1 - 2b_0) + j(1 - 2b_1)}{\sqrt{2}} \quad (4.13)$$

yielding four constellation points at $\{(\pm 1 \pm j)/\sqrt{2}\}$ with unit average power ($E_s = 1$).

The Gray coding ensures that adjacent constellation points differ by exactly one bit:

Table 4.3: QPSK Gray-coded symbol mapping: bit pairs to complex symbols.

Bit pair (b_0, b_1)	Symbol	Quadrant
00	$(+1 + j)/\sqrt{2}$	I
01	$(+1 - j)/\sqrt{2}$	IV
11	$(-1 - j)/\sqrt{2}$	III

Bit pair (b_0, b_1)	Symbol	Quadrant
10	$(-1 + j)/\sqrt{2}$	II

Demodulation applies independent sign decisions on each component: $\hat{b}_0 = \mathbb{1}(\text{Re}(\hat{x}) < 0)$ and $\hat{b}_1 = \mathbb{1}(\text{Im}(\hat{x}) < 0)$. The spectral efficiency is 2 bits/symbol, double that of BPSK.

Theoretical QPSK BER. For uncoded QPSK over an AWGN channel, the BER per bit equals the BPSK BER at the same E_b/N_0 because each I/Q component carries an independent BPSK stream [24]:

$$P_b^{\text{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_b^{\text{BPSK}} \quad (4.14)$$

The advantage of QPSK is doubled throughput for the same BER and per-bit energy.

4.5.2 Constellation Diagram Summary

Figure 4.2 presents the constellation diagrams for the modulation schemes appearing in this thesis. The progression from BPSK (2 points on the real axis) through QPSK (4 points on the unit circle) to denser grids illustrates the fundamental trade-off between spectral efficiency and noise robustness: higher-order constellations pack more bits per symbol but reduce the minimum Euclidean distance between points, increasing the BER at a given SNR. Only QPSK is evaluated in this thesis; the others are shown for orientation.

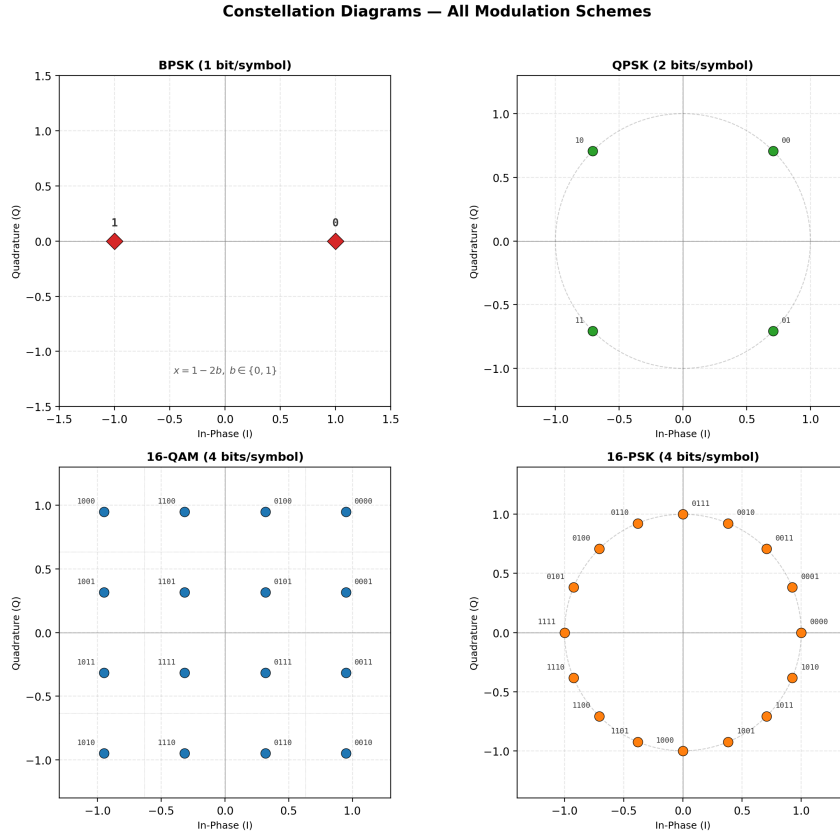


Figure 4.2: Constellation diagrams. (a) BPSK: 2 real-valued symbols at ± 1 . (b) QPSK: 4 Gray-coded symbols on the unit circle. (c) 16-QAM: 16 Gray-coded symbols on a 4×4 rectangular grid with decision boundaries (dotted lines). (d) 16-PSK, shown for completeness only; it is not studied in this work. All constellations are normalised to unit average power.

4.5.3 I/Q Splitting for AI Relay Processing of Complex Constellations

$$\hat{x}_R = f_\theta(\text{Re}(y_R)) + j \cdot f_\theta(\text{Im}(y_R)) \quad (4.15)$$

Justification and caveat. For rectangular constellations (QPSK, QAM), the I and Q components carry independent information and are corrupted by independent noise, so per-axis denoising with a shared real-valued function is a natural and parameter-efficient decomposition. It is, however, a *restriction* of the hypothesis class to axis-separable functions, not an equivalence to joint 2D processing: per-axis errors accumulate independently. For QPSK, where each axis carries a single binary decision, the restriction costs nothing; for denser constellations it becomes a structural limitation that a joint two-dimensional relay formulation would be needed to remove, which is left to future work.

Relay-specific handling:

Table 4.4: Relay-specific complex signal handling strategies and rationale.

Relay type	Complex signal processing	Rationale
AF	Amplifies complex signal directly	Power normalization ($\ y\ ^2$) is valid for complex vectors
AI relays	I/Q splitting (process Re and Im separately)	Real-valued networks; independence of I/Q in rectangular constellations

Limitation beyond binary-per-axis constellations. I/Q splitting works here because QPSK places a single binary decision on each axis, which is exactly what a tanh-output relay trained on $\{\pm 1\}$ represents. Constellations with more than two levels per axis (16-QAM and denser) break that correspondence and would require either modulation-specific retraining or a joint two-dimensional relay formulation; neither is evaluated here, and both are identified as future work.

Chapter 5

Core Experiments: The Canonical Setup

5.1 Experiments summary

Scope relative to the system model. All experiments in this chapter are conducted under the canonical system model defined in Chapter 4. In particular, the channel remains memoryless, the noise is white, transmission is uncoded, and relay processing is evaluated on a per-symbol basis. Consequently, the objective of this chapter is to compare relay functions under matched-model conditions rather than to address channel estimation, sequence detection, or channels with memory. Such effects are introduced only later in Chapter 6 as deliberate departures from the canonical assumptions in order to study model mismatch and robustness.

Following the single-model convention fixed in Chapter 1, this chapter evaluates the relay strategies on the *canonical setup only*: a two-hop SISO link, i.i.d. Rayleigh fast fading on both hops, complex baseband, QPSK, uncoded BER. BPSK appears in this chapter solely on the AWGN calibration channel, where the closed forms live (Table 4.1). Two experiments constitute the core: the canonical relay comparison (E2) and the parameter-normalization and complexity study (E3), which together answer every canonical hypothesis (H1–H4). The principal contribution (Chapter 6) is the unknown-channel study (E6), which tests hypothesis H6. All other departures from the canonical setup are identified as future work in Chapter 7. The experimental configurations are summarised in the table below.

Each experiment subsection reports the key results inline (BER tables and figures); supporting material and model hyperparameters are collected in the appendices (Appendices 9.1–9.3).

Table 5.1: Summary of experimental configurations

Exp.	Topology	Modulation	Channel / Equalization	Focus
E2	SISO	QPSK	Rayleigh (canonical)	Canonical relay comparison (H1, H2)
E3	SISO	QPSK	Rayleigh (canonical)	Parameter normalization & complexity (H3, H4)
E6	SISO	BPSK, QPSK	Unknown (ISI; composite; blind; partial-posterior); control: Rayleigh	Extension: unknown-channel failure of AF/DF and learned mitigation (H6)

5.2 Canonical Relay Comparison (SISO, Rayleigh, QPSK)

Goal: Evaluate the classical and AI-based relay strategies on the canonical setup of this thesis. Eight of the nine implemented relays are reported; the cGAN is excluded from the re-run that produced these numbers.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** Supervised (MLP, Hybrid), Generative (VAE, cGAN), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN activation:** ReLU hidden, tanh output

Conclusion: On the canonical Rayleigh channel, AI relays match but do not surpass symbol-wise DF: DF achieves the lowest or tied-lowest BER at essentially every SNR point, including the low-SNR regime, while the best neural relays (MLP, Hybrid, cGAN) track it within statistical uncertainty and clearly outperform AF. This confirms H1 (AI relays beat AF at low SNR) and H2 (DF dominance at ≥ 6 dB with zero parameters) on the canonical channel.

5.2.1 Results Tables

Table 5.2: BER on the canonical setup: QPSK over i.i.d. Rayleigh fast fading, SISO on both hops, uncoded, $10 \times 10,000$ bits per SNR point. The cGAN column is absent because the relay was excluded from the re-run that produced this table on cost grounds (roughly two GPU-hours to train against the MLP’s seconds on a CPU); the omission drops its column rather than carrying forward a value measured on the earlier, 3 dB-offset axis. “DF th.” is the closed-form two-hop composition of Eq. 2.34 (Section 2.5.2.2), included for direct comparison against the measured DF column.

SNR (dB)	AF	DF	DF th.	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2
0	0.4076	0.3337	0.3333	0.3349	0.3329	0.3359	0.4014	0.4071	0.4032
4	0.3129	0.2219	0.2216	0.2233	0.2220	0.2254	0.2758	0.2812	0.2772
8	0.1852	0.1218	0.1203	0.1229	0.1220	0.1235	0.1424	0.1448	0.1416
12	0.0822	0.0580	0.0560	0.0586	0.0579	0.0589	0.0628	0.0637	0.0621
16	0.0309	0.0245	0.0239	0.0247	0.0245	0.0247	0.0260	0.0259	0.0251
20	0.0110	0.0097	0.0098	0.0099	0.0097	0.0097	0.0101	0.0100	0.0098

Two features of Table 5.2 deserve comment. First, symbol-wise DF tracks the closed form (the “DF th.” column above): reading the abscissa as E_s/N_0 and converting to E_b/N_0 for QPSK ($k = 2$, so 3.01 dB lower), the two-hop expression $2P(1 - P)$ with $P = \frac{1}{2}(1 - \sqrt{\bar{\gamma}/(1 + \bar{\gamma})})$ gives 0.0098 at 20 dB against a measured 0.00972, and 0.0239 at 16 dB against 0.02451: agreement within one percent at every tabulated point, which validates the simulator end to end against the analysis of Section 2.5.2.2. Figure 5.1 plots this closed form alongside the measured curves, together with the single-hop floor P itself – the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – which no relay may cross and none does at any SNR.

Second, the relays divide by family rather than by paradigm. The three feedforward relays, the MLP, the Hybrid and the VAE, are statistically indistinguishable from DF from 4 dB upward and from each other throughout; the VAE reaches 0.00972 at 20 dB, matching DF exactly at the resolution of this budget. The two sequence models are the ones that separate, trailing by roughly 0.07 in BER at 0 dB before converging by 16 dB. Capacity is not the explanation, since the sequence models are the larger of the two groups; the memoryless channel simply offers no temporal structure for their inductive bias to exploit, a reading the equal-parameter comparison of Section 5.3 tests directly and confirms.

Two comments on Table 5.2. First, symbol-wise DF is the best or tied-best relay at every tabulated point, and the learned relays sit within a few thousandths of it from 4 dB upward: on a matched, known channel the learned relay matches DF rather than beating it, at 169 parameters against DF’s none. This is H2, and it is the result the rest of the thesis is measured against. Second, the two sequence models trail the feedforward relays at low SNR (0.4014 and 0.4032 at 0 dB against the MLP’s 0.3349) and converge with them by 16 dB, so their extra capacity buys nothing on a memoryless channel; Section 5.3 separates that from the parameter-count confound.

Why QPSK and not BPSK. The canonical channel is complex, so the canonical constellation is complex: QPSK places one bit on each axis the Rayleigh coefficient acts in, and the I/Q-split relays process it without modification. BPSK on this channel would require discarding the imaginary part of the equalized sample, throwing away half the plane the channel acted in, which is why BPSK is not used on this channel at all. One consequence must be kept in view when reading these BER figures against BPSK closed forms: the SNR axis here is E_s/N_0 , and QPSK carries two bits per symbol, so a given abscissa corresponds to an E_b/N_0 that is 3.01 dB lower than the same number on a BPSK axis. The constellations are not on a common axis and their error rates are not directly comparable.

5.2.2 Results Figures

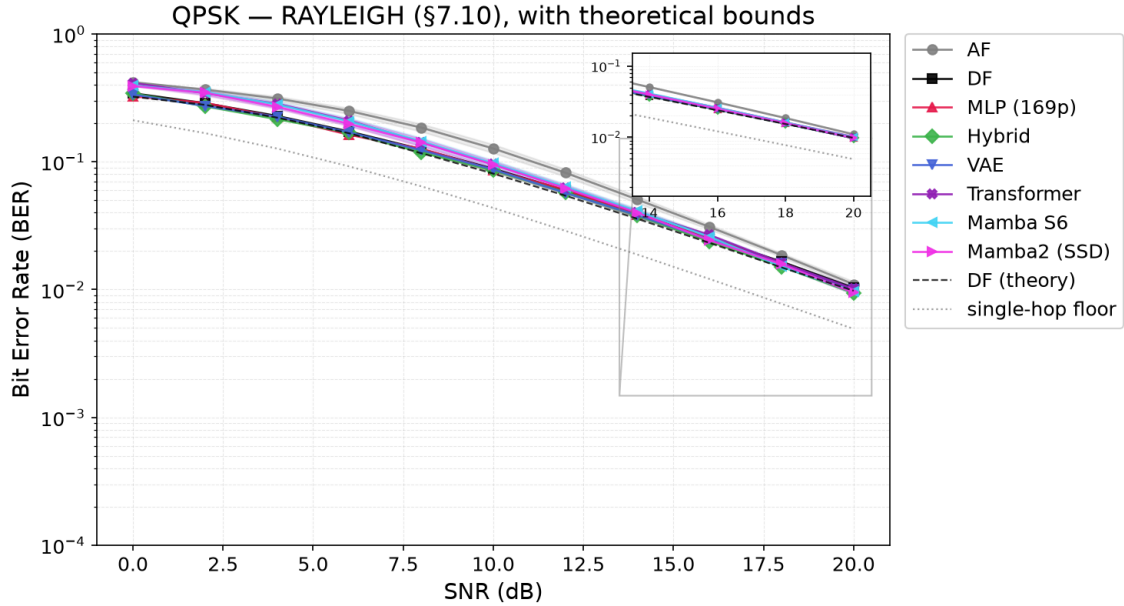


Figure 5.1: The canonical setup: QPSK over i.i.d. Rayleigh fast fading, BER vs. E_s/N_0 for the eight relays of Table 5.2, with 95% confidence intervals, plus two theoretical reference curves: “DF (theory)” (dashed), the closed-form two-hop composition, which the measured DF curve tracks within one percent everywhere; and the “single-hop floor” (dotted), the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – a physical lower bound no relay crosses. AF is uniformly worst; symbol-wise DF and the feedforward learned relays are indistinguishable from 4 dB upward; the two sequence models trail at low SNR and converge by 16 dB.

5.3 Parameter Normalization & Complexity Trade-off

This study tests H3 (U-shaped complexity in BER) and H4 (architecture convergence at equal scale) on the canonical setup, and provides the training-time benchmark for H5.

Goal: Isolate architectural inductive biases from parameter count effects and characterize the complexity-performance trade-off for relay denoising.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** All normalized to $\sim 3,000$ parameters, plus original sizes (169 to 26K) - **NN activation:** ReLU hidden, tanh output

Conclusion: A minimal 169-parameter MLP matches models roughly $140\times$ larger, and no larger model evaluated here improves on it; whether the curve eventually turns upward, and whether overfitting would be the reason, is not tested by these experiments (Section 7.2). At an equal budget of 3,000 parameters the gap between feedforward and sequence architectures is strongly SNR-dependent: it falls monotonically from 17.4% at 0 dB to 4.1% at 20 dB (Table 5.3). Parameter count therefore accounts for the comparison at high SNR, where the choice of model is nearly irrelevant, but not at low SNR, where the six equal-sized models separate cleanly into a feedforward group tracking DF and a sequence group tracking AF. Since all six carry the same budget, that separation is not explained by capacity; it is attributable to the remaining differences between the models, of which architecture is the most salient but not the only one, since depth, width, state dimension and window size were adjusted to meet the budget (Table 4.2). The generative VAE is a consistent underperformer in this configuration; as noted for Table 5.2, this may reflect the stochastic inference-time decoding used here rather than the generative paradigm itself (Section 2.3.3).

5.3.1 Results Tables

Table 5.3: Equal-parameter-budget comparison on the canonical setup (QPSK over Rayleigh). Every learned model is scaled to approximately 3,000 parameters, so parameter count is removed as an explanation for the residual differences; width, depth, state dimension and window size still co-vary with architecture (Table 4.2). AF and DF are the unparameterized references. The cGAN is excluded, as in Table 5.2.

SNR (dB)	MLP-3K	Hybrid-3K	VAE-3K	Transformer-3K	Mamba-S6 (3K)	Mamba-2 SSD (3K)
0	0.3361	0.3391	0.3424	0.4068	0.4007	0.4001
4	0.2244	0.2271	0.2291	0.2818	0.2751	0.2742
8	0.1229	0.1229	0.1267	0.1458	0.1406	0.1405
12	0.0586	0.0580	0.0604	0.0640	0.0620	0.0622
16	0.0247	0.0245	0.0251	0.0259	0.0250	0.0251
20	0.0099	0.0097	0.0101	0.0100	0.0097	0.0098

5.3.2 Results Figures

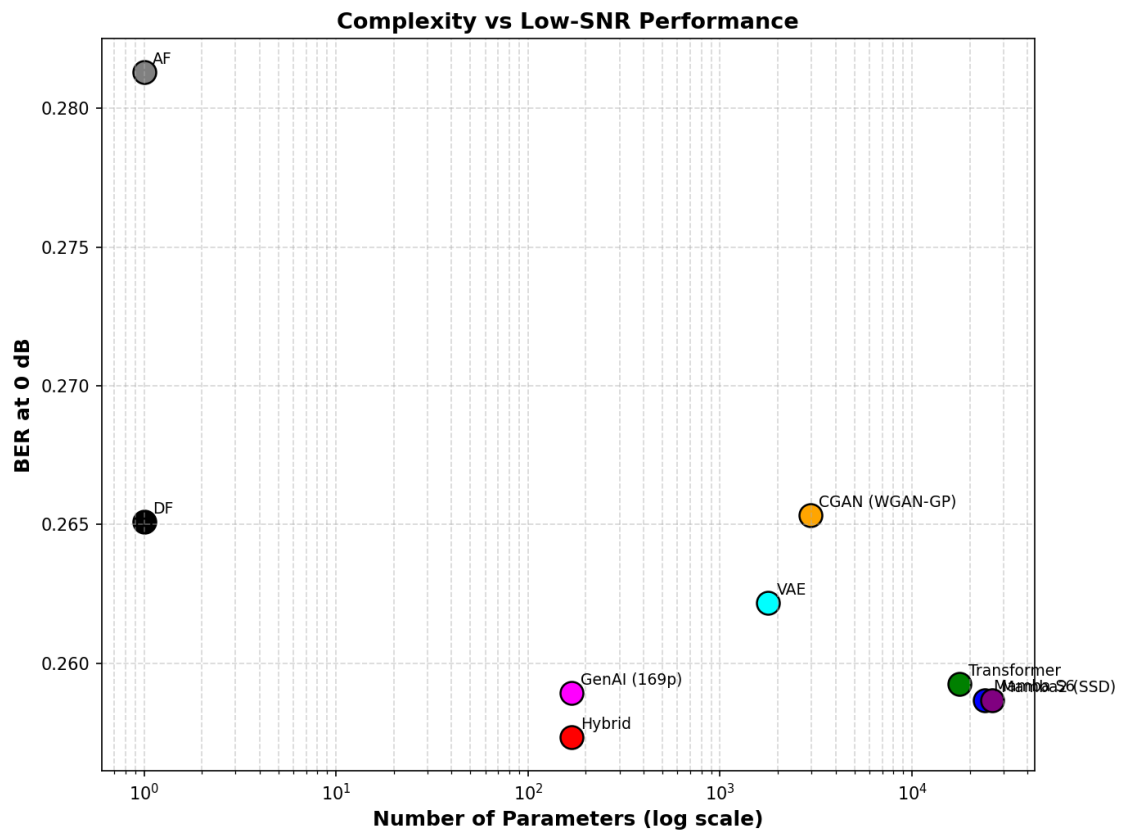


Figure 5.2: Complexity versus low-SNR performance. BER at 0 dB is shown as a function of model size.

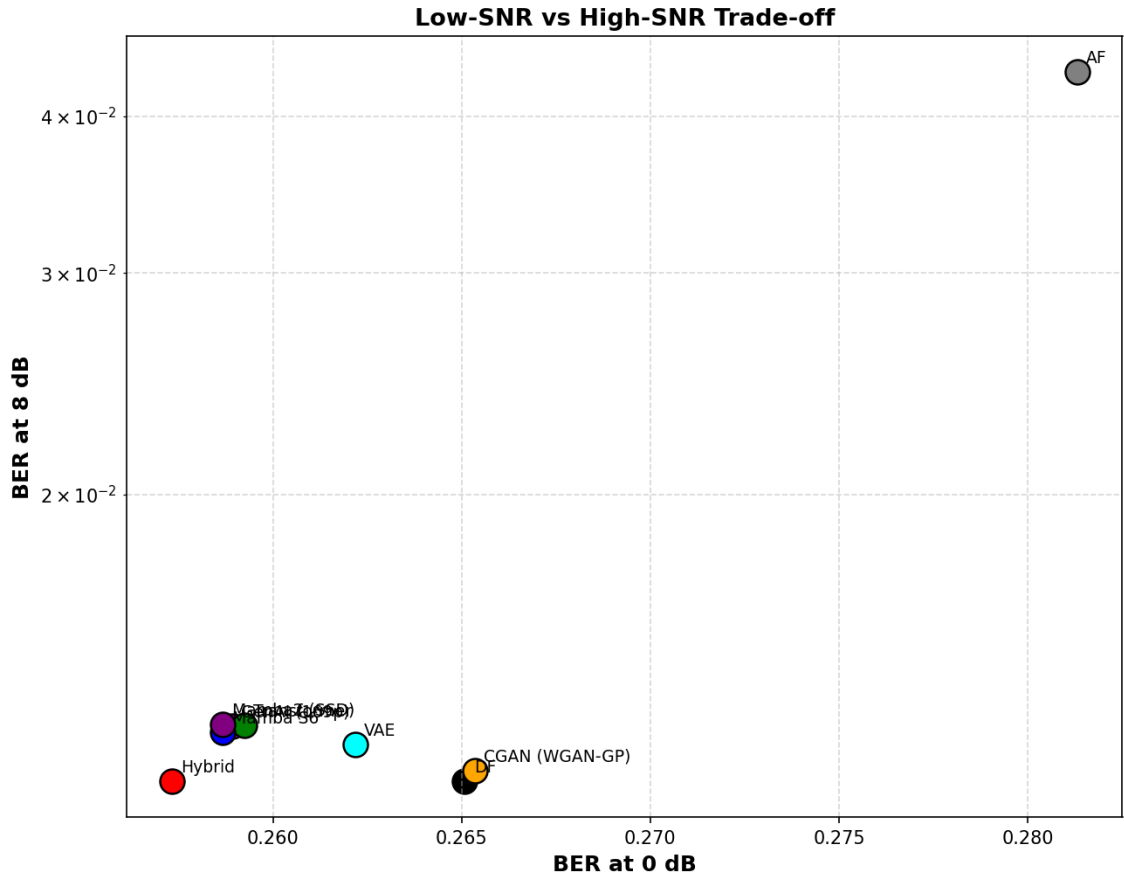


Figure 5.3: Low-SNR versus high-SNR trade-off. BER at 8 dB is plotted against BER at 0 dB.

5.4 Coded Block-DF: Measuring the Remark 4.2 Caveat

A note on constellation scope. The relay comparisons of this thesis are conducted on QPSK throughout, and 16-QAM is not studied as a relay-architecture question (Section 4.5). It does appear below as one rung of the modulation-and-coding ladder that the link-adaptation study of Section 5.4.4 selects from, where the object of study is the *rate-and-modulation choice* rather than the relay’s ability to process a denser constellation. Nothing in that study rests on how a learned relay handles four levels per axis, which is the question left to future work.

Remark 4.2 draws a distinction the rest of this thesis relies on: the DF baseline used everywhere here is *symbol-wise* (uncoded, per-symbol hard slicing), not the *block* DF of the information-theoretic relay-channel literature, and the remark explicitly cautions that “the reported DF results should not be read as bounds on coded block-DF performance.” That caveat was asserted, not measured. This section measures it: a real information-theoretic block-DF relay, built from a rate-1/2 convolutional code and a soft-decision Viterbi decoder, added as a supplementary study on the canonical channel rather than as a revision

117 of the core comparison above.

118 **Setup.** A standard rate-1/2 convolutional code (constraint length K , 2^{K-1} trellis states,
 119 zero-tail terminated per 200-information-bit frame) is applied before QPSK modulation
 120 on the canonical Rayleigh channel; `CodedDecodeAndForwardRelay` decodes the
 121 full frame at the relay via soft-decision Viterbi (squared-distance branch metric on the
 122 coherently-compensated I/Q samples), then re-encodes and re-modulates a fresh, clean
 123 codeword before forwarding – genuine block DF, as opposed to the per-symbol slicing
 124 used elsewhere in this thesis. Two coded-aware learned relays are trained on the same
 125 task for comparison: a windowed 756-parameter MLP classifier (window 21, $\approx 5 \times K$ for
 126 $K = 3$) and the Mamba-S6 architecture already used in Table 5.2 (24,084 parameters at
 127 this configuration), both predicting the transmitted QPSK symbol from a window of noisy
 128 coded observations with no explicit knowledge of the code. All results use the thesis-
 129 standard budget of 10 trials $\times$ 100,000 information bits per SNR point.

Table 5.4: Coded block-DF vs. uncoded symbol-wise DF and coded-aware learned relays, canonical QPSK/Rayleigh, rate-1/2 $K=3$ code. “coded-AF” forwards the noisy codeword unmodified (the relay does not decode), isolating the code’s own gain from any relay intelligence.

SNR (dB)	uncoded DF	coded AF	coded DF	MLP-coded	Mamba-coded
0	0.3336	0.4904	0.4210	0.4381	0.4491
4	0.2213	0.4448	0.2231	0.2807	0.3216
8	0.1204	0.2904	0.0638	0.0917	0.1175
12	0.0558	0.0544	0.0152	0.0206	0.0225
16	0.0242	0.0088	0.0043	0.0046	0.0042
20	0.0099	0.0018	0.0013	0.0011	0.0009

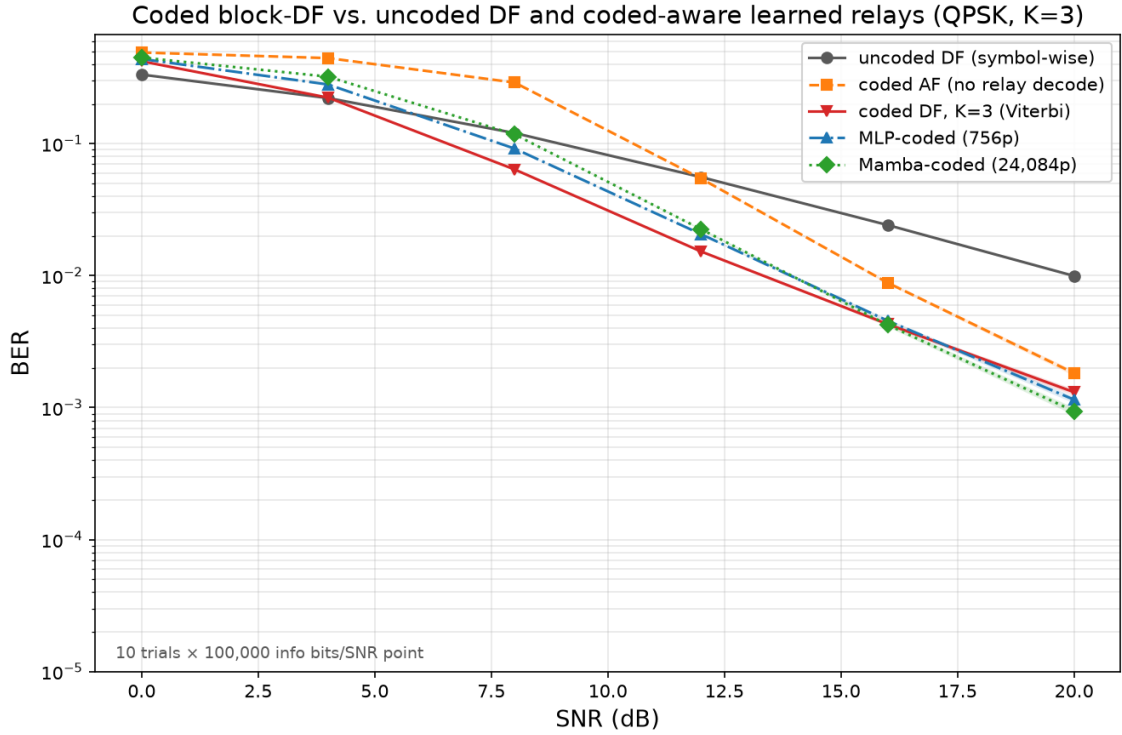


Figure 5.4: The data of Table 5.4 plotted: coded block-DF crosses below uncoded symbol-wise DF between 4 and 8 dB and stays below it thereafter; coded AF, which never decodes, never catches up. Both coded-aware learned relays track coded-DF closely from 12 dB up but are visibly worse at 4–8 dB.

Two findings, reported as measured rather than smoothed toward either a "coding always helps" or a "the learned relay wins" narrative:

The caveat holds precisely where it should, and no further. At equal E_s/N_0 , coded block-DF beats uncoded symbol-wise DF decisively from 8 dB upward – a $5.6\times$ reduction in BER at 16 dB, $7.6\times$ at 20 dB – closing the gap the remark left open. Those factors are the right comparison for a detection question but the wrong one for a link budget, because the rate-1/2 code halves the information carried per channel use; Section 5.4.3 redoes the comparison at equal throughput and the advantage largely disappears. But below roughly 6 dB, coding *hurts* even on the favourable axis: both coded AF and coded DF are worse than plain uncoded DF at 0 and 4 dB. This is the well-known convolutional-code error-propagation threshold, not an artifact: a fixed-rate code below its operating threshold produces decoding errors that cluster and propagate across the trellis, which a memoryless per-symbol slicer cannot do. The caveat is real, but it cuts both ways – coded block-DF is a strictly stronger baseline only above its own threshold, which itself has to be measured, not assumed.

At low-to-mid SNR the classical decoder is far ahead, and the learned relay's extra capacity buys nothing. Unlike the canonical memoryless channel of Table 5.2, where Section 5.3 found "the memoryless channel simply offers no temporal structure for [the

sequence models’] inductive bias to exploit,” a convolutional code *is* genuine temporal structure. It does not rescue the learned relays where the code is doing the most work: both are far behind coded-DF at 4–8 dB (Mamba-coded 0.1175 vs. 0.0638 at 8 dB). Nor does the $32\times$ -larger Mamba-S6 relay show a clear advantage over the 756-parameter MLP there, being worse at 4–8 dB (0.3216 vs. 0.2807 at 4 dB). Giving the sequence model a task with real memory to exploit therefore does not overturn the thesis’s “less is more” finding (H3): extra capacity still buys no reliable advantage over the minimal architecture.

At high SNR the ordering reverses, and it is not noise. The differences at 16–20 dB in Table 5.4 are small in absolute terms, and an earlier version of this section dismissed them as trial-to-trial variation. That was wrong, and the correction is worth stating explicitly because it changes the conclusion. Re-measuring those two points with *paired* trials – every relay driven by identical information bits and identical hop-1/hop-2 fading realizations within each trial, so the channel-draw variance that dominates an unpaired comparison cancels in the per-trial difference – and testing with the Wilcoxon signed-rank test used elsewhere in this thesis gives Table 5.5. At 20 dB both learned relays beat coded-DF in essentially every one of 100 paired trials ($p < 10^{-4}$). At 16 dB the result splits by architecture: Mamba-coded wins (68 of 100 trials) while the MLP significantly *loses* (23 of 100), both at $p < 10^{-4}$. The effects are real in both directions, and no single sentence of the form “the learned relay beats/does not beat Viterbi” describes them.

Table 5.5: Paired re-measurement of the two high-SNR points of Table 5.4, 100 paired trials $\times$ 100,000 information bits. “Diff” is the mean per-trial BER difference against coded-DF (negative favours the learned relay); “W/L” counts trials won/lost; p is Wilcoxon signed-rank.

SNR	Relay	BER	Diff vs. coded-DF	W/L	p
16 dB	coded-DF	0.004245	—	—	—
	MLP-coded	0.004437	+0.000192	23/77	$< 10^{-4}$
	Mamba-coded	0.004124	−0.000121	68/32	$< 10^{-4}$
20 dB	coded-DF	0.001362	—	—	—
	MLP-coded	0.001066	−0.000295	99/1	$< 10^{-4}$
	Mamba-coded	0.000967	−0.000395	100/0	$< 10^{-4}$

This is not a learned relay out-decoding Viterbi, and the mechanism says why. Viterbi is the maximum-likelihood *sequence* decoder for a known code on a known channel, so a learned decoder beating it at its own task would be a surprising claim requiring more than a BER table. It is not what happens here. Instrumenting the relay output directly (Table 5.6) shows coded-DF is the *better* decoder by a wide margin: it puts roughly $2.1\times$ fewer symbol errors on the air than the learned relay at both SNRs, exactly as the optimality result predicts. What separates them is what happens to those errors afterwards. Coded-

DF decodes and then *re-encodes*, so a frame it gets wrong leaves the relay as a different but perfectly *valid* codeword: the code’s redundancy has been spent and then regenerated around the error, and the destination’s decoder has no way to detect or repair it. The learned relay never re-encodes, so its (more numerous) mistakes remain isolated bad symbols inside an otherwise-valid codeword, and the destination still holds the redundancy needed to fix them. Only 16–26% of coded-DF’s relay errors are repaired downstream against 57–74% of the learned relay’s. That also explains the SNR-dependent split above: at 16 dB the $2.1\times$ raw-error penalty still outweighs the repair advantage for the MLP, and by 20 dB it no longer does.

Table 5.6: Where the errors go, 20 trials $\times$ 500 frames. “Relay symbol ER” counts symbols the relay put on the air that differ from the transmitted ones; “repaired” is the fraction of those that do not survive the destination’s decoder. The oracle relay forwards the clean codeword and so isolates the hop-2-only floor.

SNR	Relay	Relay symbol ER	Repaired	Final BER
16 dB	coded-DF	0.00507	15.5%	0.004282
	MLP-coded	0.01075	57.2%	0.004603
	oracle	—	—	0.002192
20 dB	coded-DF	0.00183	25.8%	0.001355
	MLP-coded	0.00390	74.4%	0.001001
	oracle	—	—	0.000655

5.4.1 Does a Stronger Code Help? A Constraint-Length Sweep

The $K=3$ code above is deliberately small (4 states, comparable to the smallest ISI channel studied in Chapter 6). Whether a stronger code – larger constraint length, larger free distance – simply does better is not obvious given the threshold behaviour just observed, so it is measured rather than assumed: the classical coded-DF baseline (Viterbi decode only; the learned relays are not retrained per K , each training run being too expensive to repeat six-fold) is swept over $K \in \{3, 5, 7\}$ – 4, 16, and 64 trellis states respectively, with $K=7$ the standard used in, e.g., the Voyager-era deep-space codes – on both canonical constellations.

Table 5.7: Coded-DF BER vs. constraint length K , canonical QPSK/Rayleigh, rate 1/2.

SNR (dB)	$K=3$ (4 states)	$K=5$ (16 states)	$K=7$ (64 states)
0	0.4212	0.4722	0.4892
4	0.2242	0.3009	0.3550
8	0.0633	0.0831	0.0884

SNR (dB)	$K=3$ (4 states)	$K=5$ (16 states)	$K=7$ (64 states)
12	0.0154	0.0192	0.0168
16	0.0042	0.0049	0.0039
20	0.0015	0.0013	0.0013

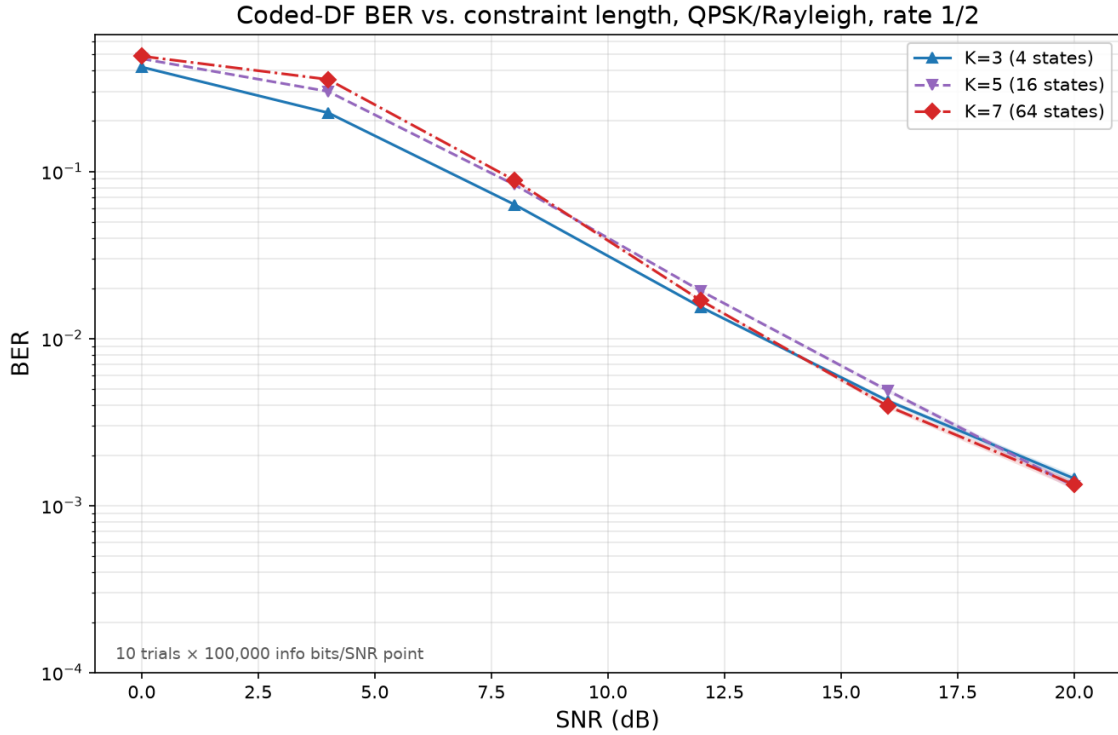


Figure 5.5: The data of Table 5.7 plotted: $K=5$ and $K=7$ sit visibly above $K=3$ at 0–8 dB and only draw level by 16–20 dB, never separating out cleanly ahead of it within this SNR range.

192 The answer is no, not monotonically, and not within the range measured here. The larger-
 193 K codes are *worse* than $K=3$ at low-to-mid SNR (0–8 dB) – a sharper error-propagation
 194 cliff for the bigger trellis, the same threshold effect visible in Table 5.4, more pronounced
 195 the stronger the code. $K=7$ pulls very slightly ahead of $K=3$ at 16 dB (0.0039 vs. 0.0042);
 196 by 20 dB all three constraint lengths are within trial-to-trial noise of each other. Within
 197 this frame length (200 information bits) and trial budget, the larger codes' greater free
 198 distance never clearly pays off at any SNR point measured, only costing more at low-
 199 to-mid SNR where the crossover matters most. This does not mean stronger codes are
 200 never worthwhile – a longer frame or a lower operating rate would very plausibly shift the
 201 picture – only that constraint length is not a free knob to turn up: like every other axis in
 202 this thesis, its benefit is a boundary to be located, not a default to be assumed.

5.4.2 Soft-Decision Relaying, and What the Liability Actually Is

The mechanism of Table 5.6 suggests an obvious repair. If the trouble with block-DF is that it commits to a codeword, replace the commitment with soft information: run BCJR (forward-backward MAP) at the relay instead of Viterbi and forward the posterior-mean symbol per trellis step, so a symbol the relay is unsure of is shrunk toward the origin rather than asserted. The same idea applies to the learned relay, whose softmax already carries a posterior it was throwing away: forwarding $\sum_k p_k s_k$ instead of $\arg \max_k p_k$ costs nothing, needs no retraining, and reuses the identical weights, so any difference isolates the read-out rule alone. Table 5.8 evaluates both against their hard counterparts and against an oracle relay that forwards the clean codeword, on paired trials.

Table 5.8: Soft- versus hard-decision relaying, canonical QPSK/Rayleigh, rate-1/2 $K=3$, 20 paired trials $\times$ 100,000 information bits. The MLP hard/soft pair shares one set of trained weights and differs only in the read-out. The oracle relay forwards the clean codeword, giving the hop-2-only floor.

SNR (dB)	hard block-DF	soft block-DF	MLP hard	MLP soft	oracle
0	0.4207	0.4142	0.4424	0.4327	0.3012
4	0.2223	0.2210	0.2923	0.2715	0.1278
8	0.0642	0.0642	0.0970	0.0849	0.0332
12	0.01544	0.01546	0.02097	0.01729	0.00762
16	0.004140	0.004141	0.004355	0.003485	0.002106
20	0.001376	0.002379	0.001069	0.000885	0.000673

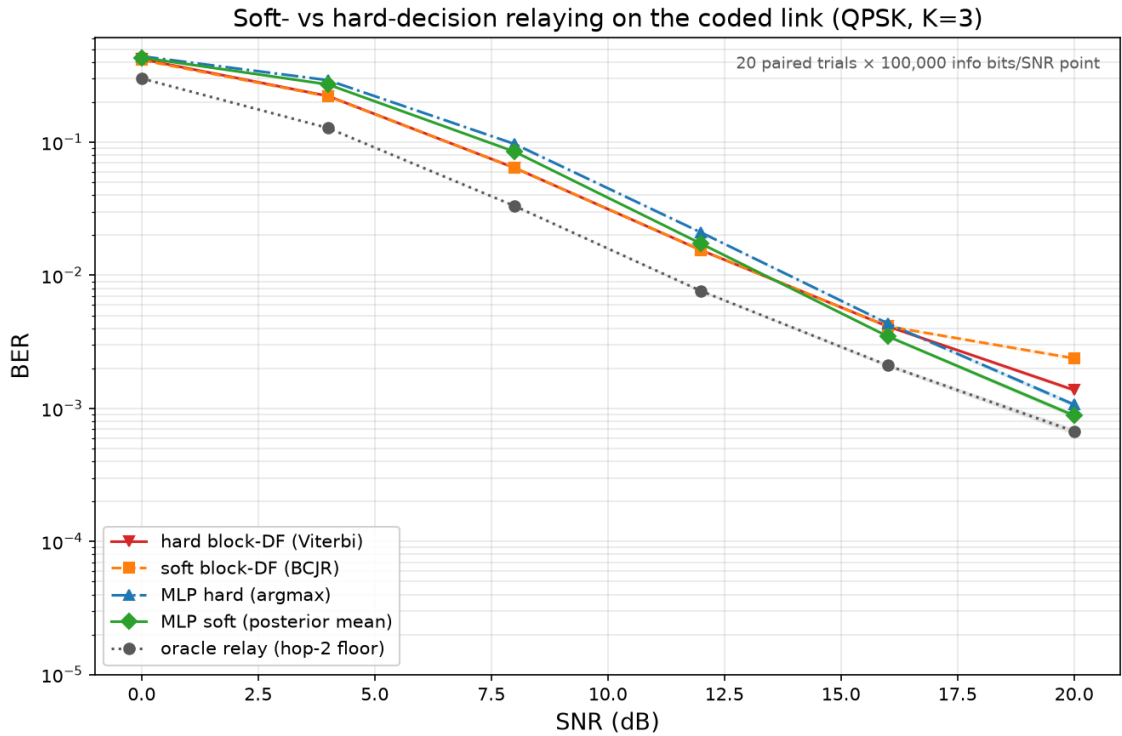


Figure 5.6: The data of Table 5.8. The soft learned relay (green) tracks the oracle floor most closely at high SNR; soft block-DF (orange) turns upward at 20 dB, an artefact of mis-calibration discussed in the text, not of soft forwarding itself.

213 **Soft read-out helps the learned relay everywhere.** At every SNR the posterior mean
 214 beats the argmax, on identical weights: 0.4327 against 0.4424 at 0 dB, 0.0849 against
 215 0.0970 at 8 dB, 0.000885 against 0.001069 at 20 dB. The gain is largest exactly where
 216 the mechanism predicts, since a relay that reports its uncertainty gives the destination's
 217 decoder more to work with than one that reports a confident guess. It is enough to change
 218 a verdict: at 16 dB the hard MLP significantly *loses* to block-DF (2 of 20 paired trials)
 219 while the soft MLP significantly *beats* it (20 of 20), and at 20 dB the soft relay comes
 220 within 0.00021 of the oracle floor that no relay of any kind can pass.

221 **Soft block-DF does not rescue block-DF, and that is the more interesting result.** The
 222 prediction going in was that BCJR would combine the code-aware cleanup of Viterbi with
 223 the reparability of a non-committing relay, and therefore beat both. It does not. It edges
 224 hard block-DF at 0–4 dB, ties it from 8–16 dB, and at 20 dB is clearly worse (0.002379
 225 against 0.001376, losing all 20 paired trials). The 20 dB degradation turns out to be a
 226 calibration artefact rather than a property of soft forwarding: the relay is given the *nominal*
 227 noise variance, but post-equalization Rayleigh noise has variance $1/(2\gamma|h|^2)$, so at high
 228 SNR, where the nominal value is small, BCJR becomes badly overconfident on deep-fade
 229 symbols. Viterbi is structurally immune, its squared-distance metric being invariant to
 230 any uniform scaling of the assumed variance. Deliberately inflating the assumed variance
 231 confirms this directly: at 20 dB a factor of two removes the entire gap (0.002394 $\rightarrow$

232 0.001367), and every larger factor up to $100\times$ holds there.

233 What that correction exposes is the substantive point. Even properly calibrated, soft
 234 block-DF only *ties* hard block-DF (0.001367 against 0.001367 at 20 dB; 0.004304 against
 235 0.004313 at 16 dB) – it never beats it. The reason is that at high SNR the BCJR posteriors
 236 saturate, so the posterior mean degenerates to the constellation point and soft block-DF
 237 reproduces Viterbi’s decisions; when a code-exploiting relay is wrong, it is confidently
 238 wrong whether or not it quantizes. The liability identified in Table 5.6 is therefore *not*
 239 hard quantization at the relay, as the soft-forwarding hypothesis assumed. It is consuming
 240 the code’s redundancy at the relay at all. Any relay that decodes the code – Viterbi or
 241 BCJR, hard or soft – has spent that redundancy by the time it transmits, and whatever it
 242 got wrong arrives at the destination unrepairable. The learned relay’s high-SNR advantage
 243 comes from the opposite discipline: it never decodes the code, only denoises symbol by
 244 symbol, so the full redundancy survives intact for the destination to spend. Soft read-out
 245 then adds a second, smaller gain on top by preserving the confidence that the destination
 246 can weigh.

247 This sharpens the thesis’s recurring claim rather than overturning it. A learned relay still
 248 does not out-decode a matched classical decoder: coded-DF remains far ahead through
 249 12 dB and, per Table 5.6, is the more accurate decoder at every SNR measured. What
 250 the high-SNR reversal shows is an architectural point about *where* in a two-hop chain the
 251 decoding should happen, and the answer that emerges – leave the code alone at the relay,
 252 denoise only, and let the destination decode once – is a system-design conclusion, not a
 253 statement about the relative power of Viterbi and neural networks.

254 5.4.3 Throughput and Latency: What the BER Tables Leave Out

255 Every table above compares relays at equal E_s/N_0 . That is the correct axis for asking
 256 which relay detects better, and the wrong one for asking which relay a link should use,
 257 because it silently ignores two costs that differ by orders of magnitude across these designs.

258 **Throughput: the coded comparison was not like-for-like.** A rate-1/2 code carries half
 259 the information per channel use. Concretely, the coded QPSK link spends 202 symbols
 260 to deliver 200 information bits (0.99 bits/symbol) where uncoded QPSK spends 100 (2.00
 261 bits/symbol). The “ $5.6\times$ better at 16 dB” quoted above is therefore bought with half the
 262 data rate, and a link designer given that trade would not read it as a $5.6\times$ improvement at
 263 all. The honest comparison holds spectral efficiency fixed, and the measurements already
 264 collected supply it: rate-1/2 16-QAM delivers $4 \times \frac{1}{2} = 1.98$ information bits per symbol,
 265 matching uncoded QPSK’s 2.00. Table 5.9 places those two side by side.

Table 5.9: Coded versus uncoded at equal spectral efficiency (≈ 2 information bits per channel symbol): uncoded QPSK against rate-1/2 $K=3$ 16-QAM, both on the canonical Rayleigh channel. Recomputed from the data of Table 5.4 and `results/coded_df_experiment.json`; no additional simulation.

SNR (dB)	uncoded QPSK	rate-1/2 16-QAM	Coding gain
0	0.3336	0.4687	$0.71\times$ (worse)
4	0.2213	0.4204	$0.53\times$ (worse)
8	0.1204	0.2722	$0.44\times$ (worse)
12	0.0558	0.0989	$0.56\times$ (worse)
16	0.0242	0.0269	$0.90\times$ (worse)
20	0.0099	0.0074	$1.35\times$ (better)

266 The result is a substantial qualification of Section 5.4. Held to equal throughput, the rate-
 267 1/2 code does not pay for itself anywhere below 20 dB, and where it finally does the gain is
 268 $1.35\times$, not the $5.6\text{--}7.6\times$ the equal- E_s/N_0 reading suggested. Coding is not free reliability:
 269 it is reliability bought with bandwidth, and on this channel, at this rate and frame length,
 270 the purchase is a poor one until the link is already operating well. This does not overturn
 271 the earlier section – coded block-DF genuinely is the stronger *detector*, and Remark 4.2’s
 272 caveat genuinely is real – but it does bound how much the caveat matters in practice.

273 **Latency: the block relays buffer three orders of magnitude more.** The relays also
 274 differ structurally in how much they must accumulate before emitting anything, which sets
 275 the delay the relay contributes independently of how fast its arithmetic is. Symbol-wise
 276 AF and DF emit per symbol. A windowed learned relay needs w symbols of look-ahead,
 277 ten here, and that figure is a property of the architecture, constant in frame length. Block-
 278 DF cannot emit until it has decoded an entire frame, 202 symbols; soft block-DF is worse
 279 still in kind rather than degree, since BCJR’s backward recursion runs from the end of the
 280 frame and so structurally forbids emitting early even in principle. Frame length is a free
 281 parameter, so this is not a fixed penalty but a scaling one: the block relays’ latency grows
 282 with the frame while the learned relay’s does not.

Table 5.10: Structural latency and measured compute cost. “Buffer” is the number of symbols the relay must accumulate before it can emit its first output. Compute is the median of 7 runs over 50,000 symbols after a discarded warm-up, single-threaded CPU; these timings are machine-dependent and are reported as informational.

Relay	Buffer (symbols)	$\mu\text{s/symbol}$	Relative to Viterbi
AF / symbol-wise DF	0	—	—
MLP hard (756p)	10	0.30	$50\times$ faster
MLP soft (756p)	10	0.22	$68\times$ faster

Relay	Buffer (symbols)	$\mu\text{s/symbol}$	Relative to Viterbi
hard block-DF (Viterbi)	202	15.07	$1\times$
soft block-DF (BCJR)	202	29.30	$1.9\times$ slower

The compute column points the same way as the latency column. The 756-parameter relay costs $0.30\ \mu\text{s}$ per symbol against Viterbi’s 15.07, a factor of 50, and the soft variant is no more expensive than the hard one because replacing an argmax with a posterior-mean inner product is the cheaper of the two operations. BCJR costs roughly twice Viterbi, as its forward, backward and marginalization passes imply. These are single-machine wall-clock figures for reference implementations, not architectural constants; the ratios are nonetheless large enough, and the structural buffering figures exact enough, that the ordering is not in question.

Taken together with Table 5.9, this reframes the coded study’s practical conclusion. Where the code pays for itself at all, it does so at 20 dB, for a $1.35\times$ gain at equal throughput, while asking for $20\times$ the buffering latency and $50\times$ the arithmetic of the learned relay that comes within a few ten-thousandths of it there. The learned relay’s case in this chapter, as in Chapter 6, rests on cost and constancy rather than on a lower error floor.

5.4.4 Adaptive Modulation and Coding: Choosing the Rate, Not Just the Relay

Every comparison above, including the equal-throughput one, holds the modulation-and-coding scheme (MCS) fixed and asks which relay achieves the lower BER. A real link does not hold the MCS fixed; it adapts. The question a link actually faces is not “which relay wins at rate R ” but “what is the highest *useful* rate this link can sustain at this SNR”, and answering it requires more than one code rate to choose from. The mother code of Section 5.4 is extended here with the standard punctured rates $2/3$ and $3/4$ (deleting a periodic subset of the rate- $1/2$ output, re-inserted as soft zeros at the decoder, so the same Viterbi decoder runs unmodified at every rate), and evaluated via bit-interleaved coded modulation across the full grid $\{\text{QPSK}, 16\text{-QAM}\} \times \{\text{uncoded}, 1/2, 2/3, 3/4\}$.

The objective is *goodput*, $G = R(1 - \text{FER})$, the information rate actually delivered once frames with any residual error are discarded, which is what any real protocol above the physical layer does. G and R are both measured per channel symbol transmitted on the active hop, matching the ergodic-capacity reference C introduced below; because the relay is half-duplex, expressing goodput instead as a fraction of the total two-slot cycle (source-to-relay plus relay-to-destination) would halve every figure in this section without changing which MCS or relay strategy is preferred, since that factor of $\frac{1}{2}$ applies identically to both. Maximizing G over the MCS grid at each SNR traces the link-adaptation envelope,

315 compared here for the two relay strategies the mechanism of Table 5.6 contrasts directly:
 316 *block-DF*, which decodes the code at the relay and so spends its redundancy before trans-
 317 mitting, against *denoise-only*, which hard-decides each bit and re-modulates without ever
 318 invoking the code, leaving the redundancy intact for the destination and requiring no train-
 319 ing and no code knowledge at the relay.

Table 5.11: Link-adaptation envelope: the MCS maximizing goodput at each SNR, block-DF versus denoise-only relaying, canonical Rayleigh channel, 10 trials $\times$ 200 frames $\times$ 200 information bits per point. C is the ergodic Rayleigh (Shannon) capacity at that SNR, $C = \log_2(e) e^{1/\gamma} E_1(1/\gamma)$ – an unconstrained-input upper bound no achievable goodput may exceed, included as a sanity ceiling rather than a target: the gap to it here is dominated by the finite constellation and coding loss, not by relay strategy. The capacity column is a *single-link* reference evaluated per active-hop channel symbol; it is not a capacity bound for the half-duplex two-hop relay system studied here, whose own bound would additionally account for the two-slot cycle and the relay constraint.

SNR (dB)	Best MCS (block-DF)	G (block-DF)	Best MCS (denoise)	G (denoise)	C (Shannon)
8	QPSK 1/2	0.0040	QPSK uncoded	0.0000	2.40
12	QPSK 1/2	0.1668	QPSK 1/2	0.1282	3.45
16	QPSK 1/2	0.5342	QPSK 1/2	0.5995	4.63
20	QPSK 3/4	0.9274	QPSK 2/3	0.9287	5.88
24	16-QAM 2/3	1.6079	16-QAM 2/3	1.5129	7.17
28	16-QAM 3/4	2.3185	16-QAM 3/4	2.2993	8.48
32	16-QAM uncoded	2.6800	16-QAM uncoded	2.6800	9.80
36	16-QAM uncoded	3.3840	16-QAM uncoded	3.3840	11.13
40	16-QAM uncoded	3.6980	16-QAM uncoded	3.6980	12.46

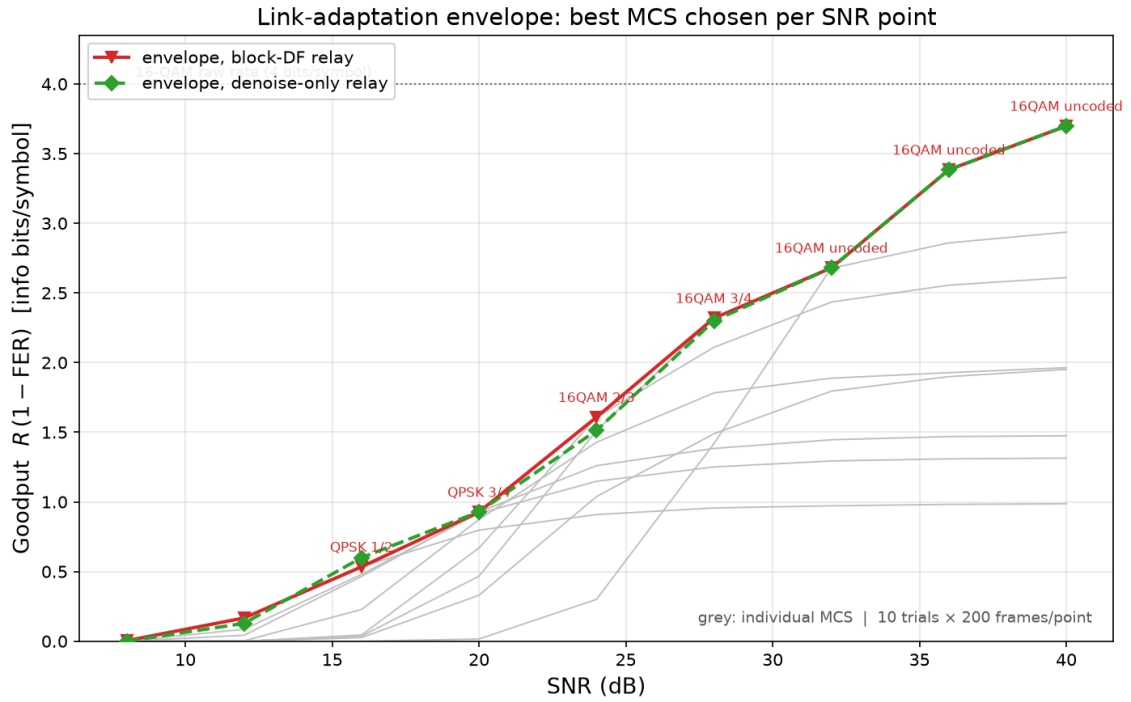


Figure 5.7: The envelope of Table 5.11 (colored) against every individual fixed MCS on the block-DF relay (grey), which is the visual case for adaptation: the envelope tracks the upper hull of the grey curves rather than any single one of them, switching MCS as SNR crosses each rate’s operating threshold. The two relay strategies’ envelopes are visually almost the same curve.

Two results follow, and the second qualifies the mechanism finding of Table 5.6 substantially.

The envelope is a real staircase, and its top step is “turn the code off”. Both relay strategies climb QPSK 1/2 $\rightarrow$ QPSK 3/4-or-2/3 $\rightarrow$ 16-QAM 2/3 $\rightarrow$ 16-QAM 3/4 $\rightarrow$ 16-QAM uncoded as SNR rises from 8 to 40 dB, each switch occurring once the next rate’s threshold has cleared. The optimum at high SNR is no code at all: once the channel is good enough, coding redundancy is pure overhead, and the highest-goodput choice is the raw 4 bits/symbol of 16-QAM with zero coding loss. This is the same lesson as the equal-throughput correction above, now reached by direct optimization over the grid rather than by hand-picking one comparison. Table 5.11’s C column confirms the envelope never approaches the information-theoretic ceiling: at 40 dB the achieved 3.70 sits at 30% of the 12.46-bit Shannon capacity there, a gap set by the fixed 16-QAM constellation (itself capped at 4 bits/symbol) and residual FER, not by anything a smarter relay strategy could close.

Once the rate adapts, which relay decodes the code barely matters. The two envelopes in Table 5.11 are close at every SNR and identical for SNR $\geq$ 32 dB, where both select 16-QAM uncoded and there is no code for the strategies to differ over. Where they do differ, the gap is small and does not consistently favor one side: block-DF leads at 8, 12,

24 and 28 dB, denoise-only leads at 16 and 20 dB, and the largest gap over the whole sweep is 0.095 information bits/symbol at 24 dB, under 6% of the envelope value there. Compare this to how much the *MCS* choice moves goodput at fixed relay strategy: at 24 dB the spread across the seven block-DF MCS options is 1.31, roughly $14\times$ the 0.095 gap between relay strategies; at 32 dB and above the MCS spread exceeds 1.7 while the relay-strategy gap is exactly zero. Rate adaptation, in other words, dominates the relay-decoding decision by more than an order of magnitude across most of this sweep.

This bounds the mechanism finding of Section 5.4.2 rather than reversing it. That the destination can repair a denoising relay's errors but not a re-encoding relay's remains true and remains the reason the two strategies differ at all – but once the system is allowed to pick its own operating rate, the room that difference has to matter shrinks to a few percent of the achievable goodput, because the adaptive link mostly avoids operating in the regime (a fixed rate pushed past its comfortable threshold) where the mechanism produces a large gap in the first place. The practical recommendation this chapter's coded study converges on is therefore rate adaptation first: get the MCS right for the channel, and only then does it become worth asking which relay strategy to run within it.

5.4.5 Capacity Under a Latency Constraint

Section 5.4.4 optimizes goodput with no bound on how long the relay may take to decide. A link with a deadline cannot make that assumption, and the question it faces instead – the achievable rate once blocklength (equivalently, latency) is fixed rather than left free – is a standard one in the coding-theory literature: the finite-blocklength capacity of Polyanskiy, Poor and Verdú [26], and the delay-limited capacity of Hanly and Tse [27] for fading channels specifically. Both frame it the same way this section does: constraining the number of channel uses (Polyanskiy et al.) or imposing a hard per-block deadline (Hanly–Tse) costs achievable rate relative to the unconstrained (Shannon, or ergodic) capacity, and by how much is exactly the object of study.

The two relay strategies differ in what they cost against such a deadline. The destination always buffers a full coded frame before it can decode, regardless of which relay handled the first hop – that cost is unavoidable for any coded MCS and is shared by both strategies. Block-DF adds a second full-frame buffer at the relay itself, since it must decode, re-encode and only then forward, so its round-trip latency is the frame length *twice*. The denoise-only relay adds only its window (Table 5.10's 10 symbols, independent of frame length or code rate), so its round-trip latency is the frame length once. Re-deriving the goodput envelope of Table 5.11 under a round-trip budget $L_{\max}$ – the best goodput each relay strategy can still deliver using only MCS options whose round-trip latency does not exceed $L_{\max}$ – therefore penalizes block-DF twice as hard per unit of code redundancy as it penalizes the denoise-only relay, using the same already-measured FER data as Table 5.11

375 and no new simulation.

Table 5.12: Link-adaptation envelope re-derived under a 150-symbol round-trip latency budget, same layout and data source as Table 5.11.

SNR (dB)	Best MCS (block-DF)	G (block-DF)	Best MCS (denoise)	G (denoise)
8	QPSK uncoded	0.0000	QPSK uncoded	0.0000
12	QPSK uncoded	0.0000	QPSK 3/4	0.0089
16	16-QAM 3/4	0.0370	QPSK 3/4	0.3719
20	16-QAM 3/4	0.4667	QPSK 3/4	0.9156
24	16-QAM 3/4	1.5096	16-QAM 2/3	1.5129
28	16-QAM 3/4	2.3185	16-QAM 3/4	2.2993
32	16-QAM uncoded	2.6800	16-QAM uncoded	2.6800
36	16-QAM uncoded	3.3840	16-QAM uncoded	3.3840
40	16-QAM uncoded	3.6980	16-QAM uncoded	3.6980

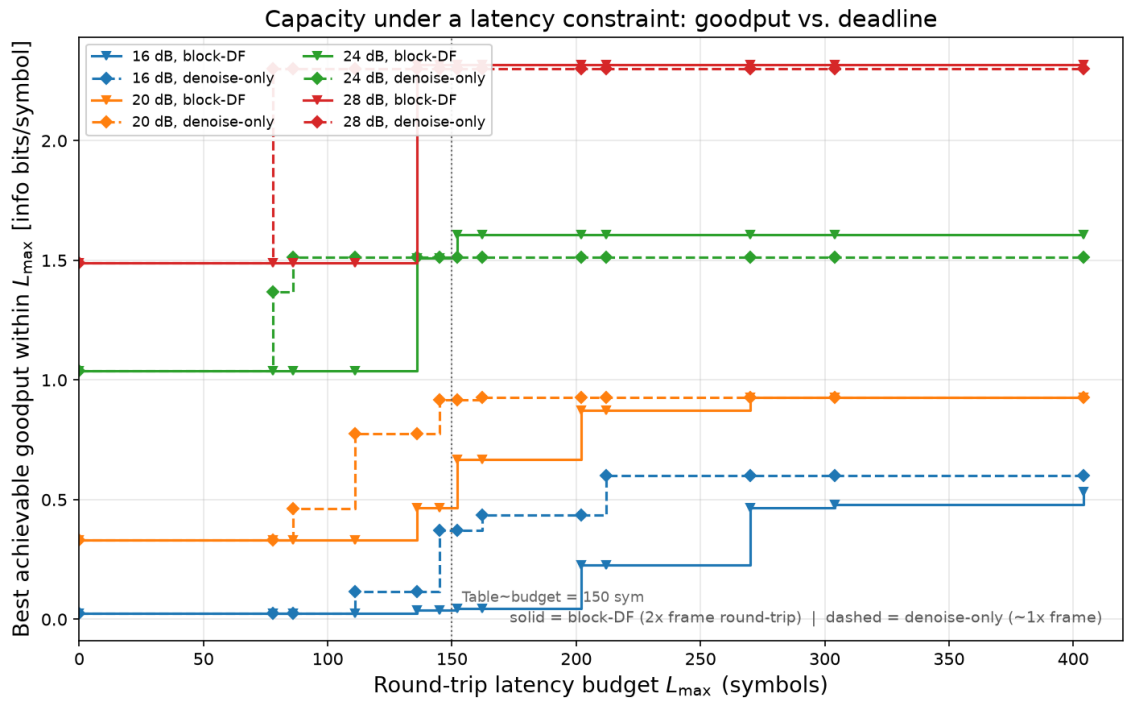


Figure 5.8: Best achievable goodput as a function of the round-trip latency budget $L_{\max}$, at the four SNRs where the two relay strategies differ (16, 20, 24, 28 dB). Each curve is a step function: goodput jumps each time $L_{\max}$ crosses an MCS's round-trip latency. The denoise-only relay's steps (dashed) occur at roughly half the $L_{\max}$ of the block-DF relay's (solid) for a comparable rate, since block-DF pays the frame cost twice.

376 **Under a tight budget, block-DF can be locked out of coding altogether.** At $L_{\max} = 150$
 377 symbols (Table 5.12), block-DF's cheapest coded option (16-QAM 3/4, 136 symbols
 378 round-trip) barely clears the budget, while its next-cheapest (16-QAM 2/3, 152 symbols)

does not – forcing it down from the unconstrained envelope’s choice at several SNRs. The denoise-only relay, whose round-trip cost for the same MCS is roughly half, still has QPSK 3/4 (145 symbols) and every 16-QAM rate available. The result is a gap the unconstrained comparison of Table 5.11 understates by an order of magnitude at 16 dB: $10.0\times$ (0.372 vs. 0.037 information bits/symbol) there, and $2.0\times$ at 20 dB, where the unconstrained envelope showed the two strategies within 6% of each other.

Tighten the budget further and the reversal spreads to SNRs where block-DF led unconstrained. At $L_{\max} = 100$ symbols, block-DF has *no* coded option left at any of these four SNRs – every one of its coded MCS choices exceeds the budget – so it is pinned to uncoded transmission throughout, while the denoise-only relay still reaches 16-QAM 2/3 or 3/4 at 20, 24 and 28 dB. At 24 and 28 dB, this is a direct reversal: Table 5.11’s unconstrained envelope has block-DF *ahead* at both points, but under the 100-symbol budget denoise-only leads by $1.46\times$ and $1.54\times$ respectively, because the MCS that gave block-DF its unconstrained edge is no longer reachable in time.

The mechanism finding of Section 5.4.2 was about which relay decodes better; Section 5.4.4 showed that once the rate is free to adapt, that question barely matters. This section adds the constraint a real deadline imposes and finds the opposite qualitative result: under a latency budget, which relay strategy is used matters again, and matters through an architectural fact – how many times the frame must be fully buffered – rather than through detection accuracy. A link with headroom to pick its own rate should do so per Section 5.4.4; a link additionally bound by a hard deadline should prefer the relay strategy that does not double-buffer, independent of which one decodes more accurately.

It is important to emphasize that all conclusions of this chapter are specific to the canonical memoryless channel model. Whether those conclusions remain valid when the underlying channel assumptions are violated is the subject of Chapter 6.

Chapter 6

Learned Relaying under Unknown and Mismatched Channels

This is the central chapter of the thesis. Everything before it is either calibration or a confirmatory result: on a matched, known channel a minimal learned relay does *not* improve on classical decode-and-forward (H2), and raising the constellation order does not change that. Here the learned relay wins, and wins decisively.

The condition under which it wins is precise, and stating it is as much a part of the claim as the win itself. **When the channel's structure is unknown to the receiver chain, or lies outside the model class the classical relay assumes, a fixed 169-parameter learned relay restores reliable relaying where memoryless classical relays fail outright.** Under three-tap intersymbol interference (S1, Table 6.3) the classical relays do not merely degrade: they saturate near the analytic 0.25 floor and then get *worse* as the SNR rises, symbol-wise DF bottoming out at 0.1839 at 8 dB and climbing to 0.2462 at 20 dB as hard decisions lock the interference in. No amount of transmit power recovers the link. The learned relay, whose length-11 window spans the channel memory, reaches 0.0123 at 8 dB and falls below 5×10^{-5} from 16 dB upward. What separates the two is the function class and not the information available: this filter is fixed, and the relay is trained on it, so it is as well informed about the channel as the genie-CSI Viterbi detector it is measured against. A memoryless slicer cannot invert a channel with memory at any transmit power, however much it is told; a windowed one can. Whether a learned relay also copes with a channel it has *not* seen is a separate question, answered at layers 3 and 4 below, where the channel is redrawn every block. That is not a marginal gain over a classical baseline; it is the difference between a working link and a broken one, and it is the thesis's answer to whether learning can defeat classical processing in relaying.

The boundary of the claim is equally sharp, and the chapter is written to establish it rather than to avoid it. The learned relay does not beat a *matched* classical receiver: given a

channel estimate, pilot-aided Viterbi maximum-likelihood sequence estimation remains ahead of it by 1–1.5 dB. The learned relay’s advantage is therefore not superior detection but the absence of a precondition. Where the channel is fixed and known to both, it is simply the cheaper detector, constant and parallelizable per symbol against the trellis’s M^{L-1} ; where the channel must be identified and cannot be, it is the one that still functions, needing neither a per-block estimate nor the channel family. Where classical processing can identify the channel it should be used; where it cannot, the learned relay is what still works.

Rather than a single comparison, the chapter is a systematic map, along every axis relevant to practical deployment, of *when* the learned relay adds value over classical processing, and, equally, when it does not.

The argument as a ladder of assumptions. The thesis as a whole is organised as a monotone sequence in which each layer removes exactly one assumption from the layer above, and the classical–learned comparison is repeated unchanged at every layer. Table 6.1 is that sequence with its outcomes. Reading it downward is the thesis’s argument in one page: classical processing is not beaten where it is well posed, and is beaten precisely where its preconditions fail.

Table 6.1: The four layers of the argument. Each removes one assumption above; the relay comparison is otherwise identical. BER figures are the representing points cited in the sections named.

Layer	Assumptions	Strongest
1	Memoryless, known channel, perfect CSI (Chapter 5)	Symbol-parameters

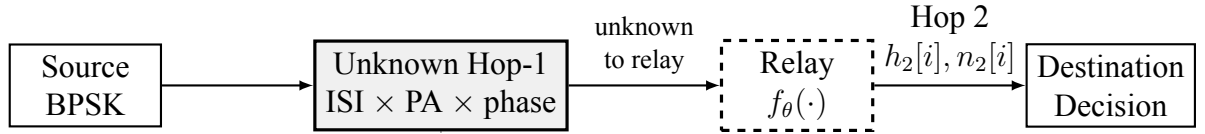
Layer	Assumptions	Strongest
2	+ channel memory (3-tap ISI), CSI still perfect (Section 6.1)	Viterbi M taps
3	+ CSI uncertainty: a finite pilot budget (Section 6.1.4)	Pilot-aide then MLS
4	+ unknown channel family and statistics, redrawn per block (Sections 6.1.3, 6.1.5)	CMA blind

⁴⁵ Two features of the ladder are worth naming before the evidence is presented. First, the op-
⁴⁶ ponent changes character at layer 2: layer 1 compares relay *functions* (AF and DF against
⁴⁷ learned relays), whereas from layer 2 onward the strongest classical option is a *receiver*,
⁴⁸ Viterbi MLSE, which is a considerably heavier and better-informed baseline. This is de-
⁴⁹ liberate, since comparing against a weaker classical option at the layers where the learned

50 relay wins would make the result worthless. Second, what degrades down the ladder is
 51 not the classical algorithms themselves but their *preconditions*: MLSE remains optimal at
 52 every layer for the channel it is given, and it loses ground only because the channel it is
 53 given grows less like the true one. The learned relay never becomes a better detector; it
 54 becomes the only one whose requirements are still met.

55 The investigation is deliberately adversarial to the learned relay. Each classical baseline is
 56 the strongest available for the impairment in question: symbol-wise decode-and-forward
 57 for the memoryless cases, conventional differential detection for unknown phase, and
 58 pilot-aided Viterbi maximum-likelihood sequence estimation, the optimal sequence detec-
 59 tor, wherever a channel estimate can be formed. The learned relay is granted no per-block
 60 channel estimate at any layer, and it never adapts online. How much it knows in advance
 61 differs by layer, and the distinction matters when reading the results. At layer 2 the ISI fil-
 62 ter is fixed, so training on it leaves the relay as well informed as the genie-CSI detector it is
 63 compared with, and what that layer isolates is the function class rather than channel knowl-
 64 edge. At layers 3 and 4 the channel is redrawn every block and the relay meets realizations
 65 it has not seen, which is where the family-agnostic claim is actually earned. That claim is
 66 scoped precisely: “family-agnostic” means the relay needs no per-block identification of
 67 *which* realization is active, having amortized the family’s parameter distribution into its
 68 weights at training time; it is trained on, and evaluated within, the same parametric chan-
 69 nel family (e.g. the 3-tap FIR structure) throughout, and generalization to a structurally
 70 different, untrained channel family is not tested here. The chapter follows the ladder in or-
 71 der. It opens with a necessary control (flat unknown channels, where nothing should fail),
 72 then takes layer 2, the impairment that does break classical relays (unknown memory),
 73 bounding it against the optimal matched receiver (Viterbi with exact taps) and against a
 74 composite multi-impairment cascade. Layer 3 relaxes the channel estimate to a finite pi-
 75 lot budget and locates the crossover in pilot count and block length; layer 4 removes the
 76 posterior altogether and compares against genuinely blind classical equalization. A final
 77 section prices the whole comparison in computational complexity. The result is a precise
 78 delineation (the learned relay never beats a matched classical receiver, but occupies a well-
 79 defined niche of family-agnostic, identification-free, constant-complexity mitigation) that
 80 constitutes the thesis’s main claim beyond the confirmatory core.

81 **Relationship to the canonical model.** The canonical setup of this thesis (Chapter 5)
 82 assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI,
 83 and uncoded transmission. The present chapter deliberately departs from those assump-
 84 tions by introducing impairments that are not part of the canonical model, including chan-
 85 nel memory (3-tap intersymbol interference), amplifier nonlinearity, unknown phase, and
 86 channel-model mismatch. Consequently, sequence-detection and equalization techniques
 87 such as Viterbi MLSE, pilot-aided channel estimation, and blind equalization become rel-



Departure from the canonical model:
 channel *memory* (3-tap ISI), nonlinearity, unknown phase.
 Noise remains white; only the *channel* violates memorylessness.

Figure 6.1: Unknown/mismatched-channel study (Chapter 6), a deliberate departure from the canonical benchmark of Figure 1.1. The Hop-1 channel is replaced by an unknown impairment (3-tap ISI, amplifier nonlinearity, and unknown phase) that is not disclosed to any relay. The noise remains white; only the channel violates the memoryless assumption, which is why sequence detection (Viterbi MLSE) becomes the relevant classical benchmark here and nowhere else in the thesis.

evant only in this chapter. The goal is not to replace the canonical study, but to evaluate whether the conclusions established under matched memoryless conditions remain valid when the underlying channel model no longer represents the true channel.

6.1 Layer 2 — Channel Memory with Perfect CSI: Classical Failure and Learned Mitigation

Goal: Test whether the learned relay’s value changes qualitatively when the channel violates the assumptions under which AF and DF are (near-)optimal: do the *memoryless* classical relays fail on an unknown channel while the minimal 169-parameter MLP mitigates by learning it from data (H6)?

Unlike the canonical experiments, where each transmitted symbol can be processed independently, the 3-tap ISI channel couples adjacent symbols in time. The sufficient statistic for detection is therefore no longer a single received sample, making sequence detection the relevant classical benchmark. To make the classical side as strong as possible, the study benchmarks against the optimal sequence detector: Viterbi MLSE applied to the 3-tap FIR/ISI channel [28] with genie CSI, and a practical variant with a 200-pilot least-squares estimate.

MLSE is introduced here solely because the channel now contains memory. In the canonical memoryless setting of Chapter 5, per-symbol detection is sufficient and sequence detection provides no additional benefit. The need for MLSE therefore arises from the deliberate introduction of ISI in this chapter rather than from the system model studied elsewhere in the thesis.

Trials.

- 110 • **Topology:** SISO (both hops)
- 111 • **Modulation:** BPSK
- 112 • **Hop-1 channel (unknown to all relays):**
 - 113 – 3-tap ISI, $h = [1.0, 0.7, 0.5]/\|h\|$
 - 114 – Control: canonical Rayleigh channel
 - 115 – Flat unknown channels (block phase, gain, per-branch asymmetry; Section 6.1.1)
 - 116 – Composite cascade of ISI, amplifier nonlinearity, and unknown phase (Section 6.1.3)
 - 117
- 118 • **Hop-2 channel:** AWGN or canonical Rayleigh, equal SNR per hop
- 119 • **Relays:**
 - 120 – AF
 - 121 – Symbol-wise DF
 - 122 – MLP (window 11, one hidden layer of 13 tanh units, 169 parameters; trained
 - 123 at $\text{SNR} \in \{5, 10, 15\}$ dB using the canonical training protocol)
 - 124 – For ISI setups only: Viterbi MLSE applied to the 3-tap FIR/ISI channel, eval-
 - 125 uated with either genie CSI or a 3-tap least-squares channel estimate obtained
 - 126 from 200 pilot symbols
- 127 • **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, SNR range 0–20 dB

128 The ISI channel admits a sharp analytic prediction: the effective slicer amplitude $h_0 \pm h_1 \pm$
 129 $h_2 \in \{1.67, 0.91, 0.61, -0.15\}$ (normalized) has one of four equiprobable patterns that
 130 *flip the symbol sign*, so symbol-wise DF has an irreducible BER floor of 0.25 (approached
 131 from below as SNR grows); AF inherits it. The floor is a property of the *memoryless relay*
 132 *class*, not the channel: Viterbi MLSE over the 4-state trellis has no floor when the channel
 133 is known.

134 6.1.1 Control: Flat (Memoryless) Unknown Channels

135 Before attributing any classical failure to “unknownness,” it must be separated from *mem-*
 136 *ory*. This subsection isolates unknown channels that are *flat* (memoryless): the receiver
 137 chain does not know the channel parameters, but the channel introduces no intersymbol
 138 coupling. Three physically motivated cases are tested, each with the parameter redrawn
 139 per block and unknown to all relays: (F1) an unknown carrier phase $\theta \sim \mathcal{U}[0, 2\pi)$ (oscilla-
 140 tor/synchronization offset), with a DBPSK source and conventional differential detection
 141 as the classical relay; (F2) an unknown positive gain $g \sim \mathcal{U}[0.3, 2.0]$ (path loss / AGC
 142 error), coherent BPSK; and (F3) an unknown per-branch amplitude asymmetry $a_+ \neq a_-$

(per-symbol saturating-PA gain), coherent BPSK. The 169-parameter MLP is trained identically to before.

These channels remain memoryless and therefore stay within the scope of the canonical model despite the parameter uncertainty. Consequently, they serve as a control that separates the effect of parameter uncertainty from the effect of channel memory: any performance change here arises from model uncertainty, not from memory.

Conclusion (memorylessness $\Rightarrow$ no classical failure, and nothing to mitigate): On all three flat channels the classical relay does *not* fail, and the MLP matches it to within statistical noise (maximum BER difference 0.0036 across every case and SNR). The reasons are structural: for F1, unknown phase is exactly what differential detection absorbs; for F2, $\text{sign}(y)$ is *scale-invariant*, so unknown positive gain is irrelevant by construction; for F3, the fixed threshold is only mildly suboptimal because the asymmetry preserves the noiseless symbol’s sign. The classical assumption, memorylessness, still holds in each case, so there is nothing for a learned relay to remove. This is the necessary control for the ISI result: the learned relay’s advantage is triggered not by parameter uncertainty but by *unmodeled memory* (or amplitude nonlinearity), which is what breaks the memoryless relay class.

Table 6.2: Flat unknown channels: classical relay vs. MLP-169 BER at 8/12/16/20 dB (mean, 10 trials). The maximum DF–MLP gap over all cases and SNRs is 0.0036, within statistical noise. Unknown parameters alone do not defeat classical relays; memory does.

Flat channel	Relay	8 dB	12 dB	16/20 dB
Unknown phase (DBPSK)	DF (diff.)	0.0649	0.0292	0.0119 / 0.0048
	MLP-169	0.0657	0.0288	0.0122 / 0.0048
Unknown gain	DF	0.0681	0.0330	0.0131 / 0.0050
	MLP-169	0.0682	0.0333	0.0132 / 0.0050
Branch asymmetry	DF	0.0760	0.0299	0.0125 / 0.0049
	MLP-169	0.0745	0.0300	0.0122 / 0.0052

Conclusion (H6: confirmed as stated, with two boundaries): Against the flat-channel control of Section 6.1.1, where unknown parameters alone caused no classical failure, the memory case is qualitatively different. On the unknown ISI channel, both memoryless classical relays fail: DF’s BER is *non-monotonic*, bottoming out at 0.184 around 8 dB and then rising toward the analytic 0.25 floor, reaching 0.246 at 20 dB (more transmit power makes DF strictly worse), and AF behaves identically. The 169-parameter MLP, whose window spans the 3-tap memory, learns a joint equalizer–detector and restores reliable relaying: BER falls monotonically below 5×10^{-5} at 16 dB (AWGN second hop), and to 2.5×10^{-3} at 20 dB (Rayleigh second hop). The Viterbi benchmarks set the first boundary

honestly: classical theory *does* solve this channel once its family is known, and genie-CSI MLSE leads the MLP by 1–1.5 dB across the clearly-separated mid-range (roughly 2–10 dB); a 200-pilot LS estimate recovers genie performance almost exactly there too. At the two extremes the ordering is not clean: near the 0 dB floor the two are within noise of each other (both near the analytic 0.25 ceiling), and with the Rayleigh second hop the gap closes again from 12 dB upward, the two tracking within 0.0001–0.0002 of each other through 20 dB (e.g. 0.0025 vs. 0.0024 there) – differences comparable to the reported standard errors rather than a resolved ordering. This does not contradict Viterbi’s sequence-ML optimality; it means the *margin* by which it holds is SNR-dependent and vanishes at both ends of this particular sweep, not that the MLP overtakes it. Two things follow, and neither is about channel knowledge, since the filter here is fixed and the relay is trained on it exactly as Viterbi is given it. The first is that the *function class* decides the outcome: a memoryless slicer cannot invert a channel with memory at any transmit power, which is why DF worsens with SNR, while a windowed relay recovers the link. The second is a complexity result: with the same channel information, 169 parameters land within 1–1.5 dB of the optimal sequence detector at a per-symbol cost that grows only linearly in the window spanning that memory, against the trellis’s exponential M^{L-1} . The stronger property, mitigating channels whose *family* is unknown or outside linear MLSE’s model class, is not demonstrated by this experiment and is established later, by the composite cascade of Section 6.1.3 and the posterior-free study of Section 6.1.5. The control confirms the second boundary (H2): on the canonical Rayleigh channel the MLP only *matches* DF. The learned relay’s distinctive value is near-MLSE reliability with no channel-family knowledge, no identification stage, and complexity that grows only linearly, rather than exponentially, in channel memory.

Table 6.3: Unknown-channel BER (mean over 10 trials). The ISI setups exhibit memoryless-relay error floors (≈ 0.25); the 169-parameter MLP restores reliable relaying within ≈ 1 –1.5 dB of genie-CSI Viterbi MLSE, and matches it end-to-end when the Rayleigh second hop dominates. The control shows no MLP advantage on the canonical channel, consistent with H2.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
Unknown ISI $\rightarrow$ AWGN	AF	0.1919	0.1785	0.1956	0.2233
	DF	0.1839	0.2010	0.2279	0.2462
	MLP (169p)	0.0123	0.0002	<5e-5	<5e-5
	Viterbi (genie CSI)	0.0072	<5e-5	<5e-5	<5e-5
	Viterbi (200-pilot LS)	0.0074	<5e-5	<5e-5	<5e-5
Unknown ISI $\rightarrow$ Rayleigh	AF	0.2047	0.1916	0.2030	0.2231
	DF	0.2031	0.2098	0.2316	0.2471
	MLP (169p)	0.0412	0.0150	0.0061	0.0025

Setup	Relay	8 dB	12 dB	16 dB	20 dB
	Viterbi (genie CSI)	0.0364	0.0151	0.0061	0.0024
	Viterbi (200-pilot LS)	0.0362	0.0151	0.0062	0.0024
Control: canonical Rayleigh	AF	0.1050	0.0559	0.0281	0.0136
	DF	0.0687	0.0299	0.0124	0.0050
	MLP (169p)	0.0691	0.0307	0.0133	0.0058

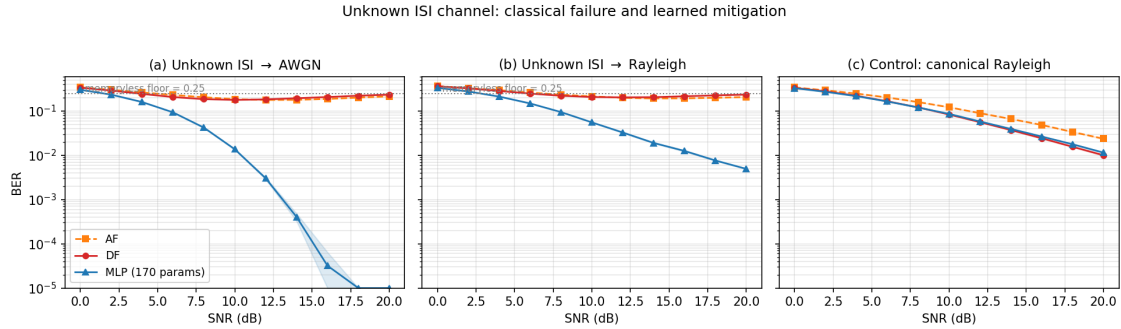


Figure 6.2: Unknown-channel relaying, BER vs. SNR with 95% CI bands. (a)–(b) On the unknown ISI channel, the memoryless AF/DF relays are pinned near the analytic 0.25 floor (DF non-monotonic in SNR); the 169-parameter MLP restores monotonically decreasing BER within ≈ 1 – 1.5 dB of genie-CSI Viterbi MLSE, which a 200-pilot LS estimate matches almost exactly. With the Rayleigh second hop (b), MLSE and the MLP are statistically indistinguishable end-to-end. (c) Control: on the canonical Rayleigh channel the MLP only matches DF, consistent with H2.

6.1.2 Does the Result Generalize to QPSK?

The unknown-ISI study above uses BPSK throughout. The canonical setup of Chapter 5 is QPSK, so this subsection checks that the failure mode is a property of the channel rather than of the constellation it was first observed on. The 0.25 floor derived in Section 6.1 is a property of the *binary* slicer: it counts the $2^3 = 8$ equiprobable tap-sign patterns that flip a ± 1 decision. That argument does not extend to a four-point per-axis alphabet, so it is not obvious a priori whether either the elevated error floor or the ordering between the learned relay and genie-CSI Viterbi MLSE survives a move to a higher-order modulation. This subsection repeats the unknown-ISI study with Gray-coded QPSK in place of BPSK, keeping every other element fixed: the same 3-tap ISI channel $h = [1.0, 0.7, 0.5]$ (normalized) on Hop 1, AWGN or canonical Rayleigh on Hop 2 at the same nominal per-hop SNR (identical convention, $\gamma = 10^{\text{SNR}_{\text{dB}}/10}$, memory-bank/techContext.md), and the same relay set generalized to the complex alphabet: AF, symbol-wise DF (hard per-axis decision), a 193-parameter 4-class MLP-QPSK classifier (window 11, softmax cross-entropy, trained at $\text{SNR} \in \{5, 10, 15\}$ dB), and genie-CSI Viterbi MLSE generalized to the complex QPSK trellis ($4^2 = 16$ states, complex branch metrics). Evaluation uses 10 trials $\times$ 100,000 bits

per SNR point over the same 0–20 dB range. No closed-form floor is derived for QPSK, so the curves below are reported as measured, not fit to a predicted asymptote. Because the QPSK relays are applied jointly to the complex signal rather than decomposed into two independent BPSK problems, and the per-relay power normalization is computed on total (I+Q) power, the resulting AF/DF operating points are not required to, and empirically do not, numerically match the BPSK figures of Table 6.3 at the same SNR label; no such match is claimed.

Table 6.4: QPSK unknown-channel BER (mean over 10 trials). AF and DF plateau at an elevated, non-monotonic BER across the SNR range tabulated; the 193-parameter MLP-QPSK restores monotonically decreasing BER and, unlike the BPSK case, ends up *below* genie-CSI Viterbi MLSE from about 2 dB upward.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
QPSK: ISI $\rightarrow$ AWGN	AF	0.2437	0.2115	0.2024	0.2098
	DF	0.2193	0.2024	0.2074	0.2211
	MLP-QPSK (193p)	0.1292	0.0827	0.0601	0.0508
	Viterbi (genie CSI)	0.1307	0.0852	0.0670	0.0618
QPSK: ISI $\rightarrow$ Rayleigh	AF	0.2715	0.2320	0.2137	0.2116
	DF	0.2528	0.2194	0.2145	0.2239
	MLP-QPSK (193p)	0.1737	0.1066	0.0707	0.0553
	Viterbi (genie CSI)	0.1749	0.1089	0.0774	0.0662

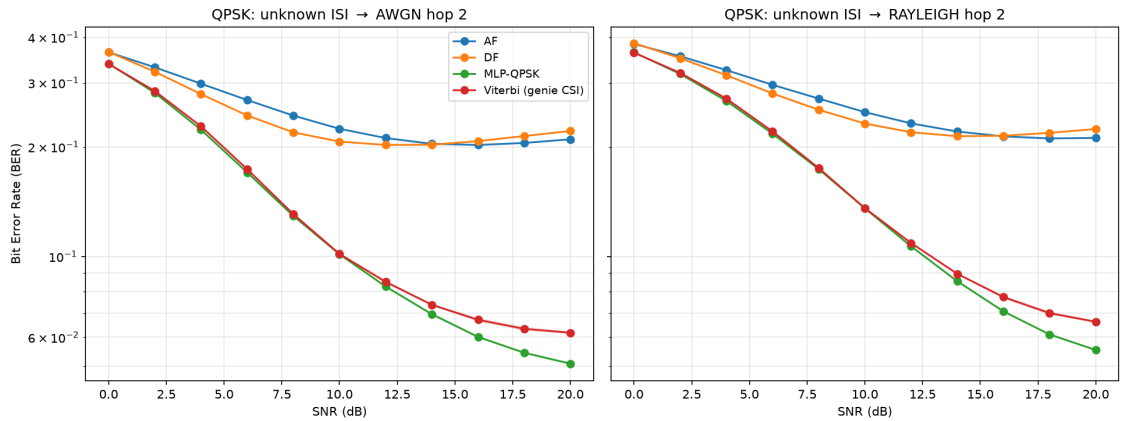


Figure 6.3: QPSK unknown-channel relaying, BER vs. SNR (10 trials, 100,000 bits/point). As with BPSK, AF and DF plateau far above the MLP and Viterbi curves. Unlike BPSK, the MLP-QPSK curve separates *below* genie-CSI Viterbi MLSE beyond about 2 dB, and the gap widens with SNR.

Conclusion (the ISI failure mode generalizes; the BPSK ordering does not): The qualitative pattern of Section 6.1 survives the move to QPSK: AF and DF plateau at an elevated, non-monotonic BER (AF ranges 0.2024–0.2715 and DF ranges 0.2024–0.2528 across the

219 four tabulated points, broadly similar in magnitude to BPSK’s memoryless-relay floor of
 220 0.25 though no equivalent closed form is derived here), while the MLP restores monotonically
 221 decreasing BER, confirming that unmodeled channel memory, not modulation order,
 222 is what breaks the memoryless relay class (H6). What does *not* generalize is the BPSK
 223 ordering between the learned relay and the optimal classical benchmark. In the BPSK
 224 study, genie-CSI Viterbi MLSE beat the MLP by 1–1.5 dB throughout; here, the two are
 225 statistically indistinguishable at 0 dB and MLP-QPSK is at or below Viterbi at every SNR
 226 from 2 dB upward, with the gap widening to 0.0508 vs. 0.0618 (AWGN) and 0.0553 vs.
 227 0.0662 (Rayleigh) at 20 dB. This is not a violation of MLSE optimality: Viterbi MLSE
 228 minimizes the *sequence* error probability under the assumed channel model, whereas the
 229 bit error rate is what a bit-wise MAP (BCJR/APP) detector minimizes. Those are distinct
 230 criteria on any channel with memory — for BPSK no less than for QPSK, since the dis-
 231 tinction is between sequence-level and bit-level objectives rather than a consequence of
 232 how many bits a trellis branch carries — so sequence-optimality never by itself guaran-
 233 tees BER-optimality at either modulation order. The 193-parameter MLP-QPSK is instead
 234 trained directly on a per-symbol softmax cross-entropy objective, which targets the per-
 235 symbol posterior rather than the sequence. That criterion mismatch may contribute to the
 236 observed ordering, but it was not evaluated in this work: separating it from the other can-
 237 didate explanations requires the BER-optimal BCJR/APP benchmark identified as future
 238 work in Section 7.5, together with a decomposition of the Viterbi output into single- ver-
 239 sus double-bit symbol errors. Until those are run the mechanism is a conjecture, not an
 240 established explanation, and the reversal is reported here as a measurement rather than as
 241 something the thesis accounts for. Either way, the practical conclusion sharpens rather
 242 than weakens H6 for higher-order modulation: on this unknown-memory channel, the
 243 fixed-complexity learned relay is not merely a near-optimal option at constant cost, as in
 244 the BPSK case, but the best-performing option measured here among either family, at the
 245 same constant per-symbol cost quantified in Section 6.1.6.

246 6.1.3 A Composite (Mixed) Channel: Mismatch with a Full Posterior

247 The preceding studies vary one impairment at a time. Real links stack several. This final
 248 study cascades three of them into a single unknown channel that *no* classical relay is jointly
 249 matched to: a DBPSK stream passes through 3-tap ISI, then a saturating power-amplifier
 250 nonlinearity, then an unknown block phase, before additive noise. Each classical base-
 251 line is the strongest available for *one* constituent impairment, so the comparison is honest
 252 rather than a straw man: conventional differential detection (matched to the phase), and
 253 pilot-LS Viterbi MLSE followed by differential decoding (matched to the ISI-plus-phase,
 254 but blind to the amplifier nonlinearity). The 169-parameter MLP and a 1,153-parameter
 255 MLP receive raw I/Q windows and no knowledge of any constituent. This composite chan-

nel departs from the canonical model in two ways: it introduces explicit channel memory through ISI and nonlinear distortion through the amplifier. The resulting benchmark therefore tests robustness to both memory and model mismatch simultaneously. **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, the same budget as the main unknown-ISI study (Section 6.1); 95% CIs use the $1.96 \sigma / \sqrt{n}$ convention throughout. The hop-1 impairment here is fixed rather than redrawn per block, so bits per trial and trials both reduce the reported uncertainty.

Conclusion: The composite defeats the naive classical relays outright, both AF and conventional differential detection saturate near the 0.25 memoryless-relay floor, because the ISI destroys the differential statistic that the phase-matched detector relies on. The 169-parameter MLP again restores reliable relaying, reaching 2.5×10^{-3} at 20 dB from raw I/Q with no impairment model. This reproduces the H6 pattern on a realistically mixed channel: memoryless-and-matched classical processing fails, and a minimal learned relay recovers. Two familiar boundaries also reappear. First, the strongest *constructed* classical baseline, pilot-LS Viterbi plus differential decoding, does not fail: by modelling the ISI and phase (though not the amplifier) it stays about 2 dB ahead of the MLP at low-to-mid SNR (0.067 vs. 0.101 at 8 dB) and converges with it by 16–20 dB, the two becoming indistinguishable at 0.0025 each at 20 dB. The learned relay’s value here is *model-agnosticism* rather than superiority over a matched classical receiver: it reaches near-Viterbi reliability with no analytic model of any of the three cascaded effects, whereas estimate-then-MLSE must assume a linear FIR channel and therefore cannot represent the amplifier at all. This cascade is itself a fixed channel that the relay is trained on, so it speaks to the model class the two methods can express, not to generalization across channels; that is the subject of Section 6.1.5, where a fresh channel is drawn for every block. Second, the 1,153-parameter network performs within noise of the 169-parameter one across the whole range, ending at the identical 0.0025 at 20 dB, with the small mid-SNR differences in both directions (0.0487 vs. 0.0531 at 10 dB for the larger network, 0.1020 vs. 0.1014 at 8 dB for the smaller) falling inside the confidence intervals. Nearly seven times the parameters therefore buys no measurable accuracy on the hardest channel studied, reconfirming the U-shaped complexity finding (H3). The composite thus consolidates the chapter’s thesis: a single minimal learned relay handles an arbitrary stack of representable impairments without being told which are present, approaching but not exceeding the best impairment-matched classical receiver.

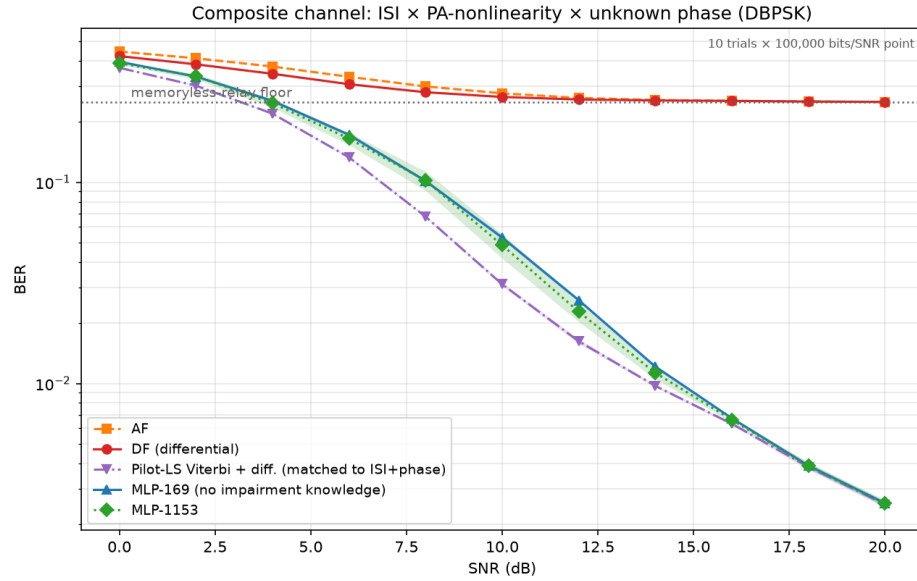


Figure 6.4: Composite channel (ISI $\times$ PA-nonlinearity $\times$ unknown phase, DBPSK; mean over 10 trials, 100,000 bits/point, 95% CI bands). Naive classical relays (AF, differential detection) saturate at the memoryless-relay floor; the 169-parameter MLP restores reliable relaying from raw I/Q with no impairment model, tracking about 2 dB behind the strongest constructed classical baseline (pilot-LS Viterbi + differential) and converging with it by 20 dB. The 1,153-parameter MLP adds nothing (H3).

6.1.4 Layer 3 — CSI Uncertainty: The Pilot-Budget Crossover

This is layer 3 of the ladder in Table 6.1: the perfect-CSI assumption is dropped, and nothing else changes. Every classical result so far has been handed either exact channel taps or a 200-pilot estimate good enough to match them (Section 6.1.3), which is what a full posterior buys. Here the posterior is made *partial*: the pilot budget is finite, and the classical receiver must work from a correspondingly noisy estimate. Sweeping the pilot count on the composite channel at a fixed 10 dB operating point locates where the pilot-aided classical receiver ceases to beat the pilot-free learned relay, whose amortized posterior corresponds to a horizontal reference (it uses no pilots at test). **Evaluation:** 50 trials $\times$ 20,000 bits per operating point (1,000,000 bits), matching Section 6.1.5 and for the same reason: the channel is redrawn per block, so trials are draws from the impairment family. In panel (b) each trial is further subdivided into independent blocks of the swept length, so every block length is measured over the same total bit budget and over many channel draws rather than a single one.

Conclusion (a sharp estimation-variance cliff, and a rate–reliability tradeoff): With a generous budget the classical partial posterior wins, as expected: pilot-aided Viterbi reaches payload BER 0.0285 at 800 pilots and degrades only gently down to 0.0335 at 20 pilots, comfortably below the MLP’s pilot-free 0.0487. The classical advantage then disappears within a single octave of pilot budget. At 10 pilots Viterbi has already lost its

edge, 0.0545 against the MLP's 0.0487, and at 5 pilots it collapses to 0.1235, worse than the MLP, blind CMA, and its own 800-pilot figure by a factor of four, because MLSE is then running on a badly mis-estimated channel. The crossover therefore sits between 20 and 10 pilots per block. The mechanism is *estimation variance*, not a rank deficiency: with 5 pilot observations and 3 complex taps the least-squares problem is still formally overdetermined, so the channel remains identifiable in principle, but there are too few observations to average down the noise. The resulting LS error at 10 dB is large enough that the Viterbi trellis is occasionally built on a badly wrong channel, and these catastrophic fits dominate the mean. The 95% confidence interval at that point makes this explicit (0.1235 ± 0.0304 , against ± 0.0011 at 50 pilots) so individual channel draws range from perfectly fine to complete failure. The learned relay, by contrast, is flat across the entire sweep at 0.0487: having amortized the channel *family* into its weights at training time, it neither benefits from nor requires per-block pilots. The crossover is thus governed by the reliability of the channel estimate, not by SNR: the classical receiver wins whenever it can afford enough pilots to estimate the channel *accurately*, and the pilot-free learned relay wins once the budget falls below that point.

The practical force of this depends on block length, because a fixed pilot *count* is a growing pilot *fraction* as blocks shorten. Panel (b) sweeps block length at the same operating point with the classical receiver spending its minimum viable ten pilots per block, and adds blind CMA as the second pilot-free reference. The payload BERs of Viterbi and the MLP are now statistically indistinguishable at every block length, both flat at approximately 0.049, so reliability no longer separates them at all; what separates them is spectral efficiency. At a 1000-symbol block the ten pilots cost 1% of the block, whereas at a 40-symbol block, representative of low-latency or fast-varying links, they cost 25%, so the classical receiver carries data on only 75% of the block against the learned relay's 100% for identical error performance.

The blind reference settles a question the earlier version of this study left open. Blind CMA is also pilot-free, so the learned relay is not unique in avoiding pilots; the distinction claimed for it was that it needs *zero per-block adaptation*, a single fixed forward pass, whereas CMA must converge its adaptation loop inside each block. Measuring CMA per block confirms this directly: its payload BER is 0.1723 at $L = 40$ and only improves to 0.1653 at $L = 1000$, against the 0.0645 it achieves when given a 20,000-symbol block to adapt over. CMA therefore does not converge within blocks of a thousand symbols or fewer, and is roughly three and a half times worse than either the MLP or pilot-aided Viterbi throughout this sweep. The partial-posterior picture therefore refines H6's boundary once more, and more sharply than before: against a classical receiver that can afford accurate channel estimation the learned relay achieves no better BER, but at short block lengths it matches that receiver's BER while removing its pilot overhead entirely, and at

these block lengths it is the only pilot-free option that stays reliable, because the classical pilot-free alternative has no time to converge: CMA sits at 0.1723 on 40-symbol blocks and is still at 0.1653 at 1,000. This is a statement about short blocks specifically, and it does not carry over to Section 6.1.5, where blocks are long enough for CMA to converge and it comes within a factor of 1.3 of the learned relay.

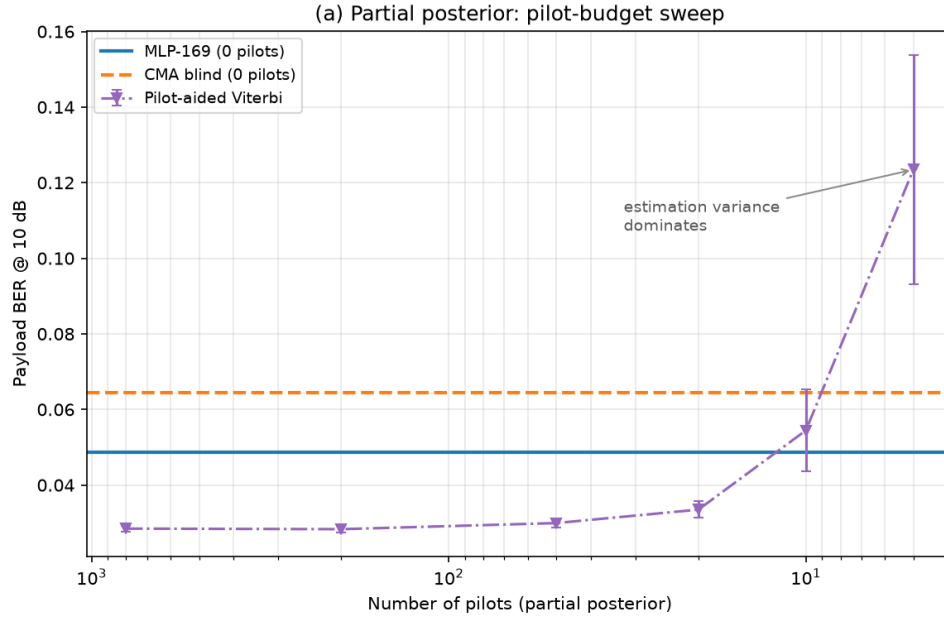


Figure 6.5: Pilot-budget sweep at 10 dB (mean over 50 channel draws, 20,000 bits/point). Pilot-aided Viterbi outperforms the pilot-free MLP down to approximately 20 pilots per block, loses its edge by 10 pilots, and collapses at 5 pilots when the least-squares channel estimate becomes too noisy to construct a reliable trellis (0.1235 ± 0.0304). In contrast, the MLP remains flat across the sweep because it operates with an amortized posterior and requires no pilots.

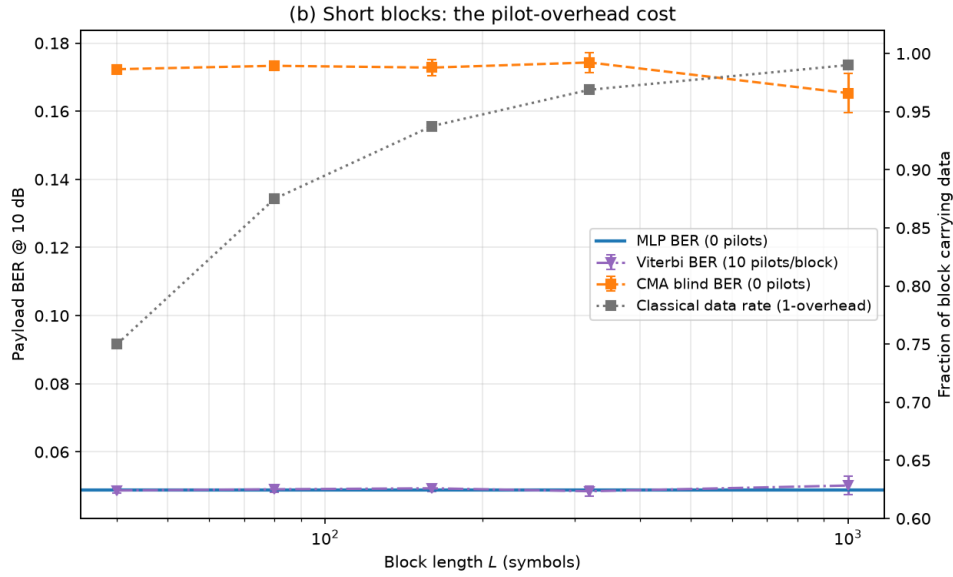


Figure 6.6: Block-length sweep with 10 pilots per block (mean over 50 channel draws, 20,000 bits/point, subdivided into independent blocks of the swept length). Pilot-aided Viterbi and the pilot-free MLP are indistinguishable in payload BER at every block length (both ≈ 0.049), but the classical receiver’s pilot overhead grows from approximately 1% at $L = 1000$ to 25% at $L = 40$. The other pilot-free option, blind CMA, cannot converge within blocks this short and sits roughly three and a half times worse throughout. The MLP therefore matches the pilot-aided receiver’s reliability at no overhead, and is the only pilot-free method that remains reliable for short packets.

6.1.5 Layer 4 — Unknown Channel Family: The Posterior-Free (Blind) Regime

This is layer 4, the bottom of the ladder. Section 6.1.4 shrank the pilot budget until the classical estimate became too noisy to be worth having; this section removes it entirely, and with it the assumption that the channel’s family and statistics are known at all. The Viterbi baseline of Section 6.1.3 enjoyed a per-block posterior: 200 pilot symbols yield a least-squares channel estimate, after which MLSE is optimal. Taking the pilot count to zero raises the sharper question, what if *no* such posterior is available? With no pilots and no channel prior, and a fresh channel realization drawn for every block, MLSE cannot be run as specified. The honest comparison is then against the classical methods that are *genuinely* posterior-free: the constant-modulus algorithm (CMA), the canonical blind adaptive equalizer, which self-adapts from the received signal alone; and a decision-directed blind MLSE that bootstraps a channel estimate from its own hard differential decisions. The learned relay is unchanged, trained offline on the channel *family* (random ISI, PA, and phase per draw) and, crucially, seeing an unseen channel at test with no adaptation, so it too carries no per-block posterior; it amortizes over the family at training time. Unlike the canonical study, the challenge here is not symbol-level denoising on a memoryless channel but inference on an unknown channel with memory and no per-block posterior

information. **Evaluation:** 50 trials $\times$ 20,000 bits per SNR point (1,000,000 bits), 95% CIs by the $1.96 \sigma / \sqrt{n}$ convention. The trial count is deliberately higher than the chapter’s usual ten: unlike the fixed impairment of Section 6.1.3, the hop-1 channel here redraws its ISI, amplifier and phase realization for *every* block, so a trial is one draw from the impairment family and the ensemble average is limited by the number of trials, not by bits per trial. Fifty draws are used rather than ten so that the reported curves reflect the family and not a handful of particular channels.

A correction to the blind equalizer. The constant-modulus algorithm used here was originally implemented with a fixed (unnormalized) step size. Because the CMA error term is cubic in the equalizer output, a fixed step is a positive-feedback loop: the implementation happened to remain bounded over the short blocks first used, but diverges to numerical overflow on longer ones, after which the equalizer output is noise rather than an estimate. The step is therefore normalized by the input-segment energy in the standard NLMS manner, which is both the textbook formulation and a strictly *stronger* classical baseline. All CMA results below use the corrected equalizer; the uncorrected one is not reported, since its outputs past the divergence point measure an implementation fault rather than the algorithm.

Conclusion (learning does not beat blind classical, but is stable where blind classical is not): A well-designed blind classical method still works in this regime: the corrected CMA converges smoothly to BER 3.3×10^{-3} at 20 dB, and the family-trained MLP reaches 2.6×10^{-3} . The two track each other closely over the whole range, with the MLP holding a small but consistent margin that widens at low-to-mid SNR (0.0937 vs. 0.1182 at 8 dB). This is the correct answer to “what if there is no posterior?”: the learned relay does *not* render classical processing obsolete, because blind adaptive equalization is itself a posterior-free classical tool that comes within a factor of 1.3 of the learned relay at high SNR. The learned relay’s advantage is instead one of *robustness and cost*. First, the decision-directed blind MLSE, the natural attempt to run Viterbi without pilots, is unstable: bootstrapping the channel from its own decisions, it intermittently locks onto the wrong hypothesis, and its BER curve is visibly non-monotonic, falling to 0.0374 at 16 dB before rising again to 0.0498 at 20 dB. Its 95% confidence interval at 8 dB is ± 0.0348 , roughly seven times the MLP’s ± 0.0046 and two and a half times CMA’s ± 0.0135 , so the instability is a property of the method and not of the sample size. Second, CMA must iterate to convergence on each received block, whereas the MLP is a single fixed forward pass with no per-block adaptation loop and no convergence to monitor; Section 6.1.4 showed this distinction becoming decisive once blocks are short. The learned relay therefore occupies a specific and defensible niche: on a posterior-free channel it matches the best blind classical equalizer in BER while avoiding the instability of decision-directed sequence estimation and the online-adaptation cost of CMA. It buys *amortized, one-shot, stable* inference over an

407 impairment family, not a lower error floor than classical signal processing can achieve.

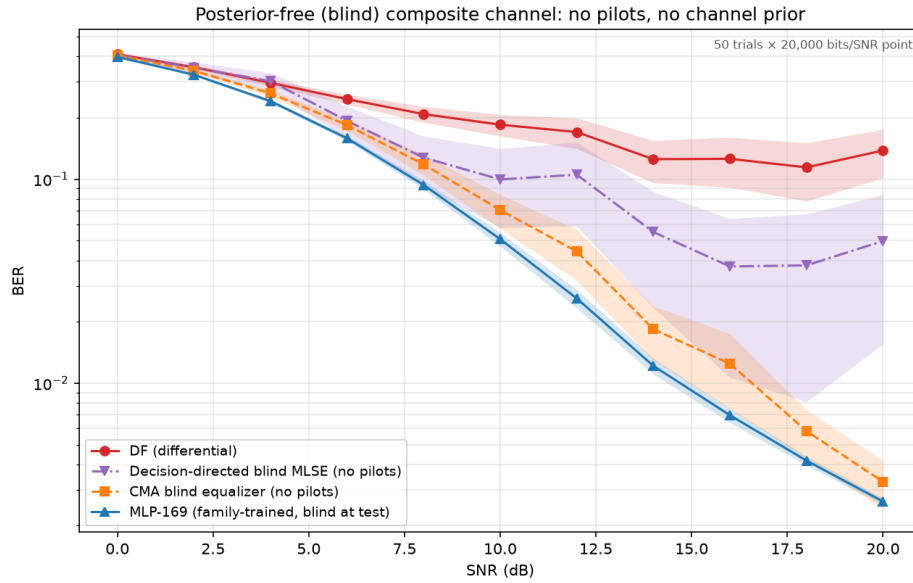


Figure 6.7: Posterior-free composite channel (no pilots, no channel prior, fresh realization per block; mean over 50 channel draws, 20,000 bits/point, 95% CI bands). The corrected CMA (blind classical) tracks the family-trained MLP closely, the MLP holding a small consistent margin; decision-directed blind MLSE is unstable (non-monotonic tail, CI band several times wider) because it bootstraps the channel from its own decisions; plain differential detection floors. Without a posterior, learning only narrowly outperforms the best blind classical equalizer, but does so with one-shot, stable inference and no online adaptation.

408 6.1.6 Complexity: The Cost Side of the Trade-off

409 Across the unknown-channel studies the pilot-aided Viterbi receiver, where it applies,
 410 holds a small BER edge over the learned relay. The learned relay's countervailing claim
 411 is complexity, which this subsection quantifies rather than asserts. Two axes matter: the
 412 per-symbol operation count and its scaling, and measured inference time.

413 **Per-symbol cost and scaling.** MLSE maintains M^{L-1} trellis states for a constellation
 414 of size M and channel memory L , performing an add–compare–select over M^L branch
 415 transitions per symbol; its cost therefore grows *exponentially* in memory and modulation
 416 order. The 169-parameter MLP performs one fixed forward pass, about 330 floating-point
 417 operations per symbol. That figure is a constant of the *deployed* architecture, not a quantity
 418 independent of the channel: the input window has to span the memory, so its length grows
 419 with L , and the output dimension grows with the constellation, from a single real output
 420 at $M = 2$ to the 193-parameter QPSK relay used earlier in this chapter, and further still
 421 for denser constellations. Once those are fixed, the per-symbol cost is the same for every
 422 symbol and every channel realization, which is the property the comparison below relies
 423 on. The contrast with MLSE is therefore one of *growth rate*, not of a constant against a
 424 variable: the network's cost rises roughly linearly in window length and output dimension,

425 the trellis’s exponentially in M^L .

426 For the channel actually studied (BPSK, $L = 3$; 4 states) Viterbi is in fact the cheaper
 427 of the two at roughly 64 operations per symbol, so the learned relay is not universally
 428 leaner. But the scaling diverges immediately: at $M = 4$, $L = 5$ the trellis has 256 states
 429 ($\sim 8,000$ operations/symbol), and at $M = 16$, $L = 5$ it has 65,536 states (~ 8.4 million
 430 operations/symbol), whereas a network resized for that same channel — an 11-sample
 431 window and a 16-way output head — is on the order of 10^3 operations by direct count
 432 of its layer dimensions, leaving a ratio in the thousands. The measured 330 applies to
 433 the BPSK, $L = 3$ case; the larger figure is arithmetic from the architecture, not a timing
 434 measurement. The learned relay’s advantage is thus not lower cost on a small channel but
 435 *constant* cost as the channel scales, exactly where classical MLSE becomes intractable
 436 and practical receivers must fall back on suboptimal reduced-state or decision-feedback
 437 equalizers.

438 **Measured inference time.** On identical signals (BPSK, $L = 3$), the MLP is $30\text{--}90\times$ faster
 439 in wall-clock terms for the reference implementations used here (a vectorized NumPy ma-
 440 trix multiplication versus an interpreted Python trellis loop) despite Viterbi’s lower nom-
 441 inal operation count. Part of that ratio is interpreter overhead rather than the algorithm
 442 itself: an optimized C or SIMD Viterbi would narrow the wall-clock gap substantially,
 443 although the per-symbol data dependency of the trellis scan is real and caps how far it can
 444 be parallelized. This parallelism gap is architectural and widens on hardware with wide
 445 SIMD or tensor units. The combined picture sharpens the thesis’s central trade-off: the
 446 learned relay accepts a small BER gap to a matched classical receiver in exchange for cost
 447 that is constant in channel complexity and structurally parallelizable, a trade that becomes
 448 favourable precisely as modulation order and channel memory grow.

449 The experiments of this chapter deliberately extend the thesis beyond the canonical mem-
 450 oryless model. The resulting conclusions should therefore be interpreted as robustness
 451 results under model mismatch rather than as replacements for the matched-model conclu-
 452 sions of Chapter 5. Together, the two parts of the thesis delineate where learning pro-
 453 vides value: not under matched memoryless conditions, where classical methods remain
 454 strongest, but when the assumptions underlying those methods are violated.

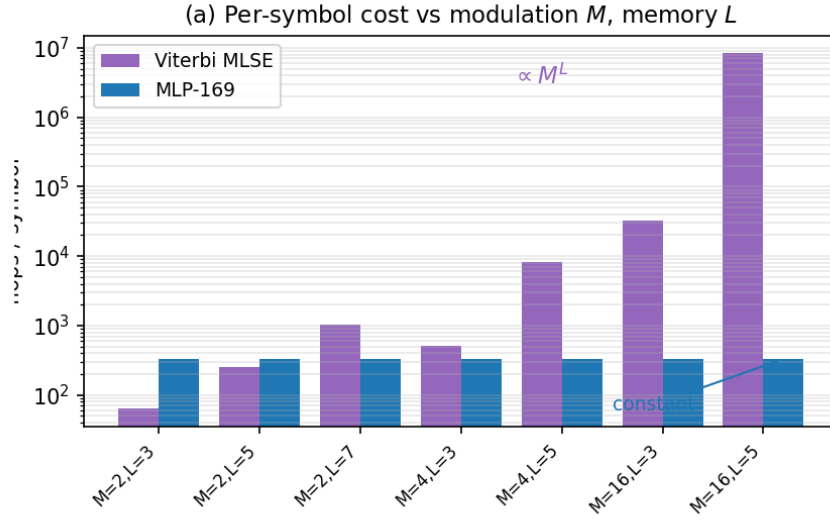


Figure 6.8: Per-symbol computational cost versus modulation order M and channel memory L . Per-symbol operations: Viterbi grows as M^L (intractable by $M=16$, $L=5$), while the network grows only linearly in window length and output dimension (~ 330 flops for the deployed BPSK, $L=3$ relay). On the small studied channel (BPSK, $L=3$) Viterbi is actually cheaper; the MLP wins on *scaling*, not on the small case.

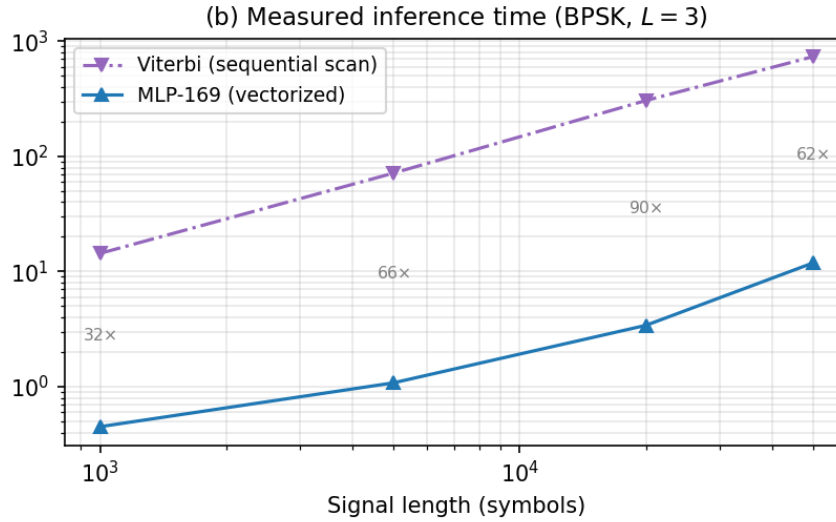


Figure 6.9: Measured inference time for BPSK with $L = 3$: the MLP is 30–90 $\times$ faster as a vectorized matmul versus Viterbi’s interpreted sequential trellis scan.

Chapter 7

Discussion and Conclusions

Scope of the discussion. The conclusions discussed in this chapter draw on two distinct parts of the thesis. Chapter 5 evaluates relay strategies under the canonical memoryless channel model, where transmission is uncoded, the noise is white, and processing is performed on a per-symbol basis. Chapter 6 then deliberately departs from that model by introducing impairments such as intersymbol interference (ISI), nonlinear distortion, and unknown phase. Accordingly, statements concerning channel memory, sequence detection, Viterbi MLSE, pilot-based estimation, or blind equalization should be understood as applying only to the extension study of Chapter 6, not to the canonical memoryless setup. The canonical core (Chapter 5) settles hypotheses H1–H4 and establishes the baseline finding that, on a matched known channel, a minimal learned relay does not improve on classical decode-and-forward. The thesis’s principal contribution (Chapter 6) settles H6 and is where the substantive new claim lies: a systematic delineation of when a learned relay adds value under unknown and mismatched channels. The interpretation below treats the two on equal footing, with the unknown-channel results carrying the main argument.

7.1 Interpretation of Results

The experimental results reveal several consistent patterns across the canonical setup and its two scoped extensions. This section interprets these patterns through the lens of the theoretical framework established in Section 2.5 and evaluates each research hypothesis.

7.1.1 Low-SNR Advantage of Neural Relays over AF (H1: Confirmed)

Low SNR (0-4 dB): AI advantage over AF. At low SNR, AI-based relays consistently outperform AF on the canonical channel, confirming H1. The theoretical explanation lies in the structure of the MMSE-optimal (posterior-mean) relay estimate. After coherent compensation, each axis of the canonical QPSK-over-Rayleigh link reduces to the same

26 real-valued single-sample problem as BPSK over AWGN (Section 4.1.2); the derivation
 27 below is carried out in that equivalent real form, applied identically per axis:

$$f^*(y) = \mathbb{E}[x|y] = \tanh\left(\frac{y}{\sigma^2}\right) \quad (7.1)$$

28

29 At low SNR (e.g., 0 dB, $\sigma^2 = 1$), this is a gentle sigmoid: $f^*(y) = \tanh(y)$. The neural
 30 network can approximate this smooth function accurately, producing a **soft estimate** that
 31 preserves information about the confidence of the decision. By contrast:

- 32 • **DF** applies the hard-decision function $f_{\text{DF}}(y) = \text{sign}(y)$, which discards all soft
 33 information. At low SNR, many received samples lie near the decision boundary
 34 ($|y| \approx 0$), and DF assigns them full confidence (± 1) regardless of the actual uncer-
 35 tainty. This information loss leads to excess errors on the second hop.
- 36 • **AF** applies $f_{\text{AF}}(y) = Gy$, which is linear and preserves the noisy signal structure but
 37 amplifies the noise by the same factor as the signal. The effective SNR degradation
 38 is given by $\text{SNR}_{\text{eff}} = \gamma^2/(2\gamma + 1) \approx \gamma/2$ at high SNR: a 3 dB penalty.

39 The AI relay implements a **non-linear soft mapping** that is intermediate between these
 40 extremes: it suppresses noise (like DF’s regeneration) while preserving soft information
 41 near the decision boundary (unlike DF’s hard decision). This explains why learned non-
 42 linear relay processing can outperform AF and, under selected channel conditions, also
 43 outperform DF at low SNR.

44 Among the AI methods, the low-SNR advantage over AF is realized by the *minimal* feed-
 45 forward relays, not by the larger sequence models. On the canonical Rayleigh channel
 46 (Table 5.2, 0 dB) the Hybrid, MLP and VAE relays reach 0.3329, 0.3349 and 0.3359 re-
 47 spectively, essentially matching symbol-wise DF’s 0.3337 and improving substantially on
 48 AF’s 0.4076, while the higher-capacity Transformer, Mamba-S6 and Mamba-2 (SSD) re-
 49 lays sit at 0.4014, 0.4071 and 0.4032 – that is, at essentially AF’s level, capturing almost
 50 none of the available advantage. The advantage over AF is therefore *not* driven by pa-
 51 rameter count: the smallest network already captures it, consistent with the complexity
 52 ordering discussed under H3 and with the convergence observed in the equal-parameter-
 53 budget comparison (H4, Section 4.4).

54 7.1.2 Classical Dominance at High SNR (H2: Confirmed)

55 **Medium-to-high SNR (≥ 6 dB): Classical dominance.** At medium and high SNR, DF
 56 matches or exceeds all AI methods with exactly zero parameters and zero training time.
 57 The theoretical explanation is straightforward: at high SNR, $\sigma^2 \rightarrow 0$ and $f^*(y) \rightarrow$
 58 $\text{sign}(y) = f_{\text{DF}}(y)$. The symbol-wise DF relay *exactly implements* the limiting form of

the posterior-mean estimate in this regime, while any neural network approximation introduces a small but non-zero reconstruction error due to the finite precision of the learned mapping and the tanh output saturation (which approaches but never reaches ± 1).

The effect is visible analytically in the AWGN calibration limit, at 10 dB the two-hop DF BER is $2Q(\sqrt{10}) \cdot (1 - Q(\sqrt{10})) \approx 0.002$, and empirically on the canonical Rayleigh channel, where DF attains the lowest or tied-lowest BER at every point from 6 dB upward. The AI models cannot beat DF in this regime, for the uncoded per-symbol setting, $\text{sign}(y)$ is the MAP symbol decision and hence optimal among per-symbol relay functions (Remark 4.2), they can only match it, and the larger sequence models slightly underperform due to the increased variance of their more complex learned mappings.

7.1.3 Robustness Across the Fading Ensemble

Robustness across fades. The relay ranking on the canonical channel is stable across the full evaluated SNR range, even though each transmitted symbol experiences an independent fading realization. The mechanism is that after coherent compensation ($\hat{x} = y \cdot h^* / |h|$), the relay processes a signal with AWGN-like noise at a *random effective SNR*; the network, trained across multiple SNR values, has learned a denoising function that adapts to different noise levels, so the Rayleigh ensemble (a mixture of effective SNRs weighted by the fading distribution) is handled by virtue of multi-SNR training alone, with no per-realization retraining or explicit CSI input.

7.2 The “Less is More” Principle (H3: Partially Supported)

One of the most practically useful findings is that relay BER does not improve with model size over the range evaluated: the Minimal MLP architecture (169 parameters) matches the performance of models with 100–140 $\times$ more parameters (3K–24K), and no larger evaluated model improved on it under the training protocol used. The stronger reading of H3 — that the curve turns upward because excess capacity causes *overfitting* — is not established by the evidence collected here. Each model was trained once, with no seed replication, no train/test-gap measurement and no held-out validation curve, so a size effect cannot be separated from run-to-run variance or from an optimization difficulty that more capacity happens to introduce. The claim carried forward is therefore the weaker, directly measured one: larger evaluated models did not improve BER. What follows in this section is offered as a candidate explanation for that observation, not as a demonstrated mechanism.

This result has important theoretical and practical implications:

7.2.1 Information-Theoretic Perspective

The canonical memoryless relay-denoising task maps a window of $2w + 1$ real-valued noisy observations to a single real-valued clean estimate, applied per axis to the canonical QPSK symbol (equivalently, on the real BPSK/AWGN calibration channel). Each real axis carries exactly 1 bit of information, and the Bayes-optimal estimator $f^*(\mathbf{y}) = \tanh(\mathbf{1}^T \mathbf{y} / \sigma^2)$ (for i.i.d. noise across the window) is a one-dimensional function of the sufficient statistic $\sum_i y_i$. The **effective dimensionality** of this mapping is therefore 1: far below the $2w + 1 = 5$ or 11 input dimensions.

As a heuristic, the minimum description length (MDL) principle suggests matching model complexity to the effective dimensionality of the target mapping. Since the target here is a one-dimensional monotone function of the sufficient statistic, even the 169-parameter network has capacity far in excess of what the mapping requires; this is offered as an intuition for why the smallest model suffices, not as a formal MDL bound.

7.2.2 Bias-Variance Analysis

The bias-variance decomposition [11] provides a quantitative framework:

$$\text{MSE}(\hat{x}) = \underbrace{(\mathbb{E}[\hat{x}] - f^*(y))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{x} - \mathbb{E}[\hat{x}])^2]}_{\text{Variance}} + \underbrace{\sigma_{\text{irred}}^2}_{\text{Noise floor}} \quad (7.2)$$

- **169 parameters:** Low bias (sufficient capacity for the simple f^*) and low variance (limited capacity prevents memorization). The model operates at the optimal bias-variance trade-off.
- **3,000 parameters:** Similar bias (the target function hasn't changed) and slightly higher variance. Performance is nearly identical to 169 params.
- **24,001 parameters (Mamba-S6):** Despite having $140\times$ more parameters than the MLP, it does not improve on it; on the canonical channel it is worse at low SNR (Table 5.2). The marginal utility of the additional parameters is at best vanishingly small.

This decomposition is a framework for interpreting the observation, not a measurement of it: bias and variance were not estimated separately, which would require training each configuration repeatedly across seeds and comparing training against held-out error. An earlier version of this section additionally attributed “clear overfitting” to an 11,201-parameter Maximum MLP; that configuration is not part of the reported experiments and no such measurement exists, so the claim has been withdrawn rather than restated.

7.2.3 Practical Implications

1. **Occam's Razor for relay AI.** The canonical relay denoising task has inherently low complexity: the per-axis mapping from noisy to clean symbols is fundamentally simple, and the neural network needs only enough capacity to learn a non-linear threshold function with some contextual smoothing. Adding parameters beyond this minimal requirement bought no measurable accuracy anywhere in this thesis.
2. **Deployment feasibility.** A 169-parameter model requires approximately 0.7 KB of memory (169 float32 values) and trains in seconds on a CPU. This makes AI-based relay processing viable even on severely resource-constrained embedded relay nodes, such as IoT devices or sensor network relays. The model can be stored in on-chip SRAM and executed without any external memory access.
3. **Generalization.** The 169-parameter MLP handles the full Rayleigh fading ensemble, every effective SNR the fades produce, with a single set of weights and no per-condition retraining, an instance of generalization that would be harder to achieve with a larger, more specialized model.

7.3 Practical Deployment Recommendations

Based on the comprehensive evaluation, we propose the following deployment strategy:

Table 7.1: Practical deployment recommendations: recommended relay strategy by operating regime.

Operating Regime	Recommended Relay	Rationale
Low SNR (0-4 dB)	MLP Minimal / Hybrid / channel-specific selection	Neural relays often help; exact best choice is channel-dependent, with DF remaining strong on some fading scenarios
Medium SNR (4-8 dB)	Hybrid (MLP + DF)	Automatic switching at empirically chosen crossover
High SNR (>8 dB)	DF	Zero parameters; lowest BER among the evaluated per-symbol relays
Resource-constrained	MLP Minimal (169 params)	0.7 KB memory, <5 s training

Operating Regime	Recommended Relay	Rationale
Best overall (BPSK/QPSK)	Hybrid	Combines AI advantage with DF reliability

The Hybrid relay is recommended as the default deployment choice because it automatically selects MLP processing at low SNR and DF at high SNR, achieving near-optimal performance across the entire SNR range with minimal complexity.

7.3.1 Latency and Adaptivity Considerations

Two practical questions bear on whether a learned relay can be deployed with minimal impact: the latency it adds relative to AF/DF, and its ability to adapt to varying channel conditions without retraining.

Latency. Two distinct latency components must be separated. The first is *buffering* latency, incurred by the sliding window $y_R[i - w : i + w]$. In the half-duplex protocol studied here this component is absorbed by the protocol itself: the relay buffers the entire listening-phase block before the transmission phase begins (Remark 4.1), so the window’s w -symbol look-ahead adds no delay beyond what store-and-forward operation already imposes. Only in a streaming or ultra-low-latency protocol, outside the scope of this thesis, would the symmetric window cost w symbols of look-ahead that the memoryless AF/DF baselines ($w = 0$) do not pay. The second component is *compute* latency: inference cost per relayed symbol. Here the U-shaped complexity result (H3) is decisive: since the 169-parameter MLP matches the performance of models $100 - 140\times$ larger, the deployed network reduces to two small matrix–vector products per symbol, a per-symbol cost that is orders of magnitude below a single channel use’s duration on any modern embedded processor, and negligible in absolute terms even if it cannot literally match the single comparison of DF’s $\text{sign}(y)$. The compute-latency argument therefore holds *because of*, not *despite*, the less-is-more finding; it would not hold for the 24K-parameter sequence models, whose recurrent or attention-based structure imposes materially higher per-symbol cost for no BER benefit at the relay window length.

The coded study of Section 5.4.3 adds a third component that the canonical comparison never had to consider, and it is the largest of the three: *buffering*. A block relay cannot emit until it has decoded a whole frame (202 symbols there), and the soft BCJR variant cannot even in principle, its backward recursion running from the end of the frame. A windowed learned relay needs only its w -symbol look-ahead, ten symbols, and that figure is constant in frame length where the block relays’ grows with it. The measured arithmetic points the same way ($0.30\ \mu\text{s}$ per symbol against Viterbi’s 15.07), so on the coded link the learned relay’s advantage in delay and cost is structural rather than marginal, even where

172 its BER advantage is small or absent.

173 **Adaptivity.** A single network, trained across the range of simulated SNRs, handles the
 174 entire canonical fading ensemble without retraining or explicit CSI input (Section 7.1.3):
 175 after coherent phase compensation, each fading realization presents the relay with an
 176 AWGN-like observation at a random effective SNR, and multi-SNR training already cov-
 177 ers this mixture. Two honest qualifications apply. First, this adaptivity is an advantage
 178 over AF, whose fixed gain is tuned to an assumed SNR, but not over symbol-wise DF,
 179 whose $\text{sign}(y)$ slicer is equally condition-agnostic; adaptivity therefore does not alter
 180 the H2 conclusion. Second, adaptation has been demonstrated only within the canonical
 181 Rayleigh ensemble; generalization to other channel families (Rician, bursty interference,
 182 phase noise, hardware non-linearities) remains untested and is identified as future work.

183 7.4 Limitations

184 Several limitations of this study should be acknowledged, as they define the boundary
 185 conditions under which the conclusions hold:

- 186 1. **Modulation scope.** All experiments use QPSK, the canonical constellation. Higher-
 187 order constellations are not investigated: 16-QAM and denser grids break the per-
 188 axis relay formulation and require a joint N -class 2D relay whose output dimension
 189 scales with M , which is left to future work.
- 190 2. **Perfect CSI.** We assume perfect channel state information at each receiver. In prac-
 191 tice, channel estimation errors would affect the coherent compensation step and
 192 hence relay performance. Imperfect CSI introduces a model mismatch between the
 193 assumed and actual channel, which could differentially impact the relay strategies:
 194 AI relays might be more robust (having learned from noisy data) or less robust (hav-
 195 ing not been exposed to estimation errors during training).
- 196 3. **Static offline training.** AI relays are trained once on synthetic data and deployed
 197 without adaptation. While multi-SNR training covers the canonical fading ensem-
 198 ble (Section 7.1.3), and the unknown-channel study (H6) shows a fixed network can
 199 learn a *stationary* unknown impairment, this protocol does not exploit online infor-
 200 mation; adaptive or continual training would be required if the deployment channel
 201 drifts over time.
- 202 4. **Single relay.** The framework considers a single relay node. Multi-relay cooperation
 203 introduces relay selection, power allocation, and cooperative diversity dimensions
 204 not captured in the two-hop model. With multiple relays, the system-level optimiza-
 205 tion (which relay to use, how to combine multiple relay observations) becomes as
 206 important as the per-relay processing function studied here.

- 207 5. **Two-hop only.** Extension to multi-hop networks with 3+ relays introduces noise
208 accumulation (for AF) and error propagation (for DF) that grow with the number
209 of hops. AI relays might show greater advantage in multi-hop settings where noise
210 accumulation is more severe.
- 211 6. **Fixed window size.** The relay window size ($w = 2$ or $w = 5$) is fixed and rel-
212 atively small. Larger windows could provide more context for denoising, particu-
213 larly in channels with temporal correlation (e.g., block fading). A systematic study
214 of window-size optimization is deferred.
- 215 7. **No coding, in the core comparison.** The core comparison operates without error-
216 correcting codes, so the DF baseline there is its symbol-level variant (Remark 4.2).
217 Sections 5.4–5.4.5 measure a coded, information-theoretic block-DF baseline, its
218 soft-decision counterparts, a rate-adaptive envelope over a modulation-and-coding
219 grid, and that envelope re-derived under a round-trip latency budget, directly rather
220 than leaving any of this as an assertion, but that study is confined to one code family,
221 one frame length, and one channel model, and the relay and destination each decide
222 once with no iterative exchange between them.
- 223 8. **Single training run per model; unequal training budgets across architectures.**
224 Each learned relay is trained once, from one random initialization, rather than across
225 multiple seeds with mean/standard-deviation/confidence-interval reporting of train-
226 ing variance; the confidence intervals reported throughout this thesis are Monte
227 Carlo intervals over the *evaluation* channel and bit realizations for a fixed trained
228 model, not over the training procedure itself. Separately, the architectures do not
229 all receive identical training budgets (e.g. the cGAN trains for 200 epochs against
230 100 for the other relays, Section 4.2), a confound the normalized-parameter study
231 (Section 4.4) does not remove, since it equalizes parameter count and input window
232 but not epoch count. Consequently, the U-shaped complexity finding (H3) and the
233 claim that architecture is a secondary factor relative to capacity (Section 5.3) should
234 be read as observations from the single trained instance of each architecture stud-
235 ied here, not as claims that have been shown robust to training-seed variance or
236 fully isolated from training-budget differences; a multi-seed re-run with equalized
237 budgets is the direct way to close this gap and is not performed in this thesis.

238 7.5 Future Work

239 Several directions warrant further investigation:

- 240 1. **Time-selective fading.** The present study addresses channels that are static within
241 a block. Extending the learned relay to time-selective fading with continuously

- 242 varying unknown phase (e.g. Jakes/Clarke Doppler) is a natural next step, likely
 243 requiring explicitly phase-aware architectures (complex-valued state, learned phase
 244 unwrapping, or a training objective aligned with noncoherent detection rather than
 245 symbol MSE) evaluated against prediction-based noncoherent receivers.
- 246 2. **Denser constellations (64-QAM, 256-QAM).** The extension shows joint N -class
 247 2D classification eliminates the 16-QAM floor; scaling the formulation to 64 or 256
 248 classes may require deeper networks or hierarchical (coarse-to-fine) classification,
 249 and whether the U-shaped complexity curve (H3) shifts rightward with constellation
 250 density is a key open question.
 - 251 3. **Imperfect CSI.** Introduce channel estimation errors to assess robustness of AI relay
 252 processing under realistic conditions.
 - 253 4. **A BER-optimal (BCJR/APP) benchmark for the unknown-ISI channel.** Sec-
 254 tion 6.1 benchmarks against genie-CSI Viterbi MLSE, the optimal *sequence* detec-
 255 tor. Sequence-ML and bit-BER-optimal detection are distinct criteria on any chan-
 256 nel with memory, at both modulation orders studied here, so the benchmark is not
 257 BER-optimal in either case; for Gray-coded QPSK there is the additional effect
 258 that a wrong trellis path can differ from the truth by one or two bits with unequal
 259 probability. A BCJR/APP detector run over the same 3-tap trellis, which directly
 260 minimizes per-bit error probability rather than sequence error probability, would
 261 test whether the classical benchmark's margin over the learned relay changes once
 262 it is BER-optimal rather than sequence-optimal, and is the measurement needed to
 263 settle the mechanism conjectured in Section 6.1.2; this was not run here.
 - 264 5. **Online learning.** Develop relay strategies that adapt their parameters during oper-
 265 ation, tracking time-varying channel conditions.
 - 266 6. **Multi-relay networks.** Extend to cooperative relay selection and multi-hop routing
 267 with AI-optimized relays at each node.
 - 268 7. **Other channels and MIMO.** Evaluate the canonical conclusions on additional
 269 channel families (Rician line-of-sight fading, and AWGN treated as a full operat-
 270 ing point rather than the bounded calibration-plus-companion role it has here) and
 271 on a multi-antenna second hop, where destination equalization interacts with relay
 272 processing and the additivity of the two gains becomes a testable question.
 - 273 8. **End-to-end learning.** Train the entire communication chain (modulation, relay,
 274 demodulation) jointly using autoencoder-based approaches, and compare against
 275 the modular classical-plus-learned-relay cascade studied here.
 - 276 9. **Coded block-DF baseline (measured; open items remain).** Remark 4.2 scopes
 277 the core comparison's DF baseline to the uncoded, symbol-level variant. Section 5.4

measures the coded, information-theoretic block-DF baseline this item once only proposed, together with the soft-information relays (BCJR at the relay, posterior-mean read-out of the learned relay) and an adaptive-rate envelope over a modulation-and-coding grid. The substantive finding is mechanistic: a relay that decodes the code has spent its redundancy before transmitting, so a wrong decode arrives as a valid-but-wrong codeword the destination cannot repair, whereas a relay that only denoises leaves the redundancy intact. Rate selection then dominates the relay-decoding question by an order of magnitude, unless a latency deadline is imposed, at which point block-DF’s double buffering reopens the gap. What remains open: one code family, one frame length, one channel model; the soft block-DF relay was given the nominal noise variance rather than per-symbol CSI, which caused its 20 dB mis-calibration; iterative (turbo-style) relay–destination exchange was not evaluated; and the latency analysis uses structural buffer counts rather than a 10^{-5} -class reliability target, so it does not establish ultra-reliable low-latency performance.

10. **Longer relay windows with state-space architectures.** Future work should explore whether longer relay windows (e.g., 128–512 symbols) combined with a chunk-parallel state-space architecture can improve both BER performance and processing speed simultaneously; the training-time crossover between sequential and chunk-parallel formulations, removed from this thesis’s scope, is the natural companion measurement.

7.6 Conclusions

This thesis presents a comprehensive comparative study of nine relay strategies: two classical (AF, DF) and seven AI-based (MLP, Hybrid, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD). All evaluated on one canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER; BPSK appears only on the separate AWGN calibration channel). The study addresses three identified research gaps (Section 3.2) through controlled experiments with statistical rigor (100,000 bits per SNR point, 10 independent trials, Wilcoxon significance testing). The following table summarizes the hypothesis outcomes:

Table 7.2: Research hypotheses, their statements, and experimental outcomes.

Hypothesis	Statement	Result
H1	Some AI relays outperform AF at low SNR	Confirmed: the best neural relays beat AF at 0–4 dB on the canonical channel

Hypothesis	Statement	Result
H2	DF lowest-BER among per-symbol relays at high SNR	Confirmed: DF matches or beats all AI methods at ≥ 6 dB with 0 parameters
H3	U-shaped complexity curve (BER)	Partially supported: 169 params matches 24K, and larger evaluated models did not improve BER under the training protocol used; the overfitting mechanism is not established, as each model was trained once and no training-variance or train/test-gap analysis was performed
H4	Architecture convergence at equal parameter budget	Confirmed at high SNR, rejected at low: at 3K params the six architectures span 4.1% BER at 20 dB but 17.4% at 0 dB, splitting into feedforward and sequence groups. Equalizing the budget required changing width, depth, state dimension and input window jointly (Table 4.2), so this equalizes capacity, not the other architectural degrees of freedom

Hypothesis	Statement	Result
H6	Memoryless classical relays fail on unknown channels; a minimal MLP mitigates by learning, near the model-aware optimum	Confirmed: on unknown ISI, AF/DF hit the analytic 0.25 floor (DF non-monotonic in SNR) while the 169-parameter MLP reaches $\text{BER} < 5 \times 10^{-5}$, within $\approx 1\text{--}1.5$ dB of genie-CSI Viterbi MLSE and indistinguishable from it under the Rayleigh second hop; the same architecture mitigates a composite $\text{ISI} \times \text{PA} \times \text{phase}$ cascade with no impairment model, tracking ≈ 2 dB behind pilot-aided Viterbi and matching blind CMA when no posterior exists; on the canonical control it only matches DF (H2), the advantage is bounded to structural model-class mismatch representable at minimal capacity

307 The main conclusions, in order of significance, are:

- 308 1. **AI relays beat AF at low SNR (0-4 dB) but not DF.** On the canonical channel,
309 the best neural relays clearly outperform AF in the low-SNR regime while tracking
310 symbol-wise DF within statistical uncertainty; DF remains lowest or tied-lowest at
311 every SNR point. The theoretical explanation is that neural networks approximate
312 the MMSE-optimal soft-threshold estimate $f^*(y) = \tanh(y/\sigma^2)$, whose end-to-end
313 benefit over the hard slicer is limited in this uncoded setting.
- 314 2. **DF attains the lowest per-symbol BER at medium-to-high SNR (≥ 6 dB) with**
315 **zero parameters.** The classical decode-and-forward relay requires no training and
316 no parameters yet achieves the best performance among the evaluated per-symbol

relays when channel quality is sufficient for reliable first-hop demodulation. At high SNR, the posterior-mean estimate converges to $\text{sign}(y)$, which is exactly the symbol-wise DF operation.

3. **No AI architecture separates itself at original parameter counts.** On the canonical channel the best neural relays (MLP, Hybrid, cGAN) perform within statistical uncertainty of one another, and classical DF matches or exceeds all of them across the SNR range. The selective state space model's $O(n)$ complexity and adaptive filtering mechanism make it an efficient sequence-model choice, but no architectural inductive bias yields a decisive BER advantage on this task.
4. **Architecture matters less than parameter count at equal scale — but only at high SNR.** When all models are normalized to approximately 3,000 parameters (Table 5.3), the spread across the six architectures falls monotonically with SNR: 17.4% at 0 dB, 9.3% at 12 dB, 4.1% at 20 dB. In the high-SNR regime the original claim holds and architecture is close to irrelevant. At low SNR it does not: the six models separate cleanly into a feedforward group (MLP, Hybrid, VAE at 0.336–0.342) and a sequence group (Transformer, Mamba-S6, Mamba-2 at 0.400–0.407), a gap far larger than the confidence intervals, with the feedforward models close to DF and the sequence models close to AF.

Two points are worth separating here. The split is by *family*, not by capacity, since all six carry the same parameter budget — so at low SNR architecture does matter, and the sequence models' inductive bias is actively unhelpful on a memoryless channel where there is no temporal structure to exploit. And the VAE is no longer the exception it was: at 3K parameters it sits inside the feedforward group at every SNR (0.0101 against MLP-3K's 0.0099 at 20 dB). The earlier reading, that the VAE was a consistent underperformer, does not survive re-measurement. The cause is not the corrected noise convention, which a controlled A/B rules out: training the same relay on data generated under the old convention and under the corrected one produces curves that agree to within Monte Carlo noise at every SNR. The earlier figures came from an implementation that predates the current relay module, so the discrepancy is one of code rather than of channel calibration. See the discussion of the generative paradigm in Section 2.3.3.

5. **A 169-parameter network is sufficient for relay denoising.** The Minimal MLP architecture achieves performance comparable to models 100-140× larger, demonstrating that the relay denoising task has inherently low complexity. The bias-variance analysis explains this: the target function is simple (a soft threshold), so additional parameters increase variance without reducing bias.
6. **The Hybrid relay is the recommended practical choice, given a usable estimate**

of the operating SNR. By combining AI processing at low SNR with classical DF at high SNR, the Hybrid relay achieves near-optimal performance across the entire SNR range with only 169 parameters. It selects between the two regimes from an SNR estimated out of the received signal power, so no manual per-regime configuration is needed. The recommendation is conditional in a way the experiments here do not discharge: the switching threshold is fixed at the crossover measured on the training sweep, and neither the accuracy of the run-time SNR estimate, the sensitivity of the BER to a misplaced threshold, nor any online adaptation of it was evaluated. All reported Hybrid results are therefore obtained under an accurate operating-SNR estimate, and the margin over its two constituents is small enough at the crossover that a mis-estimate there would plausibly erase it.

It is important to note that sequence-detection methods such as Viterbi MLSE appear only in the unknown-channel study of Chapter 6, where channel memory is introduced explicitly through a 3-tap FIR/ISI channel. The canonical system model of Chapters 1–5 remains memoryless and therefore requires only per-symbol processing.

The unknown-channel contribution (Chapter 6) settles the complementary question of *when the learned relay is not merely competitive but necessary*:

- **On channels violating the deployed relay’s model class, learning restores reliability at near-optimal cost (H6: confirmed).** The flat-channel control isolates the cause: against unknown phase, gain and per-branch asymmetry, all memoryless, the classical relay does not fail and the MLP only matches it (gap ≤ 0.0036), so parameter uncertainty alone is not what learning mitigates. Failure appears only with unmodeled *memory*: on unknown 3-tap ISI, AF and symbol-wise DF sit at the analytic 0.25 floor and DF actually worsens as SNR rises, while the same 169-parameter MLP learns a joint equalizer–detector and falls below 5×10^{-5} at 16 dB. The Viterbi benchmark bounds the claim: given the channel family and enough pilots, classical estimate-then-MLSE stays 1–1.5 dB ahead, so the learned relay’s value is *family-agnosticism* rather than optimality — near-MLSE reliability with no identification stage, on a composite ISI/nonlinearity/phase cascade it was never told the composition of, and, with no pilots or channel prior at all, matching blind CMA without its instability or per-block adaptation cost. Sweeping the pilot budget makes the boundary quantitative: pilot-aided Viterbi wins down to roughly ten pilots and then collapses at the identifiability threshold, while the learned relay is invariant to it. The crossover is therefore governed by identifiability and, through pilot overhead, by block length. Section 6 gives the full evidence; the complexity accounting there shows the cost side scales linearly in window and output dimension against the trellis’s M^L .

These findings demonstrate that neural network-based relay processing is a viable comple-

ment to classical approaches in the low-SNR regime, matching, though not surpassing, the per-symbol-optimal DF baseline everywhere else at negligible computational cost. The overarching insight is that **model complexity should be matched to task complexity**: for the canonical relay denoising task, minimal architectures suffice, and the choice between neural network paradigms, i.e. feedforward or sequential, matters less than proper model sizing and regularization. The practical recommendation is to deploy a Hybrid relay with a 169-parameter MLP sub-network, which combines the neural low-SNR advantage over AF with DF's zero-parameter reliability at high SNR; for 16-QAM deployments, the relay should be formulated as a joint N -class 2D classifier rather than a per-axis I/Q-split denoiser; and on links with unmodeled impairments (impairments outside the canonical memoryless model, including ISI, bias, and hardware nonlinearity), a minimal learned relay delivers near-MLSE reliability without channel identification, where the memoryless classical relays fail outright (H6).

Chapter 8

Summary

Scope of the summarized results. The results summarized below originate from two complementary studies. The canonical study (Chapter 5) evaluates relay processing on a memoryless channel with white noise and uncoded transmission. The unknown-channel study (Chapter 6) extends that baseline by deliberately introducing impairments that violate those assumptions, including intersymbol interference (ISI), nonlinear distortion, and model mismatch. Conclusions involving sequence detection, Viterbi MLSE, pilot overhead, or blind equalization therefore refer specifically to that extension rather than to the canonical system model.

This work examined classical and neural network–based relay strategies for *two-hop relay communication* using a unified simulation framework. The objective was to analyze performance–complexity tradeoffs and determine when learning-based relays provide practical advantages over analytically derived methods. The entire study is carried out on one canonical setup, fixed in advance and never departed from: a two-hop SISO link over i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER, with the relay function as the only varied axis. The study showed the following:

1. Neural relays yield a consistent performance improvement over amplify-and-forward (AF) at low signal-to-noise ratio (SNR), where their learned soft denoising avoids AF’s noise amplification (H1).
2. Classical *decode-and-forward* (DF) matched or outperformed all neural approaches at **medium-to-high SNR** while requiring no trainable parameters, confirming that DF attains the lowest per-symbol BER in this regime (H2).
3. A systematic complexity analysis demonstrated that performance **does not improve monotonically with model size** (H3).
4. A **minimal neural relay** (169 parameters) achieved performance comparable to significantly larger models; no larger model evaluated here improved on it, though the

28 protocol used (one training run per model, no train/test-gap measurement) cannot
 29 attribute that to overfitting.

30 5. When neural relays were evaluated under equal parameter budgets, architectural dif-
 31 ferences largely vanished, indicating that **model capacity**, rather than architectural
 32 class, dominates performance for this task (H4).

33 6. From a deployment perspective, a `hybrid relay` combining neural processing
 34 at low SNR with DF at high SNR emerged as the **recommended default where the**
 35 **operating SNR can be estimated reliably**, achieving near-optimal performance
 36 across the SNR range with minimal overhead; the robustness of its switching thresh-
 37 old to SNR-estimation error was not evaluated

38 The first six findings concern the canonical memoryless relay model; the remaining find-
 39 ings summarize the deliberate model-mismatch extension of Chapter 6.

40 7. Beyond the canonical memoryless model, on an **unknown channel** violating clas-
 41 sical assumptions (unmodeled intersymbol interference or biased nonlinearity), AF
 42 and DF failed to relay reliably (exhibiting an analytic BER floor of 0.25, with DF
 43 *worsening* as SNR increased) while the same minimal 169-parameter MLP restored
 44 reliable relaying by learning the impairment from data, including an arbitrary com-
 45 posite cascade of ISI, amplifier nonlinearity, and unknown phase. It tracked $\approx 1\text{--}1.5$
 46 dB behind a *pilot-aided* Viterbi receiver on pure ISI (≈ 2 dB on the composite cas-
 47 cade) and, in the fully posterior-free regime (no pilots, no channel prior), matched
 48 the best blind classical equalizer while avoiding its instability and online-adaptation
 49 cost (H6).

50 8. Sweeping the intermediate *partial*-posterior case located the crossover precisely: it
 51 is governed by the **reliability of the channel estimate** and by pilot overhead. Pilot-
 52 aided classical processing wins on long blocks at negligible cost, but the zero-pilot
 53 learned relay becomes decisive once blocks are too short, or channels too transient,
 54 for pilots to yield an accurate estimate. The learned relay’s advantage is therefore
 55 specific to *structural* model-class mismatch (memory, nonlinearity, absent pilots)
 56 rather than to unknownness per se. Together with the high-SNR result, this de-
 57 lineates precisely where learned relays add value: never above the model-aware
 58 optimum, but uniquely robust to *not knowing which model applies* and to the pilot
 59 overhead of channel identification.

60 9. A **complexity accounting** completed the trade-off: the learned relay’s per-symbol
 61 cost grows only linearly in window length and output dimension, and $30\text{--}90\times$ faster
 62 in measured wall-clock as a vectorized operation for the reference implementations
 63 compared (Section 6.1.6), whereas exact Viterbi MLSE grows as M^L and becomes
 64 intractable. The learned relay thus trades a small BER gap for cost that scales grace-

65 fully exactly where classical sequence detection does not.

66 Overall, the results indicate that learning-based relays are best used as targeted enhance-
67 ments rather than universal replacements, with simplicity, parameter efficiency, and hy-
68 brid design as key principles for practical relay systems. All conclusions hold for the
69 *uncoded*, symbol-level canonical setting studied here (Remark 4.2); extending the com-
70 parison to coded block-DF, soft-information (LLR) forwarding relays, other channel fam-
71 ilies, multi-antenna configurations, and denser constellations (64-QAM and beyond) is
72 identified in Chapter 7 as the principal direction for future work.

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Chapter 9

Appendices

9.1 Appendix A: Mathematical Notation

Symbol	Description
b	Binary bit ($\in \{0, 1\}$)
x	Modulated BPSK symbol ($\in \{-1, +1\}$)
y	Received signal
n	Noise sample
h	Fading coefficient
σ^2	Noise variance
SNR	Signal-to-Noise Ratio (linear)
G	AF amplification gain
$\mathbf{W}$	Neural network weight matrix
$\mathbf{b}$	Bias vector
P_e	Bit error rate
$Q(\cdot)$	Gaussian Q-function
β	VAE KL weight
λ	Regularization parameter
η	Learning rate
Δ	State space discretization step
$\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$	State space model matrices

9.2 Appendix B: Model Architectures and Hyperparameters

Relay	Network Topology	Train Loss (LR)	Total Params	Implementation
MLP (Minimal)	Feedforward MLP, 5-symbol sliding window	MSE, lr=0.01, 100 epochs	169	NumPy (CPU)
Hybrid	MLP + DF switching via learned SNR threshold (~ 5 dB)	MSE (MLP only), lr=0.01	169	NumPy (CPU)
VAE	Encoder–decoder latent model ($7 \rightarrow 8 \rightarrow 1$)	β -VAE ($\beta = 0.1$), Adam lr= 10^{-3}	1,777	PyTorch (CUDA)
cGAN (WGAN-GP)	Conditional GAN (generator + critic)	WGAN-GP, Adam lr= 10^{-4}	2,946	PyTorch (CUDA)
Transformer	2-layer encoder, 4 heads, $d_{\text{model}} = 32$	MSE, Adam lr= 10^{-3}	17,697	PyTorch (CUDA/CPU)
Mamba-S6	Selective SSM, 2 S6 layers	MSE, Adam lr= 10^{-3}	24,001	PyTorch (CUDA/CPU)
Mamba-2 SSD	Selective SSM with SSD kernel (2 layers)	MSE, Adam lr= 10^{-3} , clip=1.0	26,179	PyTorch (CUDA)

Table 9.2: Comparison of relay models, training objectives, parameter counts, and implementations.

9.3 Appendix C: Reproducibility

9.3.1 Testing

159 automated tests (pytest) cover the channels, modulation, relay-strategy, simulation and statistics modules with a 100% pass rate. The suite requires only NumPy and SciPy; PyTorch is optional and needed solely by the neural relays that use it.

9.3.2 Reproducibility

Random seeds are controlled at the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results.

Re-measurement of all results. Every simulation result reported in this thesis was re-measured after three faults were found in the earlier data: a 3 dB error in the noise convention of the AWGN channel, which propagated into the *training* data of every learned relay and so moved results on channels the fault never touched; three tables measured on (constellation, channel) pairings the thesis had withdrawn, each traceable to an unexamined default argument; and result files predating the implementation of the relays they reported. The published numbers are those of the re-measurement, at $10 \text{ trials} \times 10,000$ bits per SNR point.

Three written conclusions changed as a consequence and are stated in their revised form in the body: hypothesis H4 is now confirmed at high SNR and rejected at low, rather than confirmed outright; the VAE relay does not fail, and the earlier failure is attributed to a

25 superseded implementation rather than to the noise convention, which a controlled A/B
26 rules out; and joint 16-class classification is reported as restoring parity with decode-and-
27 forward rather than as achieving near-zero error. A cell-by-cell record of the changes, the
28 runtimes, and the reasoning is kept with the source as `thesis/RERUN_CHANGELOG.md`,
29 and `verify_thesis_tables.py` checks every published cell against the file that
30 produced it.

9.4 Appendix D: Normalized 3K-Parameter Configurations

Model	Parameters	Window	Hidden / Architecture
MLP-3K	3,004	11	hidden=231
Hybrid-3K	3,004	11	hidden=231 (+ DF switch)
VAE-3K	3,037	11	latent=10, hidden=(44, 20)
cGAN-3K	3,004	11	noise=8, g_hidden=(30, 30, 16), c_hidden=(32, 16)
Transformer-3K	3,007	11	d_model=18, heads=2, layers=1
Mamba-S6-3K	3,027	11	d_model=16, d_state=6, layers=1
Mamba-2-3K	3,004	11	d_model=15, d_state=6, chunk_size=8, layers=1

All 3K configurations use a window size of 11 (vs. 5 for the original MLP/Hybrid and 7 for the original VAE/cGAN); the original sequence models already used 11. The parameter counts are within $\pm 1.2\%$ of the 3,000 target.

9.5 Appendix E: Master Experiment Ledger

Every numerical table and figure in this thesis traces to exactly one committed result file, verified cell-by-cell against that file by `verify_thesis_tables.py` in the code repository (Section 4.3.4). This ledger lists the modulation, channel, and result source for each major study, grouped in the order they appear.

Table 9.4: Master experiment ledger: modulation, channel, and result-file provenance for every major study.

Study	Modulation	Channel	Tbl/Fig	Result source
Canonical bench-mark	QPSK	Rayleigh	tbl:2, fig:10	results/modulation/qpsk_rayleigh.json
Normalized 3K-param. comparison	QPSK	Rayleigh	tbl:8	results/normalized_3k/3k_rayleigh.json
Coded block-DF, K -sweep	QPSK	Rayleigh	tbl:34, 35	results/coded_df_experiment.json
Coded high-SNR paired re-test	QPSK	Rayleigh	tbl:37	results/coded_high_budget_test.json
Coded relay-error mechanism	QPSK	Rayleigh	tbl:38	results/coded_error_mechanism.json
Coded soft-decision relaying	QPSK	Rayleigh	tbl:39	results/coded_soft_decision.json
Equal-throughput & latency	QPSK, 16-QAM	Rayleigh	tbl:40, 41	results/coded_latency_throughput.json
Adaptive modulation/coding (AMC)	QPSK, 16-QAM	Rayleigh	tbl:42	results/coded_rate_adaptation.json
Capacity under a latency budget	QPSK, 16-QAM	Rayleigh	tbl:43	results/coded_latency_capacity.json
Unknown ISI (layer 2)	BPSK	AWGN / Rayleigh (2nd hop)	tbl:E6	e6_unknown_channel_results/e6_sim_ported_results.npy + e6_viterbi_*.npy
Flat-channel control	BPSK	AWGN / Rayleigh	tbl:E6flat	e6_flat_ported_results.npy
Unknown ISI, QPSK generalization	QPSK	AWGN / Rayleigh	tbl:E6qpsk	e6_qpsk_unknown_channel_results.npy
Composite impairment cascade	DBPSK	AWGN / Rayleigh	E6composite	e6_composite_ported_results.npy
Partial posterior (pilot sweep)	BPSK	AWGN / Rayleigh	E6partial	e6_partial_ported_results.npy
Blind channel (layer 4)	BPSK	AWGN / Rayleigh	E6blind	e6_blind_ported_results.npy
Four-layer summary	mixed (above)	mixed (above)	tbl:layers	combination of the above

41 The result files above are themselves generated by named, committed scripts (e.g. coded_rate_adapt
 42 produces coded_rate_adaptation.json); the correspondence between script and
 43 output file is one-to-one and is not separately tabulated here since the filenames make it
 44 explicit.

אוניברסיטת תל-אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר להנדסת חשמל

ארכיטקטורות למידה עמוקה לתקשורת ממסר דו-קפיצתית: מחקר השוואתי של אסטרטגיות ממסר קלאסיות ומבוססות רשתות נוירונים

חיבור זה הוגש כעבודת גמר לקראת התואר "מוסמך אוניברסיטה"

בהנדסת חשמל ואלקטרוניקה באוניברסיטת תל-אביב

על-ידי

גיל צוקרמן

העבודה נעשתה בבית ספר להנדסת חשמל

בהדרכת ד"ר אנטולי חינה ופרופ' ראג'א ג'יריס

ניסן תשפ"ו

תקציר

2

3 תזה זו משווה בין אסטרטגיות ממסר קלאסיות ונלמדות לתקשורת דו־דילוגית. תצורה אחת
4 קבועה לאורך המחקר המרכזי ואינה משתנה: קישור SISO עם ממסר בודד, דעיכת ריילי
5 מהירה בלתי־תלויה ושווה־התפלגות המוחלת זהה בשני הקפיצות, בסיס־רוחב מרוכב, אפנון
6 QPSK מקודד־גריי, ותמסורת בלתי־מקודדת, כאשר פונקציית הממסר היא המשתנה היחיד.
7 תשע אסטרטגיות ממסר מיושמות: הגבר־העבר (AF), פענח־העבר סימבולי (DF), רשת
8 פרספטרון רב־שכבתית (MLP), מתג היברידי, מודלים גנרטיביים (מקודד אוטומטי וריאציוני
9 ורשת יריבותית מותנית), ומודלי רצפים (Transformer, Mamba-S6 ו-Mamba-2; SSD)
10 שמונה מהם מוערכים בהשוואה הסופית, כאשר הרשת היריבותית המותנית הוחרגה מטעמי
11 עלות. כל התוצאות הן אומדני מונטה קרלו המדווחים עם רווחי סמך של 95%, והסימולטור
12 מאומת תחילה מול הסתברויות שגיאה סגורות־צורה של BPSK על ערוץ AWGN המשמש רק
13 כייחוס לכיול; פענח־העבר סימבולי בערוץ ריילי הקנוני נבדק באופן בלתי־תלוי באותו אופן, מול
14 ההרכבה הסגורה־צורה הדו־קפיצתית, ומתאים עד כדי אחוז אחד בכל ערך SNR שנמדד.

15 בתרחיש הייחוס הקנוני, הממסרים הנלמדים הטובים ביותר עולים על AF ביחס אות־לרעש
16 (SNR) נמוך ומשיגים BER דומה ל־DF הסימבולי ב־SNR בינוני וגבוה, בעוד ש־DF משיגה
17 את ה־BER הנמוך ביותר מבין הממסרים הנבחנו ללא עלות פרמטרים. אין לקרוא בכך טענת
18 אופטימליות כללית עבור DF: זוהי קביעה אמפירית על ממסור בלתי־מקודד
19 וסימבולי־לפי־סימבול בטווח ה־SNR הנמדד כאן; תוצאות מידע־תאורטיות עבור DF
20 מבוסס־בלוקים עוסקות בתמסורת מקודדת ובאורכי בלוק אסימפטוטיים, ולא הן ולא המדידות
21 המדווחות כאן קובעות שלא ניתן לעקוף את DF באמצעות ממסר מקודד או מבוסס־מידע־רך;
22 ממסר מקודד מבוסס־החלטה־קשה נמדד ישירות בהמשך, בעוד ממסר מבוסס־מידע־רך נותר
23 פתוח. מחקר מורכבות מראה שהביצועים נקבעים בעיקר על ידי קיבולת המודל ולא על ידי
24 הארכיטקטורה ב־SNR גבוה, כאשר MLP בעל 169 פרמטרים משיג את ה־BER של מודלים
25 גדולים פי שני סדרי גודל; ב־SNR נמוך הארכיטקטורות נפרדות, כאשר הממסרים
26 מבוססי־הזנה־קדימה עוקבים אחר DF בעוד מודלי הרצפים מפגרים.

27 מחקר נוסף מדווח בנפרד מכיוון שהוא דורש תצורה שהתצורה הקנונית אינה יכולה לקלוט. הוא
28 מודד ישירות את פערי הקידוד שנותרו פתוחים לעיל: קוד קונבולוציוני חיצוני בקצב $1/2$ עם
29 ממסר DF מבוסס־בלוק אמיתי עולה על בסיס ה־DF הבלתי־מקודד מעל כ־6–8 dB, אך נחות
30 ממנו מתחת לסף זה, סף שמחריף ואינו נעלם עבור קודים חזקים יותר (אורך אילוץ שנסרק עד
31 7). מול בסיס קלאסי מקודד זה, הממסרים הנלמדים נחותים בבירור ב־SNR נמוך ובינוני

ועדיפים ב-SNR גבוה, ומדידת פלט הממסר מזהה את הסיבה, שאיננה שרשת נורונים מפענחת טוב יותר מ-Viterbi: Viterbi משדר כמחצית ממספר שגיאות הסימבולים, אך DF מבוסס-בלוק מקודד מחדש לאחר הפענוח, ולכן שגיאותיו השאריות מגיעות כמילות-קוד תקפות-אך-שגויות שהיעד אינו יכול לתקן, בעוד ממסר שרק מסיר רעש משמר את יתירות הקוד וכך 57%–74% משגיאותיו מתוקנות ביעד. העברת החלטות רכות במקום קשות משפרת את הממסר הנלמד בכל ערכי ה-SNR עם אותם משקלים בדיוק, אך אינה מצילה את DF מבוסס-הבלוק, ובכך ממקדת את מקור הבעיה: צריכת יתירות הקוד בממסר, ולא הקוונטיזציה בו. יתרה מזאת, בהשוואה ביעילות ספקטרלית שווה ולא ב- E_s/N_0 שווה, הקוד בקצב $1/2$ אינו מצדיק את עלותו כלל מתחת ל-20 dB, ובנקודה שבה כן, הרווח הוא פי 1.35 בלבד תוך תשלום של פי 20 בהשהיית האגירה ופי 50 בעלות החישוב ביחס לממסר הנלמד. כאשר קצב הקוד עצמו מותאם לערוץ, על פני רשת הכוללת את שתי הקונסטלציות וקצבים מ- $1/2$ עד $3/4$ וכן שידור בלתי-מקודד, מיקסום התפוקה בפועל מראה שבחירת הקצב חשובה בסדר גודל אחד לפחות יותר מבחירת אסטרטגיית הממסר, וכי הבחירה האופטימלית ב-SNR גבוה היא כיבוי הקידוד כליל, תוך מסירת פחות משליש מקיבולת שאנון הבלתי-מוגבלת של הערוץ אף שם -- פער הנקבע על-ידי הקונסטלציה הסופית ושיעור שגיאת הפריימים השיורי, לא על-ידי אסטרטגיית הממסר. תיחום זה עצמו מותנה בכך שלקישור חופש לבחור כל אורך-בלוק: מכיוון שממסר המפענח ומקודד מחדש נדרש לאגור בלוק שלם פעמיים, לעומת פעם אחת בלבד אצל ממסר המסיר רעש בלבד, הטלת תקציב השהיה הלך-ושוב נועלת את DF מבוסס-הבלוק מחוץ לאפשרות הקידוד מוקדם יותר ופותחת מחדש את הפער בין אסטרטגיות הממסר, ואף הופכת אותו בעד פי 1.5 בנקודות SNR שבהן ההשוואה חסרת-המגבלה העדיפה את הממסר הקלאסי.

התרומה העיקרית מאורגנת כסולם של ארבע שכבות, כאשר כל שכבה מסירה הנחה אחת בדיוק מהשכבה שמעליה, בעוד ההשוואה בין עיבוד קלאסי לנלמד חוזרת על עצמה ללא שינוי. בשכבה 1, התצורה הקנונית, העיבוד הקלאסי מנצח: DF הסימבולי אינו משתפר, תכונה של הבעיה ולא של האימון, שכן בערוץ חסר-זיכרון ותואם, החלטה קשה היא פונקציית הממסר האופטימלית לכל סימבול. שכבה 2 מוסיפה זיכרון ערוץ בדמות מסנן הפרעה בין-סימבולית (ISI) תלת-מקדמי בלתי-ידוע, תוך שמירה על ידיעת הערוץ קבועה. כאן הממסרים הקלאסיים חסרי-הזיכרון נכשלים לחלוטין, כאשר שיעור השגיאה נעצר ואז עולה עם עוצמת השידור ככל שההחלטות הקשות מפיצות את ההפרעה, בעוד ממסר נלמד וממוסגר-חלון בן 169 פרמטרים משקם את הקישור; בהינתן אותו מסנן, גלאי רצף בעל-נראות-מרבית מסוג Viterbi נותר מוביל ב-1 עד 1.5 dB כך שמה שהשכבה הזו מפרידה הוא מחלקת הפונקציות, ומה שהממסר הנלמד רוכש הוא עלות קבועה לכל סימבול מול עלות טרליס האקספוננציאלית בזיכרון הערוץ. שכבה 3 מדרדרת את אומדן הערוץ לתקציב פיילוטם סופי, ויתרון המקלט הקלאסי נמצא אמיתי אך יקר: הוא מחזיק עד כעשרים פיילוטם ואובד עד עשרה. שכבה 4 מסירה את משפחת הערוץ כליל, תוך הגרלת ערוץ חדש בכל בלוק, והממסר הנלמד מוביל על פני מאזן העיור הקלאסי הטוב ביותר, תוך הימנעות מחוסר היציבות של אמידת רצף מבוססת-החלטה.

התוצאות מתחמות משטרי הפעלה במקום להכריז על מנצח. כאשר מודל הערוץ המשוער תואם לערוץ האמיתי, אין שיפור על העיבוד הקלאסי ויש להשתמש בו. כאשר הערוץ בלתי-ידוע, או שאומדנו מתדרדר מתחת לכעשרה סימבולי פיילוט, או שאין כלל אומדן זמין לכל בלוק, הממסר

70 הנלמד הוא האפשרות שממשיכה לפעול, בעלות קבועה לכל סימבול וללא שלב זיהוי -- תוך
71 הכללה על פני מימושים של משפחת הערוץ שעליה אומן, לא על פני משפחות ערוץ שמעולם לא
72 הוצגו באימון.